What inertial microcavitation data cannot settle, and how to design the experiment that would

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Abstract

Inertial microcavitation rheometry infers a material's mechanical response from the radius history of one collapsing bubble, and reports the fitted stiffness and viscosity as material properties. We ask what those reports actually support, and what experiment would turn them into measurements. The second question is the harder one and is what most of this paper is about.

The first has three answers and all are negative. The likelihood depends on the product $g\alpha$ alone, so a reported shear modulus is a statement about where the prior box cuts a ridge, moving the box swings the model ranking by fifty nats. A residual near the measurement scatter is not adequacy: lag-one autocorrelation is 0.918, the 201 samples carry the information of about ten, and a replicated lack-of-fit test rejects every candidate on every dataset. And the parameters carry a model-form error budget no data of this kind can remove, one to two orders of magnitude wider than the posterior. The identified modulus could be wrong by 30 % to 57 %, the relaxation time by a factor of three to four.

So we ask what experiment would fix it. Scoring a single design by parameter precision or by model discrimination gives criteria that disagree, and both are delicate, a criterion defined by an inner minimum is multimodal here, and every improvement to the optimizer lowers it. Posing the problem over design *measures* instead makes it convex, and the Kiefer–Wolfowitz equivalence theorem then *certifies* the answer rather than reporting that a search stopped. One criterion covers every question the experiment can ask (material parameters, the model label, the initial radius) as the same expected information gain at different partitions, so answers in nats are comparable across them. Four extensions keep the certificate: averaging the information over the scatter a setpoint actually delivers, normalising candidates by what they cost, averaging over a computed model-error bound rather than a point estimate, and making the measured discrepancy a coordinate the design can see.

What that machinery finds is that the variable everyone optimises is the wrong one. Pricing the candidates, a trace of N samples costs N frames, puts the split between bubbles and frames inside the certified problem, and that split is worth +3.4 to +4.0 nats against at most 1.34 for the geometry. A geometry is not a knob, and designing as though it were costs about a nat, on scatter widths confirmed against 437 per-event radii. Designing against the bias bound rather than the fit is worth 0.39 to 0.67. The temperature dependence closes nothing at any budget or geometry: with temperature held, the information is singular and the design is refused. Judged by the criterion, the experiments analysed here are near the best possible for the hardest of the three model axes and 24 to 41 times worse than the best for the two easiest.

The model comparison is the worked example that supplies those inputs and is reported as such. A quadratic Zener solid, strain-stiffening elasticity *and* stress relaxation, is the best-fitting candidate at every temperature and under every perturbation we applied, including a pre-registered sweep of the noise weights and four constructions of the fit target. Neither ingredient suffices alone. But the margins that make it look decisive do not survive the effective sample size the same datasets imply. And the operator held fixed throughout this literature is beaten on every dataset while being ordered by none of them, once the residual correlation is taken seriously.

Keywords: inertial microcavitation rheometry; model selection; Bayesian evidence; identifiability; optimal experimental design

1 Introduction

Inertial microcavitation rheometry infers the mechanical response of a soft material from the radius history of a single collapsing bubble [1]. A laser generates the bubble, high-speed imaging records $R(t)$, and a constitutive

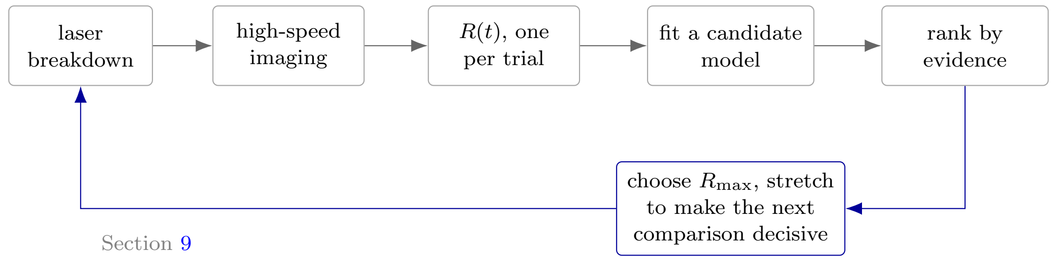

Figure 1: Where this work sits in the measurement chain. A laser nucleates a bubble, imaging datasets how its radius changes, and a material model is fitted to that curve. The fitted stiffness and viscosity are then quoted as material properties. This paper concerns the two right-hand boxes: which model to believe, and which experiment to run so that the question can be answered. Ranking on goodness of fit alone does not work here, because a model with more parameters always fits at least as well as the simpler model inside it.

model is fitted to that trace. The strain rates reached, above $10^3\,\mathrm{s}^{-1}$, are inaccessible to conventional rheometry, which is what makes the technique attractive.

The inference rests on choices that are rarely varied. A constitutive law is selected, a bubble-dynamics equation is assumed, thermal transport is included or neglected, and the resulting parameters are reported as material properties. Each choice conditions the answer, and the conditioning is seldom quantified.

What a dataset of this kind supports is considerably less than the reported parameters suggest. Most of what follows is therefore about the question that comes after: what experiment would make those parameters measurements.

What the datasets do not support. Three limits, each measured rather than asserted. The likelihood depends on the product $g\alpha$ alone, so a reported shear modulus is set by where the prior box cuts a ridge rather than by the data, and moving the box swings the model ranking by fifty nats (Section 3.5). A residual at the measurement scatter is not statistical adequacy: the residual autocorrelation is 0.918, the 201 samples carry the information of about ten, and a replicated lack-of-fit test rejects every candidate on every dataset (Section 5.1). And because every candidate is rejected, the fitted parameters carry a model-form error budget that no data of this kind can remove, one to two orders of magnitude wider than the posterior, and computable as a bound rather than left as a worry (Section 7).

What would fix it. Treat the choice of model as one more unknown to be fitted, and a single calculation says both how well an experiment separates the models and how well it pins the material. Pose that calculation over design *measures* rather than single settings and it becomes convex. The Kiefer–Wolfowitz equivalence theorem [2] then certifies the answer: a proof that no better design exists, which is unusual in this literature. The rest of the machinery is built to preserve it. Four extensions keep it: the setpoint is not the geometry, candidates do not cost the same, the parameters carry the bias bound above, and the measured discrepancy can be made a coordinate the design sees (Sections 9 to 14).

What it finds. The variable this literature optimises is not the important one. Splitting a frame budget between bubbles and samples is worth +3.4 to +4.0 nats against at most 1.34 for the geometry, and it has never been a design variable. Temperature dependence cannot be identified at any budget without varying temperature, which no design in this literature does. Judged by the criterion, the experiments analysed here are near the best possible for the hardest of the three model axes and 24 to 41 times worse than the best for the two easiest.

The comparison, as worked example. Ranking the candidates supplies what the design needs: the identified coordinates, the noise profile, the between-trial covariance and the discrepancy vector. It is reported for that purpose. A quadratic Zener solid wins at every temperature and under every perturbation applied to it. The bubble-dynamics operator held fixed at Keller–Miksis throughout this literature is beaten on every dataset. It is also not ordered against the other five once the residual correlation is taken seriously (Section 18), and by the same argument which questions are open at all is undetermined (Section 16).

Part I
What these data support

The design of Part II needs four things measured here: the coordinates the data can answer in, the noise profile, the covariance between trials, and the discrepancy vector. Each is a negative result in its own right, and each is an input there. The comparison they came out of is Part III.

2 The data and the noise model

Each dataset is a set of repeated laser-induced cavitation events at one temperature, recorded as $R(t)/R_{\max}$. Figure 2 shows the mean trace and the trial-to-trial spread. We use that spread as the measurement noise. It is measured rather than assumed, so $\chi^2/N \approx 1$ means a model reproduces the dataset to within experimental repeatability.

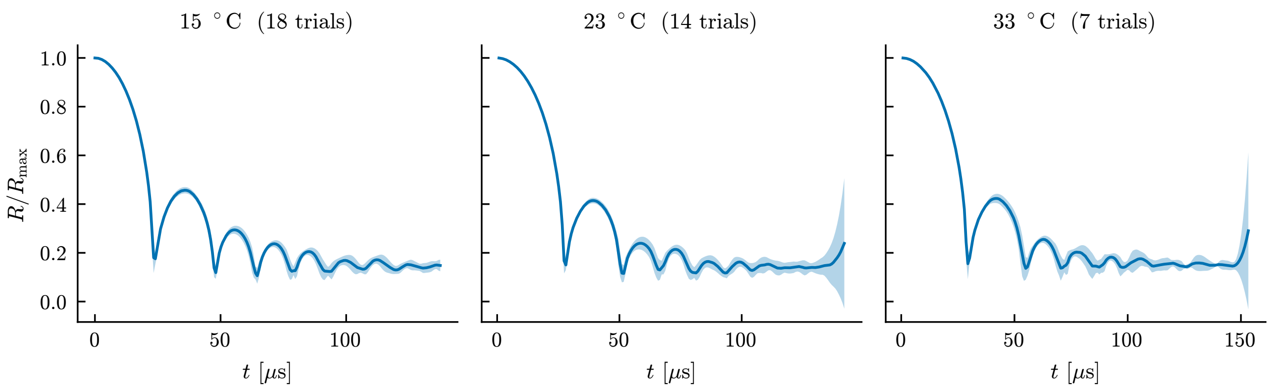

Figure 2: Measured datasets. Line is the mean over trials, band is ± one standard deviation across trials at each time. The spread is the noise model.

Two samples are excluded before fitting. A trial reading above its own maximum radius is a tracking failure. And the $t = 0$ sample, where every trial agrees exactly, is a definition rather than a measurement: giving it an error bar would invent an uncertainty and let it constrain the fit.

Each candidate is evaluated on a log-spaced grid over its own parameters, at one resolution for every model, so grid luck cannot decide the winner. The evidence is a grid quadrature over that box. It carries three further pieces: the noise scale marginalized under a half-Cauchy prior, a redundancy factor down-weighting parameters where a model merely reproduces a simpler model it contains, and a BIC-style complexity penalty.

That last piece is charged twice over and should be said plainly. Marginalising over the prior already penalises a parameter the data does not use: that is what the Occam factor of (5) is, so multiplying the integral by $N^{-k/2}$ charges dimension a second time, at 4.1 nats per parameter on the 15 °C dataset, and does so on the raw sample count where Section 22.1 puts the effective one near ten. It is a blemish rather than an error here: the candidates at the top of each dataset differ by at most two parameters against margins of 76 to 1496 nats, so removing the term, or recomputing it on the effective count, leaves the ranking and the top-five ordering unchanged on all three datasets. A comparison with smaller margins could not be treated so lightly.

Trajectories are compressible [3]. The incompressible approximation changes the residuals by a factor of two to four and must not be used for these datasets.

3 Method

Four objects carry the analysis. Each is derived here rather than quoted, because the choices between them are not neutral and two of the errors this study corrected came from treating them as interchangeable.

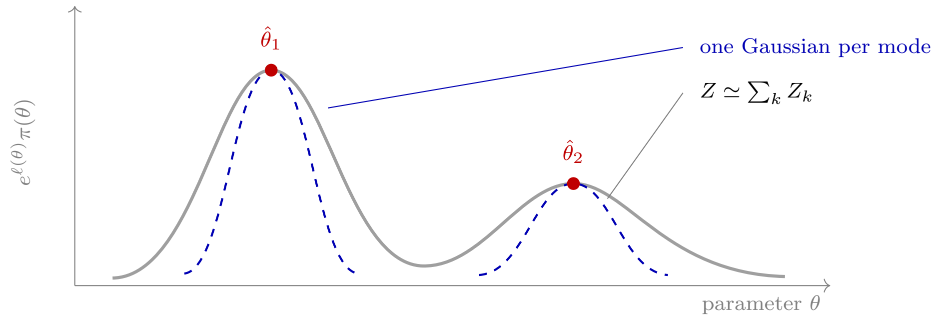

Figure 3: The Laplace expansion, and why the evidence is summed over modes. The integrand $e^{\ell}\pi$ is expanded about each maximum as a Gaussian, whose height and width give that mode's contribution Z_k in closed form; the evidence is $\sum_k Z_k$, computed as a `logsumexp` over the distinct endpoints a multistart finds. Taking only the best mode discards the rest of the integral, which is the difference between an integral criterion and a minimum: a missed mode biases the total slightly, where a missed basin corrupts a minimum entirely. This is why `fit_candidate` returns every distinct endpoint rather than its winner.

3.1 The likelihood, and what it assumes

Each dataset gives a mean radius y_i at time t_i over 18, 14 and 7 trials respectively, with the trial-to-trial standard deviation s_i as the noise estimate. Modeling the observations as independent and Gaussian about the prediction $m_i(\theta)$,

$$p(y \mid \theta) = \prod_i \frac{1}{\sqrt{2\pi}\,\sigma_i} \exp\left[-\frac{(y_i - m_i(\theta))^2}{2\sigma_i^2}\right], \qquad -2\log p = \chi^2(\theta) + \sum_i \log\bigl(2\pi\sigma_i^2\bigr). \tag{1}$$

Three assumptions are buried here and each is separately checkable. *Gaussianity* rests on y_i being a mean over trials, which is a good argument at 18 and a thin one at 7: on the 33 °C dataset the central limit theorem is doing very little work, and s_i is estimated from the same seven samples it is used to weight, a dependence this likelihood does not propagate. Conclusions resting on that dataset alone should be read accordingly. *Known* σ_i is not innocuous: using the sample spread as the true deviation understates uncertainty. Section 3.2 handles it by giving the noise scale a prior and integrating it out. *Independence* is the one that fails: the diagonal covariance in (1) says each residual carries fresh information, and Section 22.1 measures a lag-one autocorrelation of 0.918.

The deviations are not constant across the dataset. Optical tracking degrades where the wall moves fastest, so the variance is inflated by a logistic weight in the strain rate, $\sigma_i^2 = s_i^2/w_i$ with $w = w_0 + (1 - w_0)\,\mathrm{logit}^{-1}\bigl(\kappa(\dot{\varepsilon}_{\mathrm{th}} - |\dot{\varepsilon}_i|)/\dot{\varepsilon}_{\mathrm{th}}\bigr)$. This is a statement about the measurement, not the model, and it is why χ^2/N is dominated by the late dataset where the weights are large.

The three constants are a floor of 0.1, a steepness of 1 and a threshold rate of $10^5\,\mathrm{s}^{-1}$. Two things make them less load-bearing than they look. An overall factor on every variance is absorbed by the marginalised noise scale of Section 3.2, so the *level* of the weight cannot affect any comparison; only the *shape* can, because candidates place their residual in different parts of the dataset. And the shape was swept: each constant at its default and a factor of two or three either side, 27 combinations, with the range and the pass criterion fixed before the first evaluation. The quadratic Zener solid is the best-fitting candidate in all 27 on all three datasets, the tightest margin being 320 in χ^2 at 33 °C. That sweep recomputes χ^2 at each stored best fit rather than re-quadraturing 68,720 grid points per dataset twenty-seven times, so it speaks to the best-fit ordering rather than to the evidence itself.

3.2 Integrating out the noise scale

Rather than trust s_i, introduce a single scale β multiplying every deviation, $\sigma_i \to \beta\sigma_i$. Substituting into (1),

$$\log p(y \mid \theta, \beta) = -\frac{\chi^2(\theta)}{2\beta^2} - N\log\beta + \text{const}, \tag{2}$$

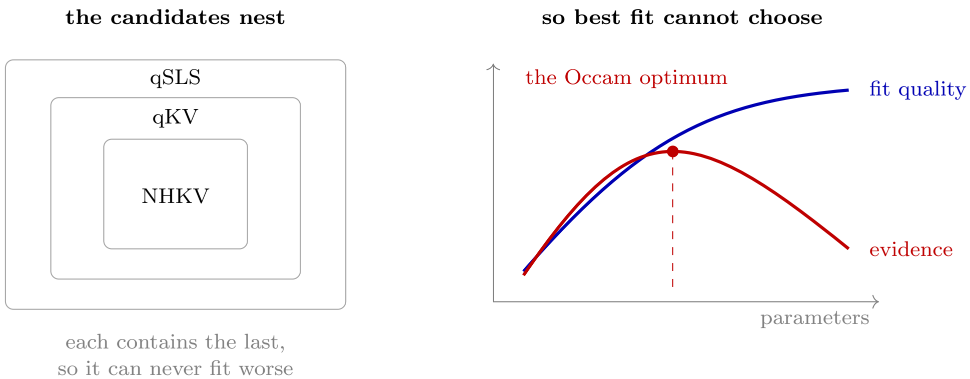

Figure 4: Why this study ranks by evidence and not by fit. The constitutive candidates nest, `NHKV` is `qKV` at zero stiffening and `qSLS` at zero relaxation time, so a larger model can never fit worse, and the best-fitting candidate is simply the most flexible one. The evidence adds the Occam factor of Figure 25, which charges a model for prior volume the data does not use, and so turns a monotone quantity into one with an interior maximum. Everything downstream depends on that penalty being computed correctly, which is why the pathology in Figure 25 matters.

which depends on the data only through χ^2 and N. That is what makes the marginalization cheap: the forward model never re-enters, so

$$\log p(y \mid \theta) = \log \int_0^\infty p(y \mid \theta, \beta)\, \pi(\beta)\, \mathrm{d}\beta \tag{3}$$

costs one pass over a quadrature grid rather than one solve per node. With the half-Cauchy prior $\pi(\beta) = (2/\pi)/(1+\beta^2)$ the integral is done numerically on $\beta \in [0.05, 10]$.

Why marginalize rather than fit β? Profiling at its best value gives $-\frac{1}{2}N\,[1+\log(\chi^2/N)]$, and marginalizing is always smaller. The gap is an Occam penalty: a model that fits only by inflating its own error bars is charged for the inflation. Fitting β would hand that flexibility over for free.

3.3 Why the evidence, and not the best fit

From Bayes' theorem, the posterior odds between two models are the prior odds times the ratio of

$$Z_M = p(y \mid M) = \int p(y \mid \theta, M)\, \pi(\theta \mid M)\, \mathrm{d}\theta. \tag{4}$$

This is the only quantity that answers "which model should I believe" without an arbitrary complexity penalty. Comparing best fits cannot: a nested model never fits worse than the model containing it, so χ^2 ranks by flexibility.

Information criteria add a penalty, $2k$ for AIC, $k \log N$ for BIC. But BIC is an asymptotic approximation to $-2\log Z$ assuming a regular, well-identified model with N large, and Section 3.5 shows this one is not identified in a direction. The integral itself assumes none of that.

The penalty appears explicitly on expanding (4) about the maximum. With $\hat\theta$ the best fit, $H = -\nabla^2 \log p(y \mid \theta)$ evaluated there, and $\theta = \hat\theta + \delta$,

$$Z_M \simeq p(y \mid \hat\theta)\, \pi(\hat\theta) \int \exp\!\left[-\tfrac{1}{2}\delta^\mathsf{T} H \delta\right] \mathrm{d}\delta = \underbrace{p(y \mid \hat\theta)}_{\text{best fit}}\ \underbrace{\pi(\hat\theta)\,(2\pi)^{p/2}\,|H|^{-1/2}}_{\text{Occam factor}}. \tag{5}$$

The Gaussian integral is exact for a quadratic $-\log p$, so (5) is the Laplace approximation and its error is governed by the third derivative. Reading the second factor as $\pi(\hat\theta) \times$ (posterior volume) makes its meaning plain: it is the fraction of the prior the posterior actually occupies, so a model needing fine tuning pays. This is the sense in which a good fit and a credible model differ, and it is what Freund and Ewoldt [4] argue should replace goodness-of-fit ranking in rheology.

One consequence matters later and is easy to miss. Because $|H|^{-1/2}$ grows as the posterior broadens, a sloppy direction *raises* the evidence. Model comparison therefore cannot detect non-identifiability, it rewards it, and the two questions must be asked separately.

3.4 The prior this study uses

Equation (4) needs $\pi(\theta \mid M)$, and a uniform box would make the answer depend on where the box was drawn. Two factors are used instead, both computed on the same grid the quadrature runs over.

The *bottleneck* maps each parameter to $u_j \in [0,1]$ in logarithms, then takes the harmonic mean $H(\theta) = p/\sum_j u_j^{-1}$. Logarithms because the parameters span decades, where a linear map would encode the bounds rather than the physics.

The harmonic mean beats the arithmetic one here precisely because it is dominated by its smallest argument. A parameter driven to the bottom of its range drags the whole prior down. The reasoning: a parameter the data pushed to a boundary is one the model did not need.

The *redundancy* weight asks whether a model merely emulates a simpler one it contains. For a candidate and each nested child, the relative stress mismatch F is mapped by $F^n/(F^n + \tau^n)$ against the resolvable scale τ. That gives a number near 0 when the parent imitates the child, approaching 1 when it does something genuinely different.

The weight is the *minimum* over children. A model that can imitate any one simpler model is redundant there, however different it looks from the rest. The children are the models a candidate declares it contains, a structural property of the candidate rather than of the comparison set, so adding a rival to the catalogue does not change anybody else's prior.

The product of the two factors is normalised to sum to one over the quadrature grid, which makes it the discrete prior (4) needs. It is therefore conditional on that grid: a different resolution is a different prior, which is why every candidate is evaluated at one resolution.

3.5 The Fisher information, and what its eigenvectors are

Differentiate $-\log p$ twice at fixed data and take the expectation over noise. The term containing $\partial^2 m$ vanishes because $\mathbb{E}[y_i - m_i] = 0$, leaving

$$F_{jk} = \sum_i \frac{1}{\sigma_i^2} \frac{\partial m_i}{\partial \theta_j} \frac{\partial m_i}{\partial \theta_k} = \left(J^\mathsf{T} J\right)_{jk}, \qquad J_{ij} = \frac{1}{\sigma_i} \frac{\partial m_i}{\partial \theta_j}, \tag{6}$$

so the Fisher information is exactly the Gram matrix of the whitened sensitivities, no approximation, given (1). Near the optimum $\Delta\chi^2 \simeq \Delta\theta^\mathsf{T} F \Delta\theta$, so the surface of constant misfit is the ellipsoid $\Delta\theta^\mathsf{T} F \Delta\theta = c$. Diagonalizing $F = V\Lambda V^\mathsf{T}$, its axes are the eigenvectors and their half-lengths are $\sqrt{c/\lambda_k}$: a small eigenvalue is a long axis, a direction along which the data barely object.

The Cramér–Rao bound gives this statistical force. For any unbiased estimator, $\mathrm{Cov}(\hat\theta) \succeq F^{-1}$, so F^{-1} is the best achievable covariance and its eigen-decomposition is the confidence ellipsoid. The standard deviation along direction k is $\lambda_k^{-1/2}$, and the ratio $\kappa = \lambda_{\max}/\lambda_{\min}$ measures how far the problem is from determining all directions equally.

Two conventions are load-bearing. Taking $\theta = \log(\text{parameters})$ makes an eigenvector a power law: moving along v sends $\Delta \log g = v_g t$ and $\Delta \log \alpha = v_\alpha t$, so $g \propto \alpha^{v_g/v_\alpha}$ and the exponent is read directly off the components. Second, sensitivities taken with respect to a scaled coordinate $u_j = (\log\theta_j - a_j)/w_j$ give $\Delta \log \theta_j = w_j \tilde v_j$, so the exponent is $(w_g \tilde v_g)/(w_\alpha \tilde v_\alpha)$. With box widths of three and four decades that is a 25% correction. Getting it wrong produced a spurious agreement earlier in this work.

Finally, F is the observability Gramian of the system augmented with $\dot\theta = 0$: identifiability and observability are the same statement. The distinction that matters is *structural* unidentifiability, where F is singular and no data can help, against *practical* sloppiness, where F is invertible but ill-conditioned. Here $\lambda_{\min} = 5.2 \times 10^{-2} \neq 0$ and the profile likelihood has a genuine minimum, so the problem is the second, which is why a different experiment can fix it.

3.6 What correlated residuals cost

Equation (1) counts N independent pieces of information. If instead the residuals have autocorrelation ρ_k at lag k, then for a sum of N correlated terms $\mathrm{Var}(\sum r_i) = N\sigma^2\big(1 + 2\sum_k (1 - k/N)\rho_k\big)$, so the information is that of

$$N_{\text{eff}} = \frac{N}{1 + 2\sum_{k\geq 1} \rho_k} \tag{7}$$

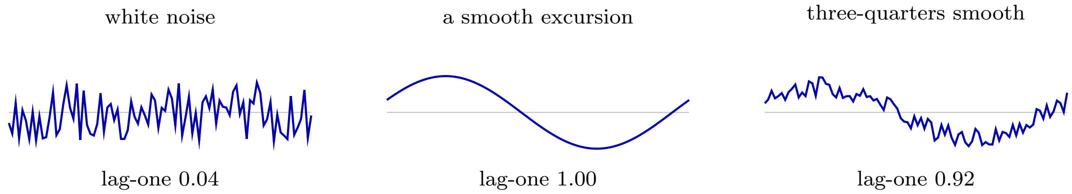

Figure 5: What lag-one autocorrelation measures, and what it does not. Sampled at the $N = 201$ points of these datasets, lag-one is close to a measure of how much of a residual is *smooth*: white noise gives 0.04, a single smooth excursion 1.00, a mixture that is three-quarters smooth 0.92. It does not say what made the residual smooth, model discrepancy, variation between bubbles and a slow calibration drift are alike in this statistic. The `qSLS` residual gives 0.918; the trial deviations from the mean, with no model fitted at all, give 0.859. Most of the smoothness the fit residual shows is therefore already in the data.

independent samples, and parameter uncertainties are understated by $\sqrt{N/N_{\text{eff}}}$. The sum is truncated at the first negative ρ_k, Geyer's initial positive sequence, because the long tail is estimated from few pairs each and contributes sampling error rather than signal.

The correct remedy is a covariance with off-diagonal terms, not a larger σ. Taking $C_{ij} = \sigma_i \sigma_j \exp(-|t_i - t_j|/\tau)$, the Cholesky factor $C = LL^\mathsf{T}$ gives the generalized least squares form

$$-2 \log p = r^\mathsf{T} C^{-1} r + \log |C| + \text{const} = \|L^{-1} r\|^2 + 2 \sum_k \log L_{kk} + \text{const}, \tag{8}$$

which is what `FieldObservation` computes. Inflating σ instead would treat structure as noise: it widens the error bars while leaving the parameter estimates biased, which is the failure Brynjarsdóttir and O'Hagan warn about.

3.7 Which design criterion, and why they disagree

An experiment is chosen by a scalar summary of F. The classical family is

$$\text{D: } \max \det F, \qquad \text{A: } \min \operatorname{tr} F^{-1}, \qquad \text{E: } \max \lambda_{\min}, \qquad \text{c: } \min c^\mathsf{T} F^{-1} c. \tag{9}$$

D minimizes the volume of the confidence ellipsoid, A its mean squared axis length, E its longest axis, and c the variance of one nominated combination. They rank designs differently because they summarize the same spectrum differently. The choice states what one wants to learn.

For identifiability the relevant fact is a scaling argument. A design change that only amplifies signal (more samples, less noise, a larger bubble) sends $F \to sF$. Then $\det F \to s^p \det F$ and $\lambda_{\min} \to s\lambda_{\min}$, so D and E both improve, while

$$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}} \quad \text{is unchanged.} \tag{10}$$

Degeneracy is a property of the eigen*vectors*, not the eigenvalues. Only a design that rotates them can remove it. That is why the E-optimal search preferred a larger bubble while conditioning preferred a gentler collapse: more signal against better separation.

The c-optimal choice carries a small trap. Nominating c as the sloppy direction *of the present design* optimizes for a direction that moves once the design changes. In this study that criterion recommended the opposite of the right answer.

When the model itself is in doubt, none of these apply: all presuppose the model whose F is being computed. T-optimality asks instead for the design maximizing the discrepancy between rival predictions, $\max_\xi \|m_1(\hat{\theta}_1; \xi) - m_2(\hat{\theta}_2; \xi)\|$, and privileges neither.

4 Identifiability of the constitutive parameters

The best-fitting candidate fits, but not every parameter it contains is measured by these data. Which parameters those are is the first thing Part II needs, because a design can only be scored in coordinates the data can answer in.

Why the trade exists

Write $\hat{g} = G/p_\infty = 1/\mathrm{Ca}$ for the nondimensional modulus and $\lambda = R_{\mathrm{eq}}/R$ for the stretch the stress law sees. The equilibrium stress the Maxwell arm relaxes toward is

$$Z_e = R^3\left[\tfrac{3\alpha-1}{2}\,\hat{g}\,P(\lambda) + 2\alpha\,\hat{g}\,Q(\lambda)\right], \qquad \begin{aligned} P(\lambda) &= 5 - \lambda^4 - 4\lambda, \\ Q(\lambda) &= \tfrac{27}{40} + \tfrac{1}{8}\lambda^8 + \tfrac{1}{5}\lambda^5 + \lambda^2 - \tfrac{2}{\lambda}. \end{aligned} \tag{11}$$

Collecting powers of α separates it into two terms with different parameter dependence,

$$Z_e = \hat{g}\,\alpha\,A(\lambda) + \hat{g}\,B(\lambda), \qquad A = R^3\left[\tfrac{3}{2}P + 2Q\right], \qquad B = -\tfrac{1}{2}R^3 P. \tag{12}$$

This is exact; it reproduces the implemented expression to 2.5×10^{-14} over random states. Only the first term carries α, so $\hat{g}\alpha$ and $\hat{g}$ are separately determined only while *both* terms contribute.

They do not, at depth. As λ grows, $Q \to \frac{1}{8}\lambda^8$ dominates every other term in either bracket, so $A \to \frac{1}{4}R^3\lambda^8$ while $B \sim \frac{1}{2}R^3\lambda^4$, and

$$Z_e \;\longrightarrow\; \tfrac{1}{4}\,\hat{g}\,\alpha\,R^3\lambda^8, \tag{13}$$

a function of the *product* $\hat{g}\alpha$ alone. The likelihood cannot then distinguish a soft material that stiffens sharply from a stiff one that barely stiffens. Holding Z_e fixed in (12) gives the trade in closed form,

$$\frac{\partial \log \hat{g}}{\partial \log \alpha} = -\frac{\alpha A}{\alpha A + B} \;\longrightarrow\; -1 \quad (\lambda \to \infty), \tag{14}$$

and the share of the signal that separates the two parameters is

$$\frac{|\hat{g}B|}{|\hat{g}\alpha A|} = \frac{|B|}{\alpha|A|} \simeq \frac{2}{\alpha\lambda^4}. \tag{15}$$

Equations (14) and (15) are what the measurements see. These datasets collapse to $\lambda = 2.215$, where (14) gives -0.981 against -0.99 from the Fisher curvature, a prediction from the constitutive law, not a fit. The profile likelihood returns -0.80 because it averages over a range of λ, and the exponent is -0.861 at $\lambda = 1.5$. Separability falls as λ^{-4}: it is 2% of the signal at $\lambda = 2.2$ and 16% at $\lambda = 1.5$.

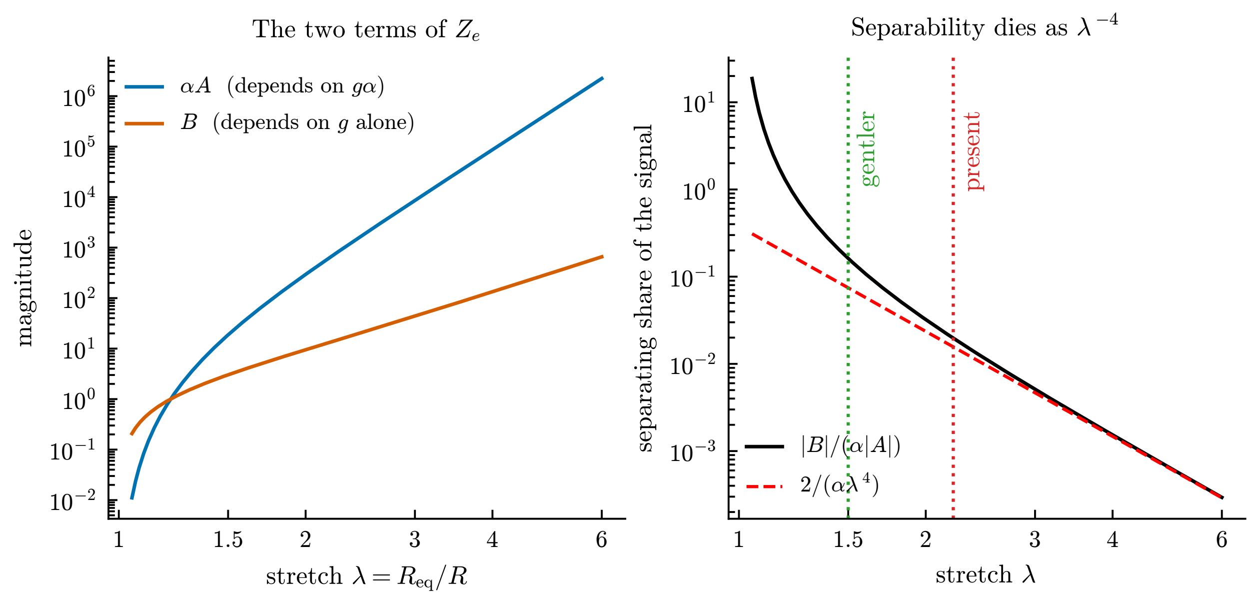

Figure 6: The two terms of (12) against collapse depth. Left: αA, which depends on the product $g\alpha$, outruns B, which carries g alone. Right: their ratio, the share of the signal that can separate the two parameters, following $2/(\alpha\lambda^4)$ once the collapse is deep. It is 2% at the depth these datasets reach and 16% at $\lambda = 1.5$.

What the Fisher information measures

For Gaussian observations with known deviations the log-likelihood is $-\frac{1}{2}\sum_i (y_i - m_i(\theta))^2/\sigma_i^2$ up to a constant, so with the whitened sensitivity $J_{ij} = \sigma_i^{-1}\partial m_i/\partial\theta_j$ the Fisher information is exactly $F = J^\mathsf{T} J$, and $\Delta\chi^2 \simeq \Delta\theta^\mathsf{T} F \Delta\theta$. Taking θ to be the logarithms of the parameters makes the eigenvectors read as power-law combinations: along an eigenvector v the ratio v_g/v_α is the exponent in $g \propto \alpha^{v_g/v_\alpha}$. This is also the observability Gramian of the system augmented with $\dot{\theta} = 0$, which is why identifiability and observability are the same statement here.

One caution, since it caught us. Sensitivities taken with respect to a unit cube $u_j = (\log\theta_j - a_j)/w_j$, rather than $\log\theta_j$ directly, give eigenvector components that carry the box widths. Then $\Delta\log\theta_j = w_j\,\tilde{v}_j$, so the exponent is $(w_g\tilde{v}_g)/(w_\alpha\tilde{v}_\alpha)$, not $\tilde{v}_g/\tilde{v}_\alpha$. With w of three and four decades the two differ by 25%.

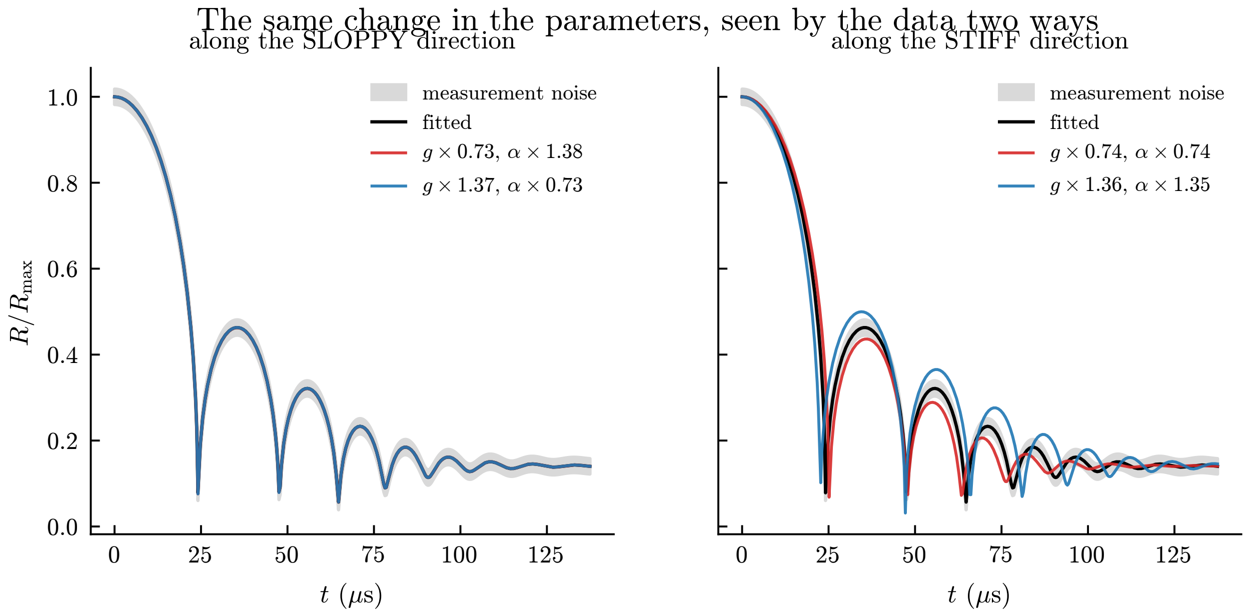

Figure 7: What an eigenvector means, shown rather than defined. The parameters are moved the same distance in log-space along each eigenvector. Along the sloppy direction a 27% change in the modulus, compensated by 38% in the stiffening, leaves the trajectory inside the measurement noise; along the stiff direction a change of the same size is obvious. The data cannot see the first and cannot miss the second.

Modulus and stiffening trade against each other. Two independent analyses agree. The Fisher information at the fitted optimum has an eigenvalue spectrum spanning six decades. The eigenvector of its smallest eigenvalue is almost purely a shear-modulus-against-stiffening direction: weights -0.796 and $+0.604$, against contributions of -0.003 and -0.032 from viscosity and relaxation time. Separately, profiling the likelihood in α, fixing it and re-optimizing everything else, gives $g \propto \alpha^{-0.80}$ across two decades. Sampling the posterior gives a third, independent measurement: its widest direction is $(-0.682, +0.002, -0.082, +0.727)$ in the logarithms of $(g, \mu, \lambda_1, \alpha)$, and $\log g$ is 98.7% anticorrelated with $\log\alpha$.

The three exponents are -0.99 from local curvature, -0.94 from the posterior covariance, and -0.80 from the profile. They agree the trade is near-reciprocal and disagree in the third digit. That is expected rather than troubling: the valley is curved, so a curvature estimate at one point and a chord across two decades measure different slopes. The correlation is the more robust statement.

So viscosity and relaxation time are cleanly determined, while modulus and stiffening are constrained only in combination. The identifiable quantity is closer to $g\,\alpha^{0.8}$ than to either alone, and a fitted α should be reported with its correlation to g rather than as an independent material property.

Figure 9 shows why the four estimates in this study disagree about g by an order of magnitude while agreeing about the fit. They are different points on the floor of one flat-bottomed trench.

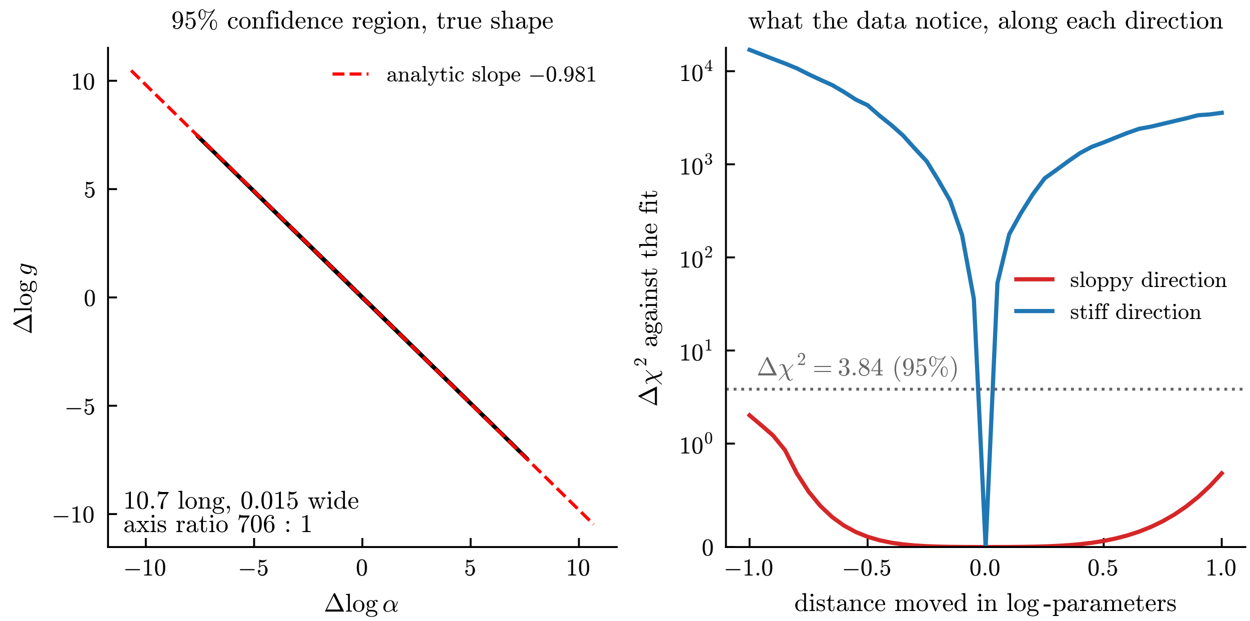

Figure 8: Left: the 95% region from the inverse Fisher information, 10.7 long and 0.015 wide in log-parameters, an axis ratio of 706, which is why it draws as a line, and the analytic slope of -0.981 lies along it. Right: the misfit measured by re-solving at perturbed parameters, not the quadratic approximation. Moving along the stiff direction costs 10^4 in $\Delta\chi^2$; moving the same distance along the sloppy one stays below the 95% threshold, so the parameters may shift by a factor of $e \approx 2.7$ without the data objecting.

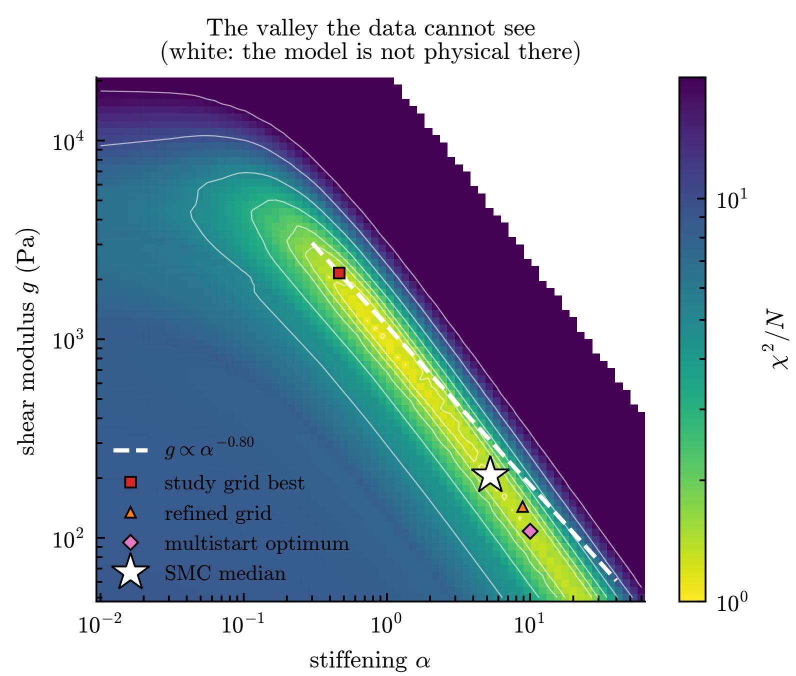

Figure 9: χ^2/N over the modulus–stiffening plane, at the fitted viscosity and relaxation time. The dashed line is $g \propto \alpha^{-0.80}$, predicted from the Fisher curvature at a single point; it lies along the floor of the valley. Four "best fits" from four methods are strung out along that floor rather than disagreeing with one another. The white region at upper right is where the model is not physical.

Widening α's own ceiling does not move it, and that is less than it looks. Sampling with the ceiling on α placed at 1, 10 and 100:

ceiling	log evidence	median α	97.5%
1	2162.9	0.85	0.97
10	2173.9	6.37	9.04
100	2172.4	5.27	8.73

Widening the prior tenfold moves the median only from 6.37 to 5.27, and the evidence by 1.5 out of 2173. The ceiling of 1 shows what genuine truncation against *that* bound looks like: the median pins against it and 11 log units of evidence are lost.

It is tempting to read the two loose rows as identification, and this document did read them that way. They do not support it, because the wall that binds is not the one being moved. The likelihood depends on $g\alpha$, the box floors g at 100 Pa, and the identified product is 1106 Pa at 15 °C, so the floor alone caps α near 11 whatever its own ceiling is set to. The two loose rows report 97.5% points of 9.04 and 8.73, just under that cap, and they agree with each other because they are limited by the same thing.

Two later sections move the wall that does bind and see exactly what that predicts. Section 18 widens the box to $g \in (10^0, 10^6)$ and five of six fits go to $\alpha = 10^3$ with g collapsed to 1.1 Pa to 3.5 Pa; Section 7 releases the same floor and reports $g\alpha$ preserved to 1.6 % across a hundredfold change in g.

Run as a sweep, that is the whole of it. A box bound is a prior truncation rather than a different likelihood, so both sweeps are regions of one surface: χ^2 on a 96×96 grid over (g, α), with the noise scale marginalised as in Section 3.2 and the viscosity and relaxation time held at their fitted values, which the spectrum in Figure 10 shows are nearly orthogonal to this direction. Sweeping the floor under g with α's box fixed:

g floor (Pa)	median α	97.5%	median $g\alpha$	$g\alpha/g_{\min}$
1	47.3	96.8	1126	1126
10	47.3	96.8	1126	113
100	4.13	11.2	1064	10.6
1000	1.05	1.13	1128	1.1

The median moves by a factor of 45 while the floor moves by 1000, and the product holds to 6 % throughout. The upper quantile is $\min(\alpha_{\max},\ g\alpha/g_{\min})$ to within a few percent in every row: at the two lowest floors α rails against its own ceiling of 100, and at the two highest against what the floor under g permits. It is never the data that stop it.

That is also why the published sweep looks reassuring. Moving α's ceiling from 10 to 100 shifts the median from 3.72 to 4.13, an 11 % wobble that reads as stability; moving the floor shifts it from 1.05 to 47.3. Same posterior, different wall. So the conclusion of the paragraph above stands unqualified: what these data identify is the product, and a reported α is a reported box.

5 What the fits actually support

A candidate that wins is not the same as a candidate that is adequate, and the difference is where the inputs to Part II come from. It measures three things, and each is used later as much as it is reported here. How much a residual near the measurement scatter entitles one to claim. What the between-trial spread is made of, which is the covariance Σ that Section 14 designs against. And whether a richer relaxation spectrum is the missing physics, which it is not. The ranking that produced these fits is Part III; the winner throughout is `qSLS`, a quadratic Zener solid carrying both strain stiffening and stress relaxation.

5.1 How good are the fits, really?

Table 4 reports the best χ^2/N found on the evaluation grid. Two questions follow, with different answers. How well does the winning model fit? And how much does a good χ^2/N entitle us to claim?

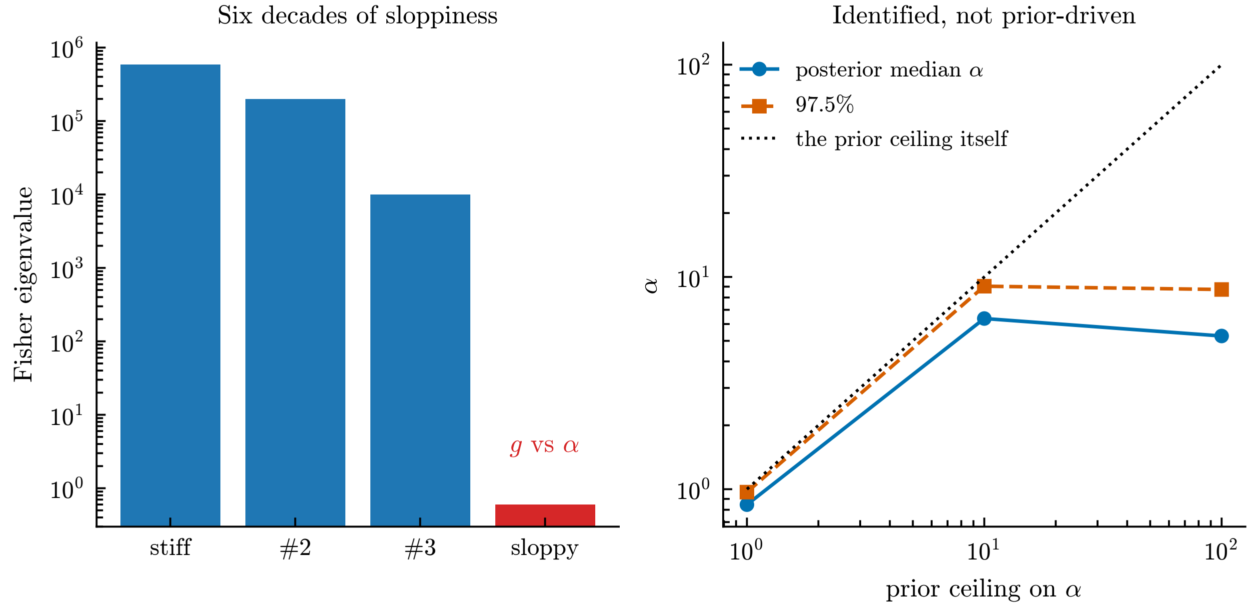

Figure 10: Left: the Fisher spectrum at the optimum, six decades between the stiffest and sloppiest directions, with the sloppiest being modulus against stiffening. Right: the posterior for α against the ceiling imposed on its prior. It tracks the ceiling while truncated and departs from it once the ceiling is loose. The panel is titled for the reading this document first took from it; the table beside it sweeps the floor under g instead and finds α tracking that bound just as faithfully, which is why the reading does not hold.

The grid understates the fit. The model-selection grid evaluates each candidate at log-spaced parameter nodes, so its best value is the best *node*, not the best fit. Refitting each model to the 15 °C dataset by multistart Gauss–Newton with analytic Jacobians (eight starts from a Latin hypercube over the wide prior box, taking the lowest) gives:

model	parameters	χ^2/N	lag-one ρ	N_{eff}	χ^2/N_{eff}	inflation
qSLS	4	**0.962**	0.918	10.3	18.9	4.43
SLS	3	1.984	0.932	7.1	56.6	5.34
qKV	3	20.18	0.948	13.4	302	3.87
NHKV	2	20.16	0.948	13.4	302	3.87

At its true optimum qSLS reaches $\chi^2/N = 0.96$ rather than the grid's 1.35, which confirms in a second way that the tabulated residuals are ceilings. The ordering is unchanged, and the gaps are large: relaxation without stiffening doubles the residual, and both memoryless models sit twenty times above it.

Stiffening buys nothing without relaxation. qKV and NHKV differ only by the stiffening exponent α, and they fit identically: 20.18 against 20.16, one part in a thousand on a residual twenty times too large. The strain-stiffening term is decisive alongside a relaxation mode and worth nothing without one.

That is sharper than the evidence ranking alone. It is not that qKV is a worse model than qSLS; its extra parameter does no work at all. It also explains why the temperature trend is carried by the interaction of the two mechanisms rather than by either alone.

But $\chi^2/N \approx 1$ does not mean the fit is within noise. The statistic χ^2/N compares the residual to its expectation under the assumption that the N residuals are independent draws of known scale. They are not. The lag-one autocorrelation of the qSLS residual is 0.918, and by the Geyer initial-positive-sequence estimator [5] of (7) the 201 samples carry the information of about $N_{\text{eff}} = 10$ independent ones. Dividing by that count instead gives 18.9.

That number is not a calibrated test. For correlated residuals the correct statistic is $r^\mathsf{T}\Sigma^{-1}r$ with the true covariance, not χ^2 rescaled by an effective count. But its message is robust, and it is the opposite of the headline's.

A residual with $\rho = 0.918$ is not noise the model failed to average away. It is *structure* the model does not contain. The fit is excellent in that no candidate does better, and not excellent in that it is inconsistent with the measurement error.

The classical test agrees, and says more. The argument above rests on lag-one autocorrelation, which Section 22.1 shows is a blunt instrument. There is a standard test for the same question and these datasets already carry what it needs: replicates. The trial spread at each time is *pure error*, costing no model, so the residual of the mean splits into that and what the model cannot follow,

$$F = \frac{J\sum_i(\bar{y}_i - \hat{y}_i)^2 \,/\, (k-p)}{(J-1)\sum_i s_i^2 \,/\, k(J-1)}, \tag{16}$$

with J trials at k times and $p = 4$ parameters. The datasets do not carry equal replication: J is 18, 14 and 7.

dataset	J	χ^2/N	pure error	lack of fit	F	corrected
15 °C	18	0.973	5.20×10^{-4}	7.36×10^{-3}	**14.2**	**23.3**
23 °C	14	1.017	1.86×10^{-3}	5.47×10^{-3}	2.9	4.8
33 °C	7	1.130	1.91×10^{-3}	6.23×10^{-3}	3.3	5.4

Degrees of freedom are 197 against 3417 at 15 °C and fall with J elsewhere, so the five percent critical value is near 1.2 throughout. Every dataset shows lack of fit. The conclusion is the one above, reached with none of the same machinery.

Two things make it worth the page. It gets the factor of J right by construction (the fit compares the *mean* of the trials against the spread *between* them, and the J in the numerator is that missing $\sqrt{J}$) and it is conservative in a direction we can quantify: 39.3 % of the trial variance is bubble-to-bubble parameter spread rather than measurement, so pure error is inflated and the corrected column divides that share out.

It also separates the datasets, which lag-one does not: those values are 0.919, 0.968 and 0.911, effectively identical. Two things drive the separation and both are real. 15 °C repeats itself about three times more tightly, and it was run with more than twice the replicates of 33 °C, replication is what buys power to detect lack of fit, so the F column correctly rewards it. Removing the replicate factor, the systematic misfit variance stands at 0.79, 0.21 and 0.47 of pure error, so the ordering survives but the spread is milder than 14.2 against 3.3 suggests.

The strongest evidence that the model is incomplete therefore comes from the dataset with the best apparatus and the most repeats, not from the worst fit.

The p-value is not usable: F assumes independent errors across time and these are correlated at 0.918. The ratio needs no such assumption, and is read here as an effect size.

Three things follow, and they should be read together.

1. *The ranking survives.* Refitting under a correlated likelihood with a Cholesky whitening of an estimated AR(1) covariance changes the qSLS–SLS evidence gap by 3%. What the correlation destroys is the absolute claim, not the comparison, and the comparison is what the study is for.

2. *The denominator is the wrong yardstick.* The scale σ_i is the trial-to-trial spread. Much of that is variation in the parameters between bubbles, not error in measuring any one of them. A χ^2 formed against it is not comparing the model to measurement precision at all.

3. *The misfit is in the early trace*, not where the tracking is hardest. Restricting the fit to a later window makes χ^2/N worse, 0.97 to 1.95, and does not whiten the residual. The structure cannot be excised by discarding the awkward part of the dataset.

The honest summary: qSLS is the best available description of these datasets by a wide margin, it is measurably incomplete, and the size of what is missing is better read from the residual's correlation than from its magnitude. What that incompleteness costs the parameters is a separate question from whether it exists, and Section 7 answers it: on these datasets the model-form budget is larger than the posterior width by one to two orders of magnitude.

5.2 What the trial spread actually is

The likelihood of Section 3.2 treats the residuals as independent draws whose scale is the trial-to-trial spread s_i. Two assumptions are buried in that, and both are wrong in ways that matter for how the numbers above should be read.

Part of the spread is the parameters, not the measurement. If the 18 trials differed only by measurement error, their deviations from the trial mean would be white and point nowhere in particular. They do neither.

The singular spectrum of the 18×201 deviation matrix puts 53.9% of the variance in one component, where white noise would give 5.9%. Projecting each deviation onto the span of $G = \partial(R/R_{\max})/\partial\theta$ for the six parameters $(R_0, R_{\text{eq}}, g, \mu, \lambda_1, \alpha)$ captures 39.3% of the variance against a chance level of $6/201 = 3.0\%$. Noise does not align with a six-dimensional subspace of a 201-dimensional space by accident.

The natural physical reading is that each bubble has its own parameters, drawn from a population: $\theta_j \sim \mathcal{N}(\theta_{\text{pop}}, \Sigma)$. The mean trace then has covariance

$$C \;=\; \frac{1}{J}\Big(G\,\Sigma\,G^{\mathsf{T}} \;+\; \sigma^2_{\text{meas}} I\Big), \tag{17}$$

low rank plus diagonal rather than diagonal.

But the dominant mode is not parameter variation, and a refit proves it. The projection above is a statement about a *linearization*: it asks whether the deviations lie in the span of G evaluated at one point. That test cannot exclude variation large enough to turn the Jacobian between trials, and it is the only test this argument had. So each trial was fitted on its own, 39 fits across the three datasets, in the identified coordinates of Section 7.

Giving every trial its own parameters does not absorb the leading mode. It keeps its shape and takes a *larger* share of the reduced residual:

dataset	mode 1 before	mode 1 after	$\lvert\cos\rvert$ between them
15 °C	0.539	0.412	0.699
23 °C	0.681	**0.772**	**0.987**
33 °C	0.720	**0.835**	**0.943**

An absorbed mode collapses; these do not. Only 15 °C shows partial absorption.

Those three axes are the MATERIAL ones, though, and the equilibrium radius was held at $R_{\max}$/stretch, one value for every trial in a dataset, so that refit could not have absorbed variation in the initial condition however large it was. That is not a pedantic gap. The mode is 98 and 97% post-collapse at 23 and 33 °C and 63% at 15 °C, and the afterbounce period is set by the gas the bubble is left with, so R_{eq} is both the quantity nucleation is least repeatable in and the one whose variation appears after the first collapse rather than during it.

Refitting with R_{eq} free over the ±11% its prior allows:

dataset	raw	material only	$+\,R_{\text{eq}}$	$\lvert\cos\rvert$ to raw
15 °C	0.539	0.412	0.338	0.683
23 °C	0.681	0.772	**0.791**	**0.988**
33 °C	0.720	0.835	**0.717**	**0.955**

At 33 °C the answer is clean: no fit rests on a bound, R_{eq} moves by 3.0% between trials, and the mode keeps its shape at $\lvert\cos\rvert = 0.955$. Freeing the initial condition does not absorb it.

At the other two datasets the test is *inconclusive rather than negative*, and the reason is worth stating instead of burying. Seven of eighteen fits at 15 °C and four of fourteen at 23 °C rest against the upper bound of 1.111: those trials wanted a larger equilibrium radius than the prior permits, and were cut off. The pinning is also one- sided, which reads as a systematic offset in the dataset's stretch rather than as scatter between bubbles, a calibration question, and a different one from this section's.

Nor is it how the bubble starts moving, which was the last axis available. Every test above varies what the bubble *is*. None varies how it *starts*, and that gap mattered: the mode sits largely after the collapse, what a bubble carries into its rebound is set by the energy it goes in with, and every fit here begins at $R_{\max}$ with zero wall velocity, asserting that the imaged maximum is a turning point of a free collapse.

Refitting each trial with an initial wall velocity free *and* the stretch free, over boxes wide enough that nothing pins:

dataset	raw	material	$+R_{\text{eq}}$	$+R_{\text{eq}}+u_0$	$\lvert\cos\rvert$ to raw
15 °C	0.539	0.412	0.338	0.340	0.678
23 °C	0.681	0.772	0.791	**0.800**	**0.987**
33 °C	0.720	0.835	0.717	**0.840**	**0.945**

Nothing moves. The leading share is unchanged to within a percent on every dataset, and the mode keeps its shape at $\lvert\cos\rvert = 0.987$ and 0.945. Nothing pinned, so the boxes were not the constraint.

There is real scatter in both quantities, u_0 spreads by 0.008 to 0.016 of the Rayleigh velocity across trials, over ranges from -0.047 to $+0.009$, and R_{eq} by 2.0 to 6.4%. It simply is not what carries the mode.

So the claim can be made at full strength rather than narrowly. *Whatever carries half to three-quarters of the trial scatter is not any quantity this model has*: not the material parameters at each trial's own optimum, not the equilibrium radius over a box that does not cut off, and not the initial wall velocity. Four absorptions attempted, four failed, and the last two were the configuration quantities that the mode's position after the collapse pointed at most directly.

It is largely the clock, and the acquisition data says so directly. Each of those tests varied a property of the bubble. The one quantity that could not be tested was the one every trace is divided by before it arrives: its own $R_{\max}$. The acquisition dataset carries it (835 events, each with a measured $R_{\max}$, a measured R_{eq} and an un-normalised trace) and two facts follow immediately.

The scatter is in the scale, not in the ratio. Per event, $R_{\max}$ has a coefficient of variation of 0.25, 0.28 and 0.23 in bins around 15, 23 and 33 °C, while the stretch $R_{\max}/R_{\text{eq}}$ has 0.052, 0.066 and 0.062 about a mean near 8.1 at every temperature. The absolute scale varies four times more than the ratio this document has been treating as the uncertain one.

And a scatter in $R_{\max}$ is a scatter in the clock. $R(t)/R_{\max}$ is a near-universal function of t/t_c with $t_c \propto R_{\max}$, so bubbles differing only in size trace one curve at different rates, and the deviation that produces is $\partial R/\partial \log t_c = -t\dot{R}$, small early, growing as phase error accumulates, and therefore concentrated after the collapse. Which is where this mode has been all along.

On the raw traces the identification is not marginal:

	events	leading share	$\lvert\cos\rvert$ to $-t\dot{R}$	chance
raw, 12 °C to 18 °C	253	0.882	**0.951**	0.075
raw, 20 °C to 26 °C	100	0.869	**0.976**	0.075
raw, 30 °C to 36 °C	84	0.861	**0.973**	0.075
processed, 15 °C	18	0.539	0.572	0.071
processed, 23 °C	14	0.681	0.581	0.072
processed, 33 °C	7	0.720	0.518	0.071

In the acquisition data the dominant mode *is* the time dilation, at $\lvert\cos\rvert$ of 0.95 to 0.98 against a chance level of 0.075, carrying seven-eighths of the variance. There is nothing to explain there.

What reaches this document is the residue. The processing already removes most of it: the timescale scatter the datasets imply is 1.0 to 2.1 percent against the 22 to 28 measured per event, a factor of 15. What survives still aligns with the dilation at 0.52 to 0.58 and carries 24 to 27 percent of the trial variance, *the largest single share anyone has attributed to this mode*, against 0.467 for the best of nine linearised parameter directions.

It is not the whole of it. Projecting the dilation out leaves a leading mode overlapping the original at $\lvert\cos\rvert$ of 0.81 to 0.85, so a fifth to a quarter is accounted for and the rest is not.

Which clock, and what each one buys. The acquisition file settles what the pipeline is doing, by its own fields: $R_{\mathrm{norm}} = R/R_{\mathrm{max}}$ to the last bit, and $\mathrm{d}t_{\mathrm{norm}}/\mathrm{d}t = 10/R_{\mathrm{max}}$ with $\sqrt{p_\infty/\rho} = 10.07\,\mathrm{m\,s^{-1}}$, so each event's time is divided by *its own* inertial scale $R_{\mathrm{max}}\sqrt{\rho/p_\infty}$. That is where the factor of fifteen went, and the file also stores `tc_norm`, the *measured* collapse time in units of that estimate, which scatters by 3.1 to 3.3%.

Rebuilding the deviation matrix under three clocks on the same events:

bin	clock	n	leading share	$\lvert\cos\rvert$ to $-t\dot{R}$	implied cv	total variance
12 °C to 18 °C	population	253	0.674	0.944	0.226	5.36
	per-event inertial		0.756	0.852	0.024	0.383
	measured collapse		0.637	0.926	0.017	**0.245**
20 °C to 26 °C	population	100	0.729	0.914	0.244	6.27
	per-event inertial		0.770	0.872	0.026	0.408
	measured collapse		0.632	0.953	0.017	**0.228**
30 °C to 36 °C	population	84	0.694	0.879	0.200	5.15
	per-event inertial		0.615	0.919	0.023	0.352
	measured collapse		0.586	0.872	0.017	**0.266**

Three things follow. *The per-event clock is worth a factor of fourteen*: a population R_{max} inflates the trial variance 14 to 15 times, and the scatter it implies, 0.20 to 0.24, matches the 0.22 to 0.28 measured directly, the identification is exact rather than suggestive. *Realigning on the measured collapse time buys a further quarter to two-fifths*, so `tc_norm` is a real residual the inertial estimate leaves behind.

And the dilation shape survives every clock. At all three the leading mode overlaps $-t\dot{R}$ at 0.87 to 0.95 and the direction carries 0.47 to 0.61 of the variance. One scalar per event cannot remove it, which is the physical content: the collapse and the rebound do not scale together. The rebound period is set by the gas the bubble is left with, so aligning on t_c aligns the collapse and leaves the afterbounce misaligned, and the afterbounce is where this mode lives.

It does, and that closes the question. Both timescales are measurable per event: the peak sits at $t_{\mathrm{norm}} = 0$ by construction, the first minimum near 0.92 and the second near 1.73, so a piecewise-linear warp taking $(\mathrm{peak}, \mathrm{min}_1, \mathrm{min}_2)$ to $(0, 1, 2)$ aligns the collapse and the afterbounce at once. They are not the same quantity: the collapse time has a coefficient of variation of 0.031–0.033 and the afterbounce period 0.056–0.069, and the two correlate at only +0.35 to +0.52.

bin	alignment	leading share	$\lvert\cos\rvert$ to $-t\dot{R}$	dilation share	total variance
12 °C to 18 °C	inertial only	0.817	0.850	0.599	0.323
	collapse only	0.609	0.756	0.408	0.107
	two timescales	0.424	**0.094**	**0.262**	**0.054**
	sham (population knots)	0.602	0.969	0.571	0.463
20 °C to 26 °C	inertial only	0.784	0.879	0.614	0.287
	two timescales	0.510	**0.133**	**0.233**	**0.059**
	sham	0.593	0.972	0.565	0.396
30 °C to 36 °C	inertial only	0.654	0.898	0.552	0.316
	two timescales	0.682	**0.297**	**0.193**	**0.065**
	sham	0.593	0.938	0.545	0.480

Aligning both timescales removes four fifths of the trial variance, ×0.168, ×0.205 and ×0.206 against the inertial clock, and, decisively, *destroys the dilation signature*: the leading mode's overlap with $-t\dot{R}$ falls from 0.85–0.90 to 0.09–0.30. What is left is no longer a clock error of any kind.

The sham is what licenses reading that as physics rather than as arithmetic. A warp with two knots per event removes variance whatever the knots are, so the same warp was run with knots drawn from the population instead of measured from each trace. It does not help, it makes things *worse*, at 0.40 to 0.48 against the inertial clock's 0.29 to 0.32, and the measured warp beats it by a further factor of seven to nine. The reduction comes from the timescales being the event's own.

On that data the mode is a clock error with two hands: a bubble has two independent timescales, the collapse set by $R_{\max}$ and the ambient pressure and the afterbounce set by the gas it is left with, the processing normalises by the first, and the mode is what the second leaves behind.

It does not carry to the datasets this document fits, and that is the finding. The acquisition file and these datasets are *different experiments*. The file is a temperature sweep of `pa5` events at $R_{\max}$ near 310 µm and stretch near 8.1; the datasets are the *gelatin* rows of the same study [6], at 277, 298 and 312 µm and stretches 7.09, 7.37 and 6.83. A mechanism shown on one is a hypothesis about the other until it is run on the other, so it was:

dataset	J	alignment	leading share	$\lvert\cos\rvert$ to $-t\dot{R}$	total variance
15 °C	18	inertial only	0.541	0.706	0.0637
		two timescales	0.635	**0.877**	0.0328
23 °C	14	inertial only	0.533	0.612	0.0580
		two timescales	0.591	**0.852**	0.0339
33 °C	7	inertial only	0.787	0.815	0.1155
		two timescales	0.642	**0.024**	0.0345

The warp halves the trial variance on all three, so it is doing something. But the dilation signature, which it destroys on the sweep data, it *strengthens* here at 15 and 23 °C, from 0.71 to 0.88 and from 0.61 to 0.85, and removes only at 33 °C. One dataset of three behaves like the sweep.

Three explanations were available (time resolution, replicate count, and the experiment itself) and two of them can be settled with the data already in hand.

Resolution is not it, and that is arithmetic. The datasets carry 201 samples over $5\,t_c$, a spacing of $0.0250\,t_c$; the sweep traces have a median spacing of 0.0315. The datasets are *finer* than the data the warp demonstrably works on.

Nor is the replicate count. If eighteen trials cannot support the warp, drawing eighteen events at random from a bin where it works should reproduce the failure, same material, same instrument, same processing, only the count changed. Over 200 draws per cell:

bin	n	$\lvert\cos\rvert$ before	$\lvert\cos\rvert$ after	$P(\text{after} > \text{before})$
12 °C to 18 °C	7	0.818	0.502	0.10
	18	0.831	0.446	0.06
	80	0.847	0.353	0.01
20 °C to 26 °C	7	0.824	0.364	0.03
	18	0.856	0.263	0.01
30 °C to 36 °C	7	0.834	0.345	0.04
	18	0.875	0.302	0.00

medians over the draws. At the datasets' own replicate counts the warp still works: it takes $\lvert\cos\rvert$ from about 0.83 down to 0.26–0.50, and it makes the signature *worse* (which is what happens on two of this document's three datasets) between 0 and 10% of the time. Observing that on two datasets of three is a one to three percent event under this explanation.

So neither resolution nor replication accounts for it, and what remains is the material.

And the two materials are named in the source. Chu et al. [6] tabulate three formulations: stiff PAAm (5% acrylamide with 0.3% bis), soft PAAm (5%/0.03%), and gelatin 5%. The acquisition file is the PAAm sweep, its events are named for their bis fractions and its stretch of 8.1 matches the tabulated 7.93 to 8.36, while the three datasets fitted here are the gelatin rows at 7.09, 7.37 and 6.83. The materials were chosen for exactly this contrast: PAAm is expected to be insensitive to temperature over the range studied, and *gelatin is expected to undergo a phase transition near* 30 °C.

That turns the discrepancy from an unexplained difference into a physical hypothesis, and the direction is right. The warp works on PAAm at every temperature, and on gelatin only at 33 °C, the one dataset above its transition, where the gel is softest and closest to a simple viscous medium. Below it, where the network is firm, the afterbounce does not behave as a second clock.

Two further facts from the same source sharpen it and are not resolved here. The gelatin bins at 24.4 and 32.8 °C were reached on the *cooling* branch after equilibration at 60 °C, while the 14.7 °C bin was reached on the warming branch from about 5 °C, so those datasets differ in thermal history as well as in temperature. And every bin carries a ±2 °C tolerance, so each is itself a small sweep.

One candidate can be eliminated, and one new limit falls out. Both $R_{\max}$ and R_{eq} are *fitted* rather than read off (the first by a fourth-order polynomial through a window at the peak, the second by averaging the tail of $R(t)$) and since the clock is $R_{\max}\sqrt{\rho/p_\infty}$, an error in the first fit is an error in that event's clock. That is the right shape for the residual dilation, and it is testable because the file stores the fit quality per event.

It is not that. The $R_{\max}$ fits are uniformly excellent, R^2 of 0.9996 on average with a worst case of 0.9944, and the dilation amplitude does not track what variation there is: |dilation| against $1-R^2$ correlates at +0.064, +0.016 and −0.220 across the three bins. The residue is not $R_{\max}$ fit noise.

R_{eq} is a different matter, and this is the new limit. The authors describe two estimators (the tail average that is stored, and the median of the post-collapse oscillations) and both are available:

bin	n	stretch, tail average	cv	stretch, median of extrema	cv
12 °C to 18 °C	253	8.211 ± 0.431	0.053	8.291 ± 0.540	0.065
20 °C to 26 °C	100	8.106 ± 0.534	0.066	8.075 ± 0.642	0.080
30 °C to 36 °C	84	8.120 ± 0.499	0.062	8.059 ± 0.541	0.067

They agree on the mean to under one percent and correlate at only +0.50, +0.66 and +0.80. Two estimators of one quantity, with comparable spreads and a correlation of a half, means a large share of the apparent scatter is the *estimator* rather than the bubble, and since both read the same trace's tail, their errors are if anything positively correlated, which makes that a conservative reading.

So the per-event stretch is not the clean replacement for a population mean that Section 19 might hope for. Having it per event removes the assumption that every bubble shares one value; it does not make any one bubble's value precise, and on this evidence a substantial part of the five-to-eight percent scatter is how R_{eq} is measured rather than how bubbles differ.

So the honest statement is narrower than the sweep alone would license. *On acquisition data of this kind the dominant trial mode is a two-timescale clock error, and aligning both timescales removes it.* On the three datasets fitted here the same alignment removes half the variance and leaves the mode more dilation-like rather than less, on two datasets of three.

The remaining explanation is the material, and it can be tested rather than left open. Resolution was eliminated by arithmetic and the replicate count by resampling, which leaves the experiment itself. That comparison does not need the acquisition file at all: the processed per-trial traces for *both* materials are distributed in one format (a normalised clock, then one $R/R_{\max}$ trace per column) and, decisively, at the same sampling step of $0.025\,t_c$. Running the identical warp, sham and diagnostic over both arms holds processing and resolution fixed by construction rather than by argument:

material	datasets	events	variance removed	|cos| before	after
gelatin	3	39	49.5%	0.691	0.716
PAAm	3	249	**71.9%**	0.923	**0.568**

In both arms the sham (each event warped with another event's knot *pair*, so the knot geometry stays realistic and only the match to the event is broken) *increases* the variance in all six datasets. None of the reduction is bought by the two fitted degrees of freedom.

The two materials therefore differ in what the warp leaves, not in whether it works. In PAAm the dominant mode begins strongly clock-aligned and de-aligns once both hands are set, which is the signature of a mode that *was* the clock. In gelatin the warp removes as much variance but what remains is as dilation-like as what it started with. With sampling, processing and analysis identical across 288 events, the difference that survives is the material, which makes gelatin's phase transition near 30 °C the thing to explain, and is consistent with 33 °C being the one gelatin dataset that behaves like the PAAm arm. #222 is now a question about a material rather than about a pipeline.

Two further facts locate it without naming it. It lives in the *rebound*: 98.0% and 97.2% of its squared length falls after the collapse window at 23 and 33 °C, while the identifiable discrepancy of Section 7 sits 65 to 69% in the *first collapse*. And it is not one curve: on a common t/t_c grid the three datasets' leading modes overlap at $|\cos|$ of 0.075, 0.252 and 0.082, against 0.050 for unrelated curves of that length. A shared curve would have been the apparatus, which no constitutive law could absorb. This is not shared either.

What the refit does give is Σ. Regressing deviations onto G inherits the g–α degeneracy of Section 3.5 and returns $\mathrm{sd}(\log g) = 159$, which is why this document previously reported Σ as not estimable. The per-trial fits estimate it directly, in the identified coordinates where no such degeneracy exists:

dataset	μ	$g\alpha$	λ_1
15 °C	14.7%	22.8%	60.2%
23 °C	33.9%	20.2%	58.9%
33 °C	19.2%	20.3%	117.1%

as coefficients of variation in log coordinates. These are overestimates, a single trial's measurement noise is not separable from the trial spread with this data, so each fit is weighted by a scale that already contains what is being estimated. They are nonetheless the first direct measurement of Σ here, and Section 14 is what they buy.

One entry has to be read with Section 19. Fitting R_{eq} per trial over a box wide enough that nothing is cut off gives spreads of 1.9, 6.7 and 3.1 percent about medians of 1.099, 1.063 and 1.032, so most of what a narrow box would have called scatter in the initial condition is a common offset in the stored stretch, and only the remainder belongs in Σ.

It is not what makes the residuals correlated. The natural hypothesis is that the lag-one correlation of the fit residual is this same effect. Whitening by (17) tests it directly:

covariance	χ^2/N	lag-one ρ
diagonal (independent)	17.54	0.758
hierarchical, (17)	21.96	**0.516**

The correlation falls by about a third, and most of it survives. χ^2 gets *worse*, so the residual sits in directions where C is small. The dominant term is model-form error, not variation between bubbles. Both effects are real and the model-error one is larger, which separates two things Section 22.1 had conflated.

Which precision χ^2/N is measured against. The same computation exposes a factor of $J = 18$ in the meaning of the headline number. The data is the *mean* of 18 trials, but s_i is the spread *between* trials, and the standard error of the mean is smaller by $\sqrt{18} = 4.24$. From the identical residual,

$$\chi^2/N = 0.974 \text{ against the trial spread,} \qquad \chi^2/N = 17.54 \text{ against the standard error of the mean.}$$

Both are meaningful, and they answer different questions. *The model reproduces the dataset to within the scatter between individual experiments* is true, and is the claim this document makes. *The model is statistically adequate* is false: against the precision the 18-trial mean actually carries, it is off by a factor of seventeen. The noise-scale marginalization of Section 3.2 absorbs the difference, the implied $\beta \approx 0.24$ lies inside its prior, so the evidence ranking is undisturbed. What changes is what a residual near one entitles one to say.

What lag-one does and does not measure. The inference above (that whitening removes a third of the correlation and the remainder is structure the model lacks) rules out variation between bubbles, and nothing more. Lag-one autocorrelation at $N = 201$ samples is close to a measure of how much of a residual is *smooth*, not of what made it smooth: sampled at this density, white noise gives 0.04, a single sine period gives 1.00, a

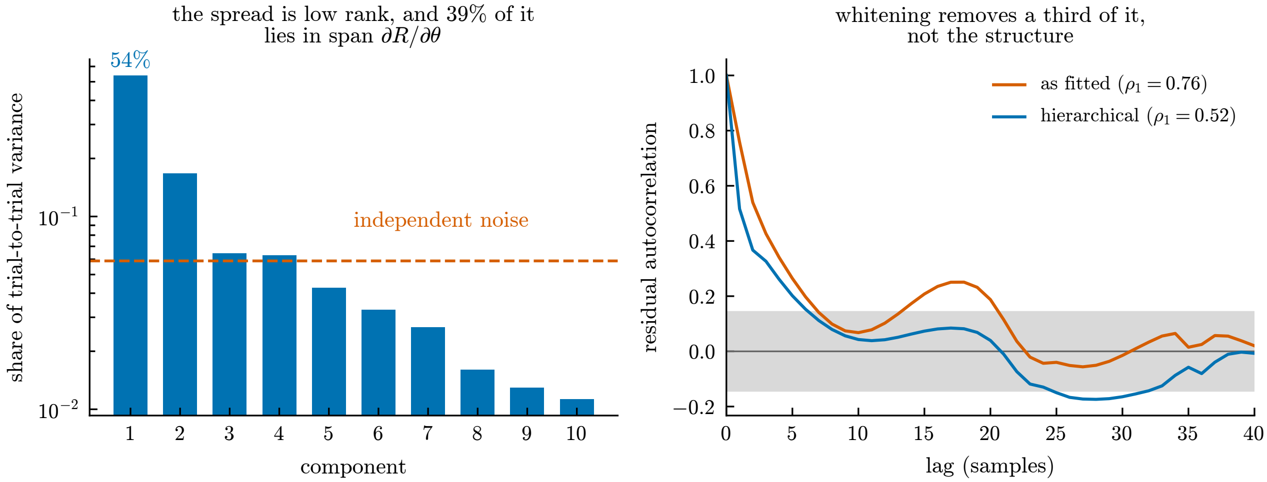

Figure 11: What the trial spread is, and what it does not explain. Left: the singular spectrum of the eighteen trial deviations. Independent measurement error would put an equal share in every component (dashed); instead the leading one carries 54%, and 39% of the total lies in the six-dimensional span of $\partial R/\partial\theta$ against a 3% chance level. Right: the autocorrelation of the qSLS fit residual before and after whitening by the hierarchical covariance (17), with the white-noise band shaded. If the correlation were caused by variation between bubbles, whitening would flatten it into that band. It removes about a third and leaves the rest, so the remainder is structure the model does not contain.

smooth random curve gives 0.999, and a mixture that is 80% smooth gives 0.944. A value of 0.918 says the residual is roughly three-quarters smooth structure. It does not say whose.

That matters because the trial deviations from the mean (pure measurement variation, with no model fitted at all) carry lag-one 0.859 (median over the eighteen trials, range 0.725 to 0.890). Most of the smoothness the fit residual shows is already present in the data before any model is involved. Whitening by the hierarchical covariance removes the part of it that lies along $\partial R/\partial\theta$, which is why it removes only a third; the rest is smooth in the data for reasons the parameter sensitivities cannot represent, and model-form error is one candidate among several rather than the established remainder.

The searches of Section 5.3 were run on the stronger reading, and their negative results are consistent with the weaker one.

5.3 What to enrich, and how it would have to be judged

The previous section leaves a specific defect: the residual carries structure the model does not contain, and it is not accounted for by variation between bubbles. The natural suspect is the assumption of a *single* relaxation time, which is strong for a crosslinked biopolymer whose relaxation is in reality distributed. Two enrichments follow from that suspicion, and they are not the same claim.

A second timescale. `TwoModeQuadraticZener` adds a second Maxwell arm sharing the elastic equilibrium, with a weight w splitting both the elastic target and the viscous forcing,

$$\dot{Z}_1 = -\frac{Z_1 - (1-w)Z_\mathrm{e}}{\mathrm{De}_1} + \frac{(1-w)\,\Phi}{\mathrm{De}_1}, \qquad \dot{Z}_2 = -\frac{Z_2 - wZ_\mathrm{e}}{\mathrm{De}_2} + \frac{w\,\Phi}{\mathrm{De}_2}, \tag{18}$$

where Φ is the forcing the one-mode law already carries. Splitting *both* terms is what makes $w = 0$ exact: the second memory is then driven by nothing, decays from zero, and stays there.

A timescale that moves. `CarreauZener` keeps one arm and lets its relaxation time depend on the shear rate. A Maxwell arm is a spring and a dashpot in series, so $\lambda = \eta/G$; if the dashpot obeys Carreau rather than Newton,

$$\mathrm{De}_\mathrm{eff} = \mathrm{De}\left[1 + (\Lambda\dot{\gamma})^2\right]^{(n-1)/2}, \tag{19}$$

and the arm relaxes faster the harder the medium is sheared. A collapse spans decades of shear rate, so this is a genuinely different claim from (18) about the same residual. Both terms of the memory equation carry $\mathrm{De}_{\mathrm{eff}}$, since an arm has one time constant; at $n = 1$ the bracket is unity at every shear rate and the law is identically the one-mode model.

Why the grid cannot judge either. Both carry six free parameters, and the quadrature used throughout is a Cartesian product costing count^p. At the 12 per axis used throughout, six axes is 5 971 968 solves against 168 072 for the entire standard set. The remedy is to expand about a fit rather than quadrature over a grid, at $1 + 2p$ solves, but doing so exposes a defect in the expansion itself.

The Occam factor must be capped at the prior. Written as in (5), the Laplace evidence contains $-\frac{1}{2}\log\det J^{\mathsf{T}}J$. A direction the data does not determine sends an eigenvalue of $J^{\mathsf{T}}J$ to zero and that term to $+\infty$: the expansion *rewards* a model for a parameter its design cannot see. This is not a small correction. On synthetic data generated from the one-mode model, the two-mode model scores

$$+29.7 \text{ nats above the one-mode model at } w = 10^{-3}, \qquad -2.6 \text{ at } w = 0.3,$$

so it wins precisely where its second arm does nothing and loses only where the arm does real work. The cause is the fitted Gaussian extending far outside the bounded cube the prior occupies. Capping each eigendirection's contribution at unity (a posterior cannot be wider than the prior it came from, Section C.7) returns -0.0000, -0.0004, -0.033 and -6.81 at $w = 10^{-3}, 10^{-2}, 0.05, 0.3$: monotone, and never paying for what the data cannot resolve. The correction is a strict generalization, agreeing with the plain form to round-off wherever every direction is already sharper than the prior.

The answer, on the 15 °C dataset. Both were fitted by multistart in the prior's unit coordinates, scored by the capped expansion, and summed over the modes the multistart found.

model	p	χ^2/N	$\log Z$	vs. qSLS	lag-one ρ	N_{eff}
qSLS	4	0.962	529.26	,	0.918	10.3
qSLS2	6	0.950	531.03	+1.78	0.918	7.2
qSLSthin	6	0.922	532.05	+2.79	0.875	8.2

The one-mode χ^2/N of 0.962 reproduces Section 5.1 exactly, by a different optimizer in different coordinates, which is the check that the fits are comparable at all.

Both enrichments are preferred, and neither margin means what it appears to. The residual autocorrelation is the defect they were reached for, and it does not move: 0.918 unchanged for the second relaxation time, $0.918 \rightarrow 0.875$ for the rate-dependent one, against N_{eff} near 8 out of 201 either way. Both $\log Z$ and χ^2/N presume independent residuals, so the likelihood overstates its own information by roughly $N/N_{\mathrm{eff}} \approx 25$; deflating a 2.79-nat margin by anything of that order leaves nothing. The second arm is not idle, it lands at 16% of the memory with a timescale 15.7 times the first, it simply does not address what is wrong. And `qSLSthin` pins its crossover at the bottom of its range, so that margin is bound-limited and the crossover is not resolved by this dataset at all.

What this says. A richer relaxation spectrum is not the missing physics, or at least not the part that matters here. Two independent enrichments, each reducing exactly to the one-mode law and each free to switch itself off, both decline to and both leave the correlation in place. That is narrower than it sounds, and Section 22.1 says why: both are *narrow* parameterisations, two discrete modes and one rate-dependent mode, while a network of unequal chain lengths debonding stochastically implies a *broad* continuous spectrum. The candidate that structure points to is a power-law or fractional kernel, and it is untested here. That is evidence about where to look next (the residual is concentrated in the first collapse, where sphericity and the far-field boundary are least secure) and it is more useful than a small positive margin would have been.

6 Separability of the model axes

With the constitutive law, the operator and the thermal treatment all uncertain, the question is not which experiment discriminates but whether the axes are separable at all. They need not be, in two ways. A difference lying in the span of $\partial R/\partial\theta$ is reproduced by refitting the material, and is then invisible at every design (Figure 13). And two axes whose remainders are nearly parallel move the evidence together, so no design can say which moved it.

axis	size	after refit	absorbed	+ initial state
dynamics (KM $\to$ Keller enthalpy/Mie–G)	8.89	**1.98**	**77.7%**	1.89
constitutive (one $\to$ two modes)	14.42	8.14	43.5%	7.72
thermal (cold $\to$ bubble + medium)	**15.99**	8.46	47.1%	7.80

in units of the dataset's noise. The last column widens the refit span to include R_0 and R_{eq}, which are held at their measured values everywhere else in this work. They take a further 4.7, 5.2 and 7.8 % of what the material left, so none of these differences is initial-state freedom in disguise. A near-zero number here is also what a broken projection would report, so the operation is checked both ways on the same vectors: handed the difference itself the basis absorbs it to 5×10^{-15}, handed a random column it absorbs 0.02 of 1.98.

The remainders sit at 46, 71 and 38° to one another. No pair is confounded, so the design problem is well posed for all three. Two things follow that were not expected.

Thermal is the *largest* lever, not a negligible one. It moves the trace most and survives refitting most. So the earlier observation, that switching it on made χ^2/N slightly worse, says the direction it moves in is not the one the dataset wants, not that it is small.

The operator pair measured here is the most absorbed of the three, and that is the mechanism behind Section 18: most of *this* operator change can be mimicked by adjusting the material.

It does not generalize to the axis, and the difference matters because it was read as a property of operators. Absorption is a property of a *pair*, and the pair above is the closest one available at the time. Measured against Keller–Miksis on the same dataset, with the same projection, the six dynamics of Section 18.1 span the range rather than clustering at its top:

against Keller–Miksis	size	after refit	absorbed
Keller enthalpy/Mie–G	8.89	**1.98**	**77.7%**
Keller enthalpy/Tait	12.85	5.81	54.8%
Herring/Tait	34.49	24.80	28.1%
Gilmore/Tait	39.30	29.33	25.4%
Lezzi–Prosperetti (2nd order)/Tait	53.21	**42.83**	**19.5%**

The second-order equation leaves 42.8 noise units where the pair this section was built on leaves 1.98, twenty-two times as much, and four fifths of it visible to the data rather than one fifth. So the operator axis is not uniquely hidden; the two operators available when that claim was made happened to be the two hardest to tell apart.

The peak wall Mach number on this dataset's own geometry is 0.110, so $O(M^2)$ is near 1.2% and a second-order correction is not a refinement: it is the largest operator difference measured here. One caution on reading the column: it is an L^2 norm over the whole dataset, so it is dominated by how far the trajectories decohere near collapse rather than by a pointwise $O(M^2)$ magnitude. It is the right yardstick against the other entries and the wrong one for a physical size.

A design against all three. Make each axis a parameter through $R(\theta,\varepsilon) = (1-\varepsilon)R_A + \varepsilon R_B$, so $\partial R/\partial\varepsilon = R_B - R_A$ is one more column of J. This puts discrimination and estimation in one information matrix and leaves the criterion concave, so the certificate of Figure 22 applies unchanged. The information is averaged over a decade of $g\alpha$, because operator information is governed by collapse depth and a measure certified at one material misranks others. The measure certifies at a gap of 10^{-9} on four support points.

axis	variance at the optimum	at the 15 °C geometry	efficiency
dynamics	6.78×10^{-3} (binding)	9.91×10^{-3}	68.4%
constitutive	5.45×10^{-4}	2.24×10^{-2}	2.4%
thermal	4.40×10^{-4}	1.05×10^{-2}	4.2%

Reported per axis rather than as one determinant. A design maximizing det could buy its whole score on the two easy axes and leave the operator undetermined without saying so. The experiments performed are near-best for the hardest axis and 24 to 41 times off for the two easiest. So a change of geometry buys a great deal of constitutive and thermal discrimination at little cost to the operator. The support wants $R_{\max}$ near 60 µm at high stretch and near 1200 µm at low stretch, against the 277 µm and stretch 7.09 performed.

7 What the model error costs

Every comparison in this paper, constitutive, operator, initial radius, thermal, is made between models that Equation (16) rejects. That does not make the comparisons worthless, since a ranking among wrong models is still a ranking, but it leaves a question no χ^2 answers: how much of a reported parameter is physics, and how much is the model being the wrong shape? The question has an answer, and it is not the one an error bar gives.

First, whether the ranking survives being wrong, which is testable and was not tested. "A ranking among wrong models is still a ranking" is a claim. Under misspecification the evidence rewards whichever model best ABSORBS the discrepancy, and absorption is precisely the mechanism (21) below shows biases the parameters, so the ranking could be measuring absorption capacity rather than physical closeness, and nothing separates those.

It separates directly. Generate from a truth the catalogue does not contain (the two Section 14 uses, thermal transport and a second relaxation mode) rank the catalogue by evidence, and separately measure which candidate is genuinely closest, defined as the residual its OWN best fit leaves against the noiseless truth.

	thermal truth			two-mode truth		
candidate	$\log Z$ margin	rank	distance	$\log Z$ margin	rank	distance
qSLS	0	1	**0.545**	0	1	**0.191**
SLS	−73.0	2	1.077	−44.1	2	0.803
NHKV	−2549	3	4.974	−2314	3	4.698
qKV	−2598	4	5.039	−2405	4	4.809

The orderings are identical: rank correlation +1.000 under both truths, and the evidence winner is the closest model in both. The demanding part of that is the bottom pair, whose distances differ by 1.3% and which the evidence still orders correctly.

Two limits on what this licenses. Both truths are drawn from physics this document already screens, so it tests misspecification of a KIND already considered and of a size Section 6 measures at 14 to 16 noise units. And the catalogue's tiers are separated by thousands of nats, so the test is not stressed where it would be most informative, between rivals a few nats apart, which is the regime Section 16 says the operator axis is actually in. What it does establish is that the defence above is not merely an assertion at the size of error these datasets show.

Half of it is invisible, and that is a theorem

Write the truth as $y = m(\theta) + \delta + \varepsilon$. Whitening by the standard error of the mean (the $\sqrt{J}$ of Equation (16), not the trial spread) and letting J hold the parameter sensitivities, the residual splits along the model's own span,

$$d = \underbrace{Pd}_{\text{absorbable}} + \underbrace{(I - P)\, d}_{\hat{\delta}}, \qquad P = J(J^{\mathsf{T}} J)^{-1} J^{\mathsf{T}}. \tag{20}$$

At a converged interior fit the normal equations give $J^\top d = 0$ exactly, so $Pd = 0$ and $\hat{\delta} = d$: the whole residual is identifiable discrepancy, and the component of δ lying in the span of J has left *no trace whatsoever*. It was absorbed into $\hat{\theta}$ while fitting. Two worlds (a biased θ with no discrepancy, and the true θ with a discrepancy in the span) predict the same data exactly. This is the unidentifiability of Kennedy and O'Hagan [7], and it is why the bias cannot be estimated from these data at any budget.

So bound it

Suppose only that the invisible part is no larger than the visible one, $\|\delta_\parallel\| \le B$. The induced bias is $\Delta\theta = (J^\top J)^{-1} J^\top \delta_\parallel$, and maximising each component over that ball gives

$$\max_{\|\delta_\parallel\| \le B} |\Delta\theta_k| \;=\; B\sqrt{\left[(J^\top J)^{-1}\right]_{kk}} \;=\; B\,\sigma_k, \tag{21}$$

since $\|J(J^\top J)^{-1} e_k\|^2 = [(J^\top J)^{-1}]_{kk}$. The worst bias is B multiples of each parameter's *own* standard deviation, the same multiple for every coordinate, so a single number covers the whole vector. Taking $B = \|\hat{\delta}\|$ is the natural choice and is not conservative: nothing in the data forbids the unseen component being larger than the seen one.

What $\hat{\delta}$ is the size of. It is worth naming what (20) is, because this document goes on to use $\hat{\delta}$ as though it were the discrepancy and it is the discrepancy's *visible remainder*. The confounding between θ and δ is Kennedy and O'Hagan [7]'s, and forcing the discrepancy orthogonal to $\partial m/\partial\theta$ is the construction Plumlee [8] uses to identify θ at all; that omitting δ biases the parameters is Brynjarsdóttir and O'Hagan [9], that the resulting estimator need not be consistent is Tuo and Wu [10], and the identifiability of the two is analysed by Arendt et al. [11]. None of that is new here. What this package can add is a number for how bad it is in this problem.

Screened the way (22) screens anything else, a timebase perturbation is 97 % absorbed by refitting the material, so its invisible component is between 5.7 and 5.9 times its visible one in norm. Two things follow for the tables above. Taking $B = \|\hat{\delta}\|$ was called natural and not conservative; for a mechanism these datasets demonstrably contain it understates (21) by nearly six, and it does so worst where an error is hardest to see. And (22) scores a candidate by how much of its *visible* part aligns with the visible residual, so a large absorbable mechanism scores low. The null result of the screen therefore says that nothing on the list resembles the visible remainder, which is weaker than saying nothing on the list is present: the clock is present, is measured in Section 22.1, and sits mid-table.

The bound exists only in the identified coordinates

Equation (20) means something only where $Pd = 0$, and in the natural axes that can never hold. Section 3.5 shows the likelihood depends on $g\alpha$ alone; measured at the fitted point, g and α are anticorrelated at -1.00000 with $J^\top J$ conditioned near 10^8 and a null direction of $(-0.707, 0, 0, +0.708)$ in log coordinates. A fit slides along that valley until it meets a wall of the prior box, and at a *constrained* optimum the gradient along the active bound does not vanish, so part of the residual stays absorbable.

Widening the box does not help, which is worth stating because it is the obvious thing to try. Released from $g \ge 100\,\mathrm{Pa}$, the pair slid until α reached its own upper bound instead: the pin moved rather than lifted, and $g\alpha$ was preserved to 1.6 % across a hundredfold change in g. A box has walls, and a fit on a flat valley finds one.

Refitting in the identified coordinates $(\mu, g\alpha, \lambda_1)$ removes the valley, and with it the obstruction:

coordinates	cond $J^\top J$	absorbable	at optimum	bound
$(\mu, g, \lambda_1, \alpha)$	9×10^5	39 % to 59 %	no	unavailable at any box
$(\mu, g\alpha, \lambda_1)$	3×10^1	0.0%	yes	(21) applies

The budget

Evaluating (21) at each dataset's own $\hat{\delta}$ and its own $(J^\mathsf{T} J)^{-1}$, in the identified coordinates the previous subsection established are the only ones it holds in:

dataset	bound	μ	$g\alpha$	λ_1
15 °C	59.3 σ	×1.58	×1.30	×3.11
23 °C	52.6 σ	×2.00	×1.55	×3.90
33 °C	24.7 σ	×1.54	×1.57	×2.98

Read as a model-form error budget: the identified modulus could be wrong by 30 % to 57 %, the viscosity by 54 % to 100 %, and the relaxation time by a factor of three to four, from model-form error alone, with no amount of data of this kind able to distinguish that from the truth. The large multiples of σ and the modest factors are consistent, and their ratio is the point: σ is small in these coordinates precisely because they are the identified ones, which is what makes the budget readable rather than infinite.

What could remove it

The bound says how much of a parameter the discrepancy could be eating. It does not say what the discrepancy *is*, and $\hat{\delta}$ is not a shapeless residual: the three datasets' $\hat{\delta}$ correlate at 0.49 to 0.84, where unrelated curves of this length would sit near 0.08, and between 65 and 69 percent of each sits in the first collapse. One curve, measured rather than guessed, is a thing a candidate physics can be screened against.

The screen is a projection, not a fit. Any enrichment supplies a direction v (the trace it would add, differenced against the current model) and only the part of it orthogonal to the existing sensitivities can remove any of $\hat{\delta}$. The rest is absorbed by refitting the parameters already present, which improves χ^2 and explains nothing. So the question is a cosine,

$$\text{removable} \;=\; \cos^2\!\big(\,(I-P)v,\;\hat{\delta}\,\big), \qquad \cos = \frac{\langle (I-P)v,\;\hat{\delta}\rangle}{\|(I-P)v\|\,\|\hat{\delta}\|}, \tag{22}$$

at one solve per candidate rather than a multistart and an evidence integral each. That inversion is the point. Fitting every candidate answers "which of these fits best", which among models Equation (16) rejects is the wrong question; (22) answers "could this be the missing physics at all", and a candidate near zero cannot be, whatever its Bayes factor does. Screen first, fit what survives.

The sign is not a detail. Most enrichment amplitudes are one-sided: a Maxwell arm's share, a stiffening coefficient and a switched-on transport term are all non-negative. A candidate whose direction *anti*-aligns with $\hat{\delta}$ therefore cannot reduce it at any admissible amplitude, and scoring $|\cos|$ reads such a candidate as the most promising one on the table. That is not hypothetical, it happened here, and it is why the second relaxation arm carries a dagger below. The reachable set is the *cone* of non-negative combinations rather than the span, so the joint figure is a non-negative least squares onto that cone and not a projection onto a subspace.

candidate	15 °C	23 °C	33 °C
initial radius	**23.7**%	**13.0**%	**13.9**%
operator: keller-enthalpy/MG	7.7%	1.8%	3.9%
operator: gilmore/Tait	1.9%	0.4%	4.8%
thermal (bubble + medium)	2.6%	2.5%	6.5%
thermal (bubble)	1.9%	2.4%	3.6%
mass transfer	0.2%	0.2%	2.4%
second relaxation arm	1.1%†	58.1%†	3.8%
all together, on the cone	25.2%	**15.1**%	21.0%
the same, read as a subspace	25.9%	**71.5**%	21.7%
$\|\hat{\delta}\|$	59.3	52.6	24.7

† anti-aligned: the cosine is negative, so no admissible amplitude removes anything. At 23 °C the cosine is −0.762, and read as $|\cos|^2$ that candidate is the best on any dataset in the table at 58.1%, while in fact it can only make the residual worse. The last two rows are the cost of getting this wrong, measured: read as a subspace the 23 °C candidates jointly remove 71.5% of the discrepancy, and on the cone they remove 15.1%. The other two datasets move by less than a point, because there the anti-aligned candidate was small anyway, the error is not uniform, and it is largest exactly where it would have mattered most.

The list omitted the first collapse, which is where the discrepancy is. Every candidate above is a bulk constitutive, transport or configuration effect, and 65 to 69 percent of $\hat{\delta}$ lies in the first collapse, where Section 5.3 names sphericity and the far-field boundary as least secure, and then screens neither. Two first-collapse candidates, on the same cone and the same footing:

candidate	15 °C	23 °C	33 °C
asphericity $n = 2$ (silhouette)	0.4%	0.6%†	0.0%
asphericity $n = 2$ (volume equivalent)	0.2%†	2.5%†	5.0%†
asphericity $n = 3$ (volume equivalent)	0.2%	1.5%†	3.3%†
initial wall velocity +0.02	**22.6%**	**37.6%**	**11.7%**
initial wall velocity −0.02	33.6%†	28.9%†	19.6%†

† anti-aligned, so no admissible amplitude removes anything; the signed pair on the last two rows is one candidate screened in both directions.

The shape modes are integrated from Plesset [12] along the fitted trajectory, $\ddot{a}_n + 3(\dot{R}/R)\dot{a}_n = A_n a_n$ with $A_n = (n-1)\ddot{R}/R - (n-1)(n+1)(n+2)\sigma/(\rho R^3)$, and screened under both imaging conventions because what a shape mode does to a REPORTED radius depends on whether the instrument returns a silhouette or a volume equivalent, an assumption, so both are shown.

Asphericity is not it. At most 5%, anti-aligned on most datasets, and smaller than thermal transport. That is worth knowing precisely because it was the obvious suspect.

The initial condition is the largest single candidate ever screened here. Every fit in this document starts the bubble at $R_{\max}$ with zero wall velocity, which asserts that the imaged maximum is a turning point of a free collapse. A laser deposits energy; it need not be. Releasing that assumption reaches 37.6% of $\hat{\delta}$, against 23.7% for the initial radius and 6.5% for the best genuine-physics entry on the original list.

Fitted rather than screened, that 37.6% does not survive. A screen bounds what a candidate could remove with its amplitude free, and the way to collect on it is to fit. Doing so with the stretch *also* free (pinning a quantity known only to five–seventeen percent would let one error masquerade as the other) the velocity does land somewhere definite and does not buy what the screen promised:

	15 °C		23 °C		33 °C	
free	χ^2/N	lack of fit	χ^2/N	lack of fit	χ^2/N	lack of fit
nothing	0.971	16.79	1.029	2.95	0.439	1.80
R_{eq}	0.623	**8.43**	0.901	**2.15**	0.379	1.60
u_0	0.724	15.38	0.372	2.86	0.278	1.57
both	0.500	,	0.324	2.30	0.229	**1.38**

χ^2/N falls everywhere, which is what free parameters do and is not the question. Against *lack of fit*, adding u_0 on top of a free stretch improves one dataset and worsens another, and 15 °C is withheld: its three prior widths fail the nesting gate of Section 19, a box containing another scoring worse than it, at 16 restarts and again at 32, so that surface is multimodal and the cell is not read.

Two things do come out of it cleanly. The velocity lands at −0.022 to −0.033 of the Rayleigh scale on every dataset, always inward, and identically at ±10% and ±30% prior width, so it is a determined quantity rather than a bound. And R_{eq} and u_0 are *separately identified*: their correlation is +0.06 to +0.13 with $J^{\mathsf{T}}J$ conditioned near 10^2, against −1.00000 and 10^8 for the g–α pair of Section 3.5. Two configuration quantities that could easily have been one direction are not.

The obvious reading of an inward velocity at the imaged maximum is that the maximum is a sampled frame rather than the true turning point. That is checkable and does not survive either: half a frame of the model's own deceleration is -0.007 to -0.008 of the Rayleigh scale, three to four times smaller than what is fitted. The sign and the order are right and the magnitude is not, so sampling is part of it at most.

Read with Section 19, the two point the same way. The largest entry on the old list was the initial radius and the largest on the new one is the initial velocity; both are the initial condition rather than the material. The discrepancy this document has been attributing to constitutive incompleteness is, on the evidence of its own screen, better aimed at how the datasets begin.

What this says. Nothing on the constitutive or transport list accounts for the discrepancy. Every genuine physics candidate is small, thermal transport between 1.9 and 6.5 percent, mass transfer below 2.4, the operator below 7.7, and taken *all together* on the reachable cone they leave between 75 and 85 percent of $\hat{\delta}$ unexplained on every dataset. The largest single entry is not physics at all but the initial radius, a configuration quantity that Section 19 shows moves the trace as much as the operator does, and even it removes at most a quarter.

Two cautions travel with the table. The figures are *upper bounds*, not forecasts: each is what a candidate could remove with its amplitude free and the base parameters responding linearly, and a fit obeys neither, so the delivered reduction is smaller. And a screen cannot propose what it was not given: these are the enrichments this package implements, and the 75 to 85 percent that survives is a statement about them rather than a proof that no mechanism exists. What it does establish is that the discrepancy Equation (16) detects is not the thermal, mass-transfer or operator physics already on the shelf, and that (21) is therefore the honest way to carry it rather than a placeholder awaiting the obvious fix.

Three consequences run through the rest of this paper. A parameter interval that omits (21) understates its own uncertainty by one to two orders of magnitude, so the posterior widths of Section 5.1 are not the error bars a rheologist wants. A model comparison is a comparison of the identifiable parts, since the absorbed parts of two models' discrepancies are invisible in both. And a design criterion computed under a rejected model optimises a quantity whose parameters carry this budget, which is the gap Section 14 names and (21) finally sizes.

An observable the screen was never given

The caution above is that a screen cannot propose what it was not given, and everything it was given is a direction in the trace. The trace is a function of time, so anything that rescales time trades against anything that dissipates energy, and Section 7 measures that trade at 97 %: a timebase perturbation is almost entirely absorbed by refitting the material, which is why it lands in the parameters rather than in $\hat{\delta}$. An observable built to be blind to that trade is a different instrument, not a different candidate.

The ratio of successive afterbounce maxima, A_{k+1}/A_k, is one. It is unchanged by any rescaling of time, it needs no derivative of a sampled trace, and it needs no model. It is also computed within a trial, so it may be averaged across trials where a trace may not, and it uses all 39 events rather than three mean curves. Each bounce is slower than the one before, so the sequence is a descending sweep in rate, and the fact that the maxima are broad is what makes it usable: reading the model on the dataset's own 201 points rather than a grid a hundred times finer moves its ratios by 0.000 to 0.001, against measured-against-model gaps of 0.03 to 0.10.

No material in this form reaches it. Fitting the qSLS directly to the ratio sequence, over a box six decades wide, takes χ^2 per bounce from 36.75 to 6.52 at 15 °C and from 11.62 to 5.33 at 23 °C, and stops there. The 33 °C dataset reaches 0.67 on 7 events, which is 1.73 at the sample size of the other two and is therefore weaker evidence rather than contrary evidence. The fitter is not the limitation: run against a sequence the model itself generated at four times its relaxation time, the same fitter returns $\chi^2 = 0.0000$ and recovers that time to three decimal places on all three datasets. The elastic modulus is not the limitation either, and not because it was bounded: the ratios are identical at 10^{-4} and 10^{-6} of the fitted value, so the objective is flat and the afterbounces simply cannot see it.

The failure has a shape, and the shape is not what it first looked like. A spherical viscoelastic model damps monotonically, because damping weakens as the bounces slow and nothing in the form allows otherwise.

Its ratio sequence is monotone in all six cases, three datasets at the trace fit and at the best achievable fit. Every measured ENSEMBLE sequence is non-monotone, and at the best fit the residual concentrates on a single bounce, at -5.0, -4.4 and -1.4 standard errors of the mean. Read directly, that is one bounce losing more amplitude than any material in this form can take from it, and the bounces concerned sit at a common frequency near 72 kHz rather than at a common index.

It does not survive a shift null, and the retraction is the result. Each trace carries its own event's clock: the first collapse sits at a normalised time whose between-event coefficient of variation is 0.017 to 0.025 on all six datasets, where a shared clock would give the 0.223 of $R_{\max}$. So a feature belonging to the bubble sits at the same bounce INDEX in every event, while a feature at a fixed frequency sits at an index that moves with that event's radius, and averaging at fixed index preserves the first and dilutes the second. Circularly shifting each event's ratio sequence destroys alignment between events while preserving each event's own dip exactly, which builds the null that separates them with no noise model, no radius and no pairing.

Table 1: The ensemble dip against the distribution produced by shifting each event's own sequence at random. Five of six datasets show nothing beyond the null, and the sixth is $p = 0.022$ on one of six tests. The apparent single-bounce anomaly is an artefact of averaging events whose deepest fall sits at a different bounce in each. Read the non-detections with the power of Section 8 attached: on the three gelatin datasets this test cannot detect an index-locked fall the size of the one observed, so their p of 0.48 and 0.78 is not evidence of absence. The retraction rests on the multiplicity of the one significant cell and on the two large PAAm datasets, where the test does have power.

	events	ensemble dip	null mean	z	p
gelatin 15 °C	18	0.1222	0.0595	+2.34	0.022
gelatin 23 °C	14	0.0976	0.1014	−0.08	0.486
gelatin 33 °C	7	0.0782	0.1174	−0.79	0.781
PAAm `PA05`	52	0.0006	0.0506	−2.18	0.997
PAAm `PA05` temp	117	−0.0393	0.0171	−4.88	1.000
PAAm `PA05003` temp	80	0.0552	0.0323	+1.37	0.094

The 72 kHz reading goes with it. That agreement across datasets was already carrying a $p = 0.016$ computed against clustered periods, and with no coherent feature to attribute there is nothing for a frequency to be the frequency OF. Two candidate mechanisms died on the way and neither needs restating at length: no shape mode can hold to 3 % when R_0 spreads 15.8 % and the modulus spreads 43 %, and every capillary degree falls with temperature where the anomaly rose. The observation those were competing to explain is now withdrawn.

What survives is the part that was never a dip. The first afterbounce damps more than the model and the second damps less, on all three gelatin datasets, at $-9.3/+7.5$, $-4.2/+6.0$ and $-3.9/+3.1$ standard errors. That is a smooth systematic across the whole sequence rather than a localised fall, and a shift null says nothing about it. It is also what carries the finding that matters here: fitting the qSLS directly to the ratio sequence, over a box six decades wide, takes χ^2 per bounce from 36.75 to 6.52 at 15 °C and from 11.62 to 5.33 at 23 °C and stops, while the same fitter run against a model-generated sequence returns $\chi^2 = 0.0000$. No fixed material in this form reaches those two sequences, and that conclusion never rested on the anomaly. It does not extend to the third: 33 °C reaches 0.67 and is fitted by a fixed material, on seven events. Section 8 runs the same test on 249 events of a second material, where three of the four datasets it can be run on behave like the first two.

And the observable itself is sound, which had to be checked. A maximum of noisy samples is biased, the bias grows as the bounces shrink, and it is systematic so averaging does not remove it. Re-extracting the measured ratios at half-widths of 2, 3 and 4 moves them by 0.000 on the first three bounces of every dataset. Injecting noise into the model shows that indifference holds only below $\sigma \approx 0.001$, an order below what a naive second-difference estimator returns, and at that level the bias accounts for 3 to 7 % of the gap. An extraction corrupted by noise cannot be indifferent to how wide a window it searches, so the measurement bounds its own error here.

8 The same measurements, on a second material

Everything in this part rests on 39 events. Three gelatin datasets carry 18, 14 and 7 repeats, and from them come the rejection of Section 5.1, the empty physics shelf of Section 5.3, the shaped discrepancy of Section 7 and the afterbounce failure of Section 7. A reader entitled to one objection should spend it here: three datasets of one material, one of them with seven repeats, is a case study, and a case study about gelatin is not a claim about the method.

The same release carries 249 polyacrylamide events on the same instrument. This section runs the three load-bearing measurements on them, unchanged. Nothing in the machinery is retuned; only the datasets are new.

What a PAAm dataset is, which took establishing

Two of the three released PAAm files are not datasets as Section 2 defines one, and using them as they stand would have decided the answer before the first fit.

`PA05_temp` is a pooled temperature sweep. Its 117 columns are three blocks of 30, 57 and 30 at 10, 21 and 33 °C, which is what the per-temperature event counts in the released auxiliary files state and what the first-collapse time confirms independently: partitioned there, the between-block F is 14.1 against a permutation null of the same partition, $p < 0.0005$. Fitting the file whole would have averaged three materials and reported the average as a lack of fit. `PA05003_temp` shows no block structure at any two-cut partition, its best F of 6.6 sitting below a best-of-all-partitions null of 8.6, so it is carried whole. Should it be pooled after all, its trial spread is inflated, which enlarges the denominator of every ratio below and makes its rejection conservative. The three files share no column, so the 249 events are distinct.

The radii are the harder gap. $R_{\max}$ and $R_{\max}/R_{\mathrm{eq}}$ are tabulated per dataset for gelatin and are not available per dataset for PAAm: the release holds 437 radii whose pairing to the trace columns is not recoverable, as Section 7 records. So every PAAm dataset is carried at the file's mean of 359 µm, with a stretch read off its own late trace. That is an assumption, and it is measured rather than defended. Refitting one dataset across the plausible range of both separates them cleanly. Over a factor of 1.35 in $R_{\max}$, χ^2/N moves by 3 % and the lack-of-fit ratio by 4 %, while μ tracks $R_{\max}$ at 1.000, 1.169, 1.339 against the 1.000, 1.176, 1.353 the dimensionless groups require and g does not move at all. The missing per-dataset radius therefore costs the reported moduli a scale factor and costs the residual-shape conclusions nothing. The equilibrium radius is the opposite: over its own uncertainty the ratio runs 12.2 to 22.9, a factor of 1.88. That is a caution for every number in this document rather than for PAAm alone, and it is taken up below.

The model is rejected there, and refitting the radius does not rescue it

Table 2 is the lack-of-fit test of Section 5.1 on both materials. It is reported without the correction that divides out the parameter share of the trial spread, since that share was measured on gelatin and would be carried rather than established here.

Why this table is not the one a classical treatment would print. The textbook statistic sums $J\sum_i(\bar{y}_i - \hat{y}_i)^2$ unweighted against the replicate pure error. These records are strongly heteroscedastic and the fit minimises the WEIGHTED sum, so the unweighted numerator is not what any fit here descends, and its $k-p$ degrees of freedom belong to no minimisation performed. That is not a formal complaint: raising the multistart budget from 10 to 40 improves the 15 °C fit from $\chi^2/N = 0.9727$ to 0.9535 and *raises* its unweighted ratio from 14.16 to 16.69. A statistic that worsens when the fit improves is measuring the wrong thing.

Weighting both sums consistently repairs it and, in doing so, removes a claim this document has been making. The pure-error term becomes $\sum_{ij}(y_{ij} - \bar{y}_i)^2/s_i^2 = k(J-1)$ identically, so $\mathrm{MS}_{\mathrm{pure}} = 1$ exactly and what remains is $F_w = \frac{k}{k-p}J\chi^2/N$. The classical test applied to a heteroscedastic record with its own weights IS $J\chi^2/N$. Section 5.1 presents these as two routes to one conclusion, "reached with none of the same machinery"; they are one route with a factor of J between them, and that sentence is withdrawn. What survives is the part that was always the point: the factor of J is not a detail, because the fit is compared against the spread BETWEEN events when the standard error of their mean is smaller by $\sqrt{J}$.

The threshold is then the only live question, and it is settled by simulation rather than disclaimed. An analytic $F(k-p,\cdot)$ critical value near 1.2 assumes independent errors; these are correlated at $\rho = 0.68$ to 0.86.

Table 2: The lack-of-fit test on both materials, weighted so its numerator is the sum the fit minimises, and judged against a null carrying each record's own autocorrelation rather than an assumption of independence. $\mathrm{MS}_{\mathrm{pure}}$ is 1.000 on every dataset by construction, which is the finding rather than a rounding: weighting by the trial spread makes the pure-error sum $k(J-1)$ identically, so the denominator carries no information and the test is $J\chi^2/N$. The last two columns refit the equilibrium radius over seven stretches from 5.5 to 8.5. Every dataset is rejected, before and after.

				null		at the preferred stretch	
dataset	events	χ^2/N	F_w	95 %	99 %	stretch	χ^2/N
gelatin 15 °C	18	0.95	17.51	1.35	1.53	6.5	0.64
gelatin 23 °C	14	1.02	14.55	1.50	1.79	7.0	0.91
gelatin 33 °C	7	0.44	3.11	1.48	1.74	7.0	0.55
PAAm `PA05`	52	0.60	31.99	1.37	1.54	7.0	0.50
PAAm 10 °C	30	0.50	15.30	1.38	1.58	7.5	0.34
PAAm 21 °C	57	0.38	22.19	1.35	1.52	7.0	0.26
PAAm 33 °C	30	0.47	14.45	1.41	1.56	7.5	0.48
PAAm `PA05003`	80	0.62	50.56	1.54	1.82	7.0	0.50

Surrogates carrying each record's own autocorrelation put the 99 % point at 1.52 to 1.82 instead. Correlation moves the threshold by a quarter to a half and changes no verdict, which is worth stating precisely because Section 5.1 raises this objection against itself and then leaves it unquantified.

Two entries move enough to be worth naming. Gelatin 23 °C reported 2.94 unweighted and read as the weakest rejection in the document; weighted it is 14.55, because the unweighted statistic understates the misfit exactly where the trial spread is largest. Gelatin 33 °C reported 3.26 from a fit that ten starts could not find; at forty starts it reaches $\chi^2/N = 0.4354$ and $F_w = 3.11$, which is the weakest rejection here and still sits above its own 99 % null of 1.74. The seven-event dataset is the one carrying the least, as it should be.

The rejection reproduces. It is worth separating two readings of how strongly, because F_w alone oversells it. Since $F_w = \frac{k}{k-p} J\chi^2/N$ exactly, the statistic scales with the replicate count by construction, so most of the gap between PAAm's 14.5 to 50.6 and gelatin's 3.1 to 17.5 is 30 to 80 events buying power to detect the same misfit rather than a worse model. The reading that does not depend on the replicate count is the χ^2/N column, and it says the opposite of what the F_w column suggests: PAAm fits BETTER, at 0.38 to 0.62 against gelatin's 0.44 to 1.02. What PAAm buys is not a larger failure but a securer detection of one.

That χ^2/N column is also the sharpest form of the argument this document has been making, and the two facts are the same fact. Every PAAm dataset fits at 0.38 to 0.62. Reported alone, against the between-event spread, as it usually is, that number says the model reproduces the data to better than experimental repeatability. The same fits are rejected at 14 to 51 against a null of 1.5 to 1.8, because the fit is being compared to the scatter of single events when it was fitted to their mean.

The last two columns close the obvious objection. The enrichment screen below makes the initial radius the largest admissible candidate on seven of eight datasets, and the sensitivity above makes the ratio a factor of two sensitive to it, so a reader is entitled to suspect that the discrepancy is a mis-set R_{eq} rather than model-form error. Both of those are linearised about one point; refitting is not. Every dataset is refitted at seven stretches from 5.5 to 8.5, all eight then have an interior χ^2 minimum, and all eight prefer a value between 6.5 and 7.5 while being carried at 6.83 to 8.01. A real share of what is reported as misfit is a pinned radius: χ^2/N at the preferred value falls by 25 % on the median dataset and by as much as a half, though on one it does not fall at all. It is not the whole share, which is the result. Carried through the same weighted statistic, every dataset at its own preferred stretch still gives F_w of 3.96 to 41.3 against nulls of 1.5 to 1.8. Refitting the radius freely does not rescue the model.

One thing in that column is worth stating rather than smoothing. Among the four datasets whose preference sits more than 0.5 from the value they are carried at is gelatin 15 °C, whose stretch is tabulated in the source rather than estimated here, so the offset is not an artefact of how the PAAm datasets were prepared.

A claim made earlier in this section has to be withdrawn here rather than quietly dropped. On the unweighted statistic gelatin 23 and 33 °C fell to 2.3 and 1.9 at their preferred stretch, and that was read as the gelatin case resting on one dataset. Weighted, they are 13.0 and 4.0: 23 °C is among the firmest rejections in the document rather than the weakest, because the unweighted numerator was discounting exactly the samples where the trial spread is largest. The argument for the PAAm datasets carrying the claim stands on their sample size, not on a weakness in gelatin that the statistic manufactured.

The shelf is empty there too, and the afterbounces mostly fail there too

Table 3 runs the other two measurements. The enrichment screen of Section 5.3 scores what share of each dataset's measured discrepancy a candidate enrichment could remove, and the afterbounce test of Section 7 asks what χ^2 per bounce a fixed qSLS can achieve against the ratio sequence over a box six decades wide.

Table 3: The enrichment screen and the afterbounce test on both materials. The screen scores every candidate together, as a cone, so the last of those columns is what no admissible enrichment reaches. The afterbounce fit is run at four seeds; the spread is the range across them. A dataset whose control did not run takes no verdict.

	enrichment screen		afterbounce fit			
dataset	reachable	left over	bounces	best χ^2	seed spread	control
gelatin 15 °C	25 %	75 %	5	6.52	0.000	passed
gelatin 23 °C	15 %	85 %	5	5.33	0.000	passed
gelatin 33 °C	21 %	79 %	5	0.67	0.000	passed
PAAm `PA05`	24 %	76 %	5	0.70	0.028	passed
PAAm 10 °C	43 %	57 %	5	8.27	0.000	passed
PAAm 21 °C	39 %	61 %	4	1.32	0.042	passed
PAAm 33 °C	24 %	76 %	4	2.43	0.000	passed
PAAm `PA05003`	26 %	74 %	5	n/a	0.007	did not run

Every candidate enrichment together leaves 57 to 76 % of the PAAm discrepancy unexplained, against 75 to 85 % on gelatin. Weaker, and the same verdict: the shelf does not hold it. The largest admissible single candidate is the initial radius on seven of the eight datasets, which is the linearised form of what the stretch refit measured directly, and it removes at most 36 %.

The afterbounce test reproduces on three of the four PAAm datasets where the fitter is validated, at 8.27, 2.43 and 1.32 against gelatin's 6.52 and 5.33. One dataset of each material is reachable by a fixed material: gelatin 33 °C at 0.67, which Section 7 already reported, and `PA05` at 0.70. So the count is two of three on gelatin and three of four on PAAm. The effect is markedly weaker on PAAm and it is the same effect.

Three cautions belong with that column. The first concerns the shape of the PAAm failure, and it must be stated more carefully than a first reading of the sequences suggests. The three temperature blocks are monotone throughout; `PA05` falls at bounce 6 and `PA05003` at bounces 3 and 5, so it is not true that PAAm never falls. What is true is that neither fall survives the shift null of Section 7, at $p = 0.998$ and $p = 0.090$, where gelatin 15 °C reaches $p = 0.022$. Pooling was the obvious suspect for the monotone blocks, since averaging datasets that do not share a condition is what manufactured the gelatin dip, and it is not the cause: the blocks are monotone individually, and the same extractor finds the dips on all three gelatin datasets.

That null is weaker than it looks, and the weakness runs against the reading above rather than for it. Injecting an index-locked fall of 0.10 into every event and re-running the test recovers it at $J = 52$ and $J = 117$ but misses it at $J = 7$, 14 and 18, so at gelatin's sample sizes the test cannot detect a dip of the size gelatin actually shows. The PAAm non-detections, at J of 30 to 80, carry more weight than the gelatin ones do. Circular shifting also manufactures a fall at the seam, which puts the null mean above the observed statistic whenever a sequence rises throughout: a large negative z means the events are individually monotone, not that a dip failed to align. The second caution is the direction of the bound. A fit returns an upper bound on the achievable χ^2, so a small value is self-evidencing while a large one rests on the optimiser having found the optimum; the synthetic-target control is the only thing licensing the second kind of claim, and no verdict is issued where it did not pass. `PA05003` is that case, and it is also the only dataset seen to settle in two different basins between runs, at 26.3 and 39.1. The control's absence and the instability appear together, which is the argument for the control rather than against it.

That control's SCOPE has to be stated with it, because it is narrower than the claim it licenses. The synthetic target is generated at λ_1 displaced by a factor of four from the starting material, so recovering it to $\chi^2 = 0.0000$ certifies the search over that displacement and not over the six decades the box spans. A material far from the start, in a basin the multistart never samples, would produce exactly what is reported here. So ”no fixed material reaches these afterbounces” should be read as ”none that this search finds, from a start the fit to the whole trace chose, with a search demonstrated to work over a fourfold displacement”. Widening the sham until it fails would measure that scope, and has not been done.

The third is a defect in how these numbers were produced, since it touches values already quoted. The multistart behind Section 7 was seeded from a string hash, which Python salts per process, so it drew different starting points on every run and its 6.52 and 5.33 were not reproducible. The seed is fixed now, and running every dataset at four seeds says the defect was harmless in effect: the spread is 0.000 to 0.042 and the gelatin values return identical. That is a measured statement rather than an assumed one, which is the only reason it is worth a sentence.

What this changes

The first three links of the chain hold on a second material at six times the sample size. The model is rejected, at χ^2/N values that would be reported as a good fit. No candidate enrichment accounts for the bulk of what is missing. No fixed material in this constitutive form reaches the afterbounce sequence on most datasets. And the one alternative explanation with a free parameter behind it, a mis-set equilibrium radius, is refitted rather than argued away and does not survive.

One claim must be narrowed rather than extended by this. Only gelatin shows a fall that survives its own null, so the part of Section 7 that reads the SHAPE of the failure is a statement about one material. It is narrowed a second time by the premise it was argued from. Section 7 asserts that a spherical viscoelastic model damps monotonically and that nothing in the form allows otherwise; that is false. A quadratic Zener at $g = 7\,\mathrm{kPa}$, $\alpha = 0$, $\mu = 0.2\,\mathrm{Pa\,s}$ and $\lambda_1 = 1.78\,\mu\mathrm{s}$ produces the ratio sequence 0.598, 0.579, 0.751, 0.916, 0.975, which falls at the second bounce and is unchanged to five digits under grid and tolerance refinement. The mechanism is the Debye loss peak at $\omega\lambda_1 = 1$: the afterbounces are a descending frequency sweep, so a sequence that crosses the peak loses more per cycle and then less. That point lies inside the box searched here. Non-monotonicity therefore does not by itself place a record outside the form, and the argument for the failure rests on the achievable χ^2 of Table 3, which is measured, and not on the monotonicity premise, which is wrong.

The correction that is not applied, and why. Table 2 is quoted without the factor of Section 5.2 that divides out the share of the trial spread lying in the span of the model's own sensitivities, and that share is now measured on both materials rather than carried. It is 28 % to 46 % on PAAm against 15 % to 46 % on gelatin, so the correction it implies runs 1.39 to 1.84 on PAAm against the 1.65 the published gelatin value gives. Applying one material's number to the other is therefore not neutral, and the uncorrected column is the one that assumes nothing.

Two things came out of measuring it. Evaluated at the point Section 5.2 uses, the share returns 39.3 % exactly, so this is the same measurement rather than a similar one. Evaluated at the same dataset's own fit from Table 2, which sits elsewhere along the $g\alpha$ ridge, it returns 46.1 %. The ridge is a degeneracy of the likelihood and was not expected to move the span of the sensitivities; it moves it by 6.8 points, so that much of the share is a property of where a fit landed rather than of the data. That is a third reason to leave the correction off.

And a test that does not transfer, because it does not work. Whitening the fit residual by the hierarchical covariance $G\Sigma G^{\mathsf{T}} + \sigma^2 I$ leaves its lag-one correlation at 0.42 to 0.88 across both materials, against 0.68 to 0.86 before, which reads as the correlation surviving the only explanation Section 5.2 offers for it. It was reported here as exactly that, and it is withdrawn.

The reading requires the whitener to remove a correlation that IS parameter variation, and nothing had checked that it does. On two of the eight records whitening RAISES the lag-one, 15 °C from 0.741 to 0.878, which is what a whitener that is not working looks like. Running the identical operation on synthetic events whose deviations are parameter variation by construction – each event simulated at its own θ_j drawn at the measured per-trial scatter, with only 0.002 of white noise on top – settles it. The parameter share of those synthetic deviations is 52, 62 and 57 %, so the mechanism dominates as designed, and the lag-one still rises under whitening on all three: $0.362 \to 0.459$, $0.641 \to 0.692$, $0.666 \to 0.834$.

So the whitener does not remove this correlation even when the correlation is entirely parameter variation, and its survival on the measured records carries no information about the source. The conclusion Section 5.2 draws from it does not hold on eight datasets; it does not hold on one either, and that section's use of the same test needs the same treatment. What survives from Section 5.2 is the part that never used the whitener: the trial spread is low rank, its autocorrelation decays faster than any AR(1), and the dominant mode does not lie in the span of the sensitivities. Those are measurements. The whitening step was the inference built on top of them, and it is the step that fails.

One number in this document should be read differently from here on. The lack-of-fit ratio depends on the assumed equilibrium radius by a factor of about two over that radius's own uncertainty, on gelatin exactly as much as on PAAm, and the multistart behind it lands in neighbouring basins between adjacent stretches. No F quoted anywhere above is pinned better than a factor of two. The rejection is not close enough to that boundary for it to matter, which is the only reason the conclusions stand as written.

Part II
Designing the experiment that would settle it

Everything above says what a dataset of this kind cannot settle. What follows is a criterion that says which experiment would, and a proof that no better one exists among the settings offered.

9 Experimental design

Everything to this point measures what these datasets cannot settle. This chapter asks what experiment would, and the order is deliberate: first the one assumption the whole chapter rests on, then the two ways of scoring a single design and why both are unreliable, then the reformulation over measures that makes the problem convex and the answer certifiable. The sections after it are extensions that keep that certificate.

One assumption carries this whole chapter

Every design below is scored at geometries nobody has run, over $R_{\max}$ from 50 to 1200 µm. The datasets span 277, 298 and 312 µm, a twelve percent range. So the chapter extrapolates by a factor of twenty-four in the one variable it optimizes, and three separate quantities have to be carried out there with it: the noise profile σ, the correlation length τ, and the discrepancy $\hat{\delta}$.

Each is carried by *phase* rather than by clock time, t/t_c, on the argument that all three are properties of the collapse and not of the apparatus. That argument is supported. It is not established, and the distinction matters enough to state once rather than three times in passing.

What supports it is that each transfer was tested for consistency across the datasets, and each passed: median-filtered σ profiles bring all three datasets to the same recommended geometry (Section 14), and the three $\hat{\delta}$ overlap at 0.44 to 0.81 against 0.05 for unrelated curves (Section 12). Transfer by absolute time fails the same test, and fails it for a statable reason, t_c spans a factor of twenty-four across the candidates, so a fixed clock time lands at a different point of every collapse.

What that evidence cannot do is test the extrapolation actually being made. The three datasets differ in *temperature*, not in size. Agreement among them says the transfer is not sensitive to the material changing; it says nothing about a bubble four times larger, because no such bubble has been recorded. Everything downstream (the certified measures, the batch tables, the run budgets) is conditional on this, and the first experiment run at a genuinely different $R_{\max}$ tests it directly and cheaply.

Designing without assuming the model

Everything above optimizes the Fisher information of one model at its fitted parameters, which is circular if that model is wrong. Two checks that do not privilege it.

The trace is information-rich; the parameterization is not. Draw 400 trajectories from the posterior and take the singular values of the centered ensemble. Eleven modes clear the trial noise floor of $0.018\,R_{\max}$. That is not a count of identifiable combinations: four parameters span at most a four-dimensional manifold, and the extra modes are its curvature. But the dataset distinguishes far more shapes than the fit extracts. The limit is qSLS, not the measurement.

Discrimination does not privilege a model. T-optimality [13] asks a different question: which design makes the candidates disagree most? The criterion is

$$T(d) \;=\; \min_{\theta_2} \left\| m_1(\hat{\theta}_1, d) - m_2(\theta_2, d) \right\| / \sigma, \tag{23}$$

and the minimization is the whole content of it. A rival that retunes its parameters to imitate has not been discriminated against. So the separation is measured against the rival's *best imitation*, never against its own fitted values.

Two errors here inflated T, both for one structural reason. T is a *minimum*, so anything stopping short of it reports discrimination that is not there. Evaluating each rival at its own fit gave 2.69 for qSLS–SLS at the present design; re-fitting with one Nelder–Mead pass gave 0.67.

Restarting and polishing changed other entries again. At 1200 μm the reported 4.92 fell to 2.38. A 0.1% change of design moved one loose evaluation from 7.25 to 1.36, the optimizer's signature, not the physics. The converged values are:

design	qSLS–SLS	qSLS–qKV	qSLS–NHKV
277 μm, stretch 7.09 (present)	0.67	4.53	4.37
277 μm, stretch 5.00	0.72	4.61	4.61
277 μm, stretch 3.00	0.74	1.72	1.72
199 μm, stretch 3.55 (E-optimal)	0.56	2.59	2.59
895 μm, stretch 4.93	**2.39**	7.13	**7.13**
910 μm, stretch 5.19	1.36	6.97	6.97
1200 μm, stretch 7.09	2.38	5.03	5.63
100 μm, stretch 20.0	1.26	**8.04**	4.17
57 μm, stretch 19.5	1.64	5.53	5.00

At the present design SLS imitates qSLS to within 0.67 noise units, inside the measurement scatter. An unrelated calculation corroborates it. A systematic offset of 0.668σ over 201 samples contributes $201 \times 0.668^2 = 89.7$ to χ^2, against a measured log-evidence gap of 101.6. Two computations sharing no machinery, agreeing to 12%.

These values are upper bounds, and the criterion is the reason. T is an inner *global* minimization, and that minimization is multimodal. Nelder–Mead and Gauss–Newton both converge, to different basins. At 895 μm Nelder–Mead returns 2.39 against Gauss–Newton's 1.52; at 1200 μm the ordering reverses.

Multistart Gauss–Newton finds a lower value in 12 of 27 cells. Analytic Jacobians make it 44× cheaper than the derivative-free version, 34 s against 25 minutes for the table. A stationarity check from the same Jacobians reports 11 cells still not at a minimum.

Every improvement to the optimizer lowers the scores, because a better-optimized rival imitates better. So the tabulated values are upper bounds on discrimination. The true separations are smaller than any of them.

This is not a defect of the implementation but of the criterion. A minimum is attained at a point, so landing in the wrong basin gives a wrong answer with no indication that anything went wrong, and enumerating the outer design space exactly does not repair it: a dense grid over an unreliable integrand produces a front that is precisely wrong. Any criterion built on an argmin, T-optimality, profile likelihood, minimax design, inherits this.

The structural fix is to replace the minimum with an integral. Treating the model label as a discrete parameter makes discrimination an expected-information-gain problem over the augmented parameter (θ, m), whose inner quantity is the marginal likelihood $\int p(Y \mid \theta, M, d)\, \pi(\theta \mid M)\, \mathrm{d}\theta$. An integral accumulates over all modes instead of requiring the best one be found. Multimodality stops being a failure mode and becomes a property of the integrand: a missed mode biases the evidence slightly rather than corrupting the criterion.

It also puts discrimination and parameter precision in one framework. The cost is that the marginal likelihood is prior-sensitive where T is not. The prior becomes part of the criterion rather than a nuisance.

The criteria do not agree. Computed correctly, the criteria want different experiments. E-optimality peaks at 199 μm and stretch 3.55. Separating qSLS from SLS or from NHKV is best at 895 μm and stretch 4.93. Separating qSLS from qKV is best at 100 μm and stretch 20, near the opposite corner of the design space. At the E-optimal design the discrimination scores are the worst in the table, so the conflict is not marginal.

The reason is physical, not numerical. Telling a material with a relaxation mode from one without needs a collapse fast enough to excite it. Separating g from α needs a gentle collapse, because the term of Z_e carrying that distinction dies as λ^{-4}. No design is both fast and gentle.

There is accordingly no single best design, and a scalar design objective is answering a question that has no scalar answer. What replaces it is the *Pareto front*: the set of designs on which no other design is better on every criterion at once. Enumerating a 14×18 grid over $(R_{\max}, \lambda)$ and applying that test leaves 12 designs undominated, spanning 50–1200 μm and stretch 4–18. Reporting that set rather than a single optimum is the honest summary of an experiment serving several questions at once; `pyimr.pareto.explore_tradeoff` computes it.

The T columns entering the domination test are upper bounds, so the front is provisional in its discrimination coordinates. The E-optimality coordinate is an eigenvalue of $J^{\mathsf{T}}J$ with no inner optimization, and carries no such reservation.

Designing over measures, and certifying the answer

Every design result above shares a defect that no amount of care in computing them repairs: each is the output of a search over a single design point, that search is not convex, and nothing in its output distinguishes a good answer from a bad one. The evidence that this is not hypothetical is in the preceding subsections. A surrogate search recovered 16% of the best conditioning actually available and reported the result without qualification; a dense grid over a criterion whose inner minimization was unreliable produced a front that was precise and wrong. Both look exactly like success.

The classical formulation removes the defect rather than mitigating it.

Designs as measures. Instead of choosing one design x from a candidate set $\mathcal{X}$, choose a probability measure ξ on $\mathcal{X}$: run a fraction ξ_i of the bubbles at setting x_i. This is not an abstraction imposed for mathematical convenience: it is what an experimenter with a run budget does anyway, and it is strictly more expressive, since a single design is the special case of a measure concentrated on one point. The information available from a run allocated by ξ is the average

$$M(\xi) \;=\; \sum_i \xi_i\, M(x_i), \qquad M(x) = J(x)^{\mathsf{T}} J(x), \tag{24}$$

because information from independent runs adds and ξ says how many of each to take.

The consequence is the whole point: $M(\xi)$ is *linear* in ξ.

Concavity, and why the first-order condition is enough. $\log\det$ is concave on the positive-definite cone, and the composition of a concave function with a linear map is concave, so $\Phi(\xi) = \log\det M(\xi)$ is concave on the simplex, a convex set. A concave function on a convex set has no local maximum that is not global, and its first-order condition is not merely necessary but sufficient. Every pathology this study ran into (wrong basins, surrogate blind spots, answers that improve whenever the optimizer does) is a symptom of non-convexity, and none of them can occur here.

The same holds for $\Phi_A = -\operatorname{tr} M(\xi)^{-1}$ and $\Phi_E = \lambda_{\min}(M(\xi))$, and for any non-negative combination of concave criteria.

The directional derivative. Move mass α from ξ toward the single design x, giving $\xi_\alpha = (1-\alpha)\xi + \alpha\,\delta_x$. By linearity $M(\xi_\alpha) = (1-\alpha)M(\xi) + \alpha M(x)$, and using $\partial \log\det A = \operatorname{tr}(A^{-1}\partial A)$,

$$\phi(x,\xi) \;\equiv\; \frac{\partial}{\partial\alpha}\log\det M(\xi_\alpha)\bigg|_{\alpha=0} = \operatorname{tr}\big[M(\xi)^{-1}M(x)\big] - p. \tag{25}$$

For a linear utility u averaging over the design, the same construction gives $\phi_u(x,\xi) = u_x - u^\mathsf{T}\xi$.

The General Equivalence Theorem. Since Φ is concave and the simplex is the convex hull of its vertices δ_x, the measure $\xi^\star$ maximizes Φ if and only if no vertex direction ascends:

$$\boxed{\xi^\star \text{ is D-optimal} \iff d(x,\xi^\star) \equiv \operatorname{tr}\big[M(\xi^\star)^{-1}M(x)\big] \le p \quad \text{for all } x \in \mathcal{X},} \tag{26}$$

with equality at every support point of $\xi^\star$. This is the theorem of Kiefer and Wolfowitz [2], and it is what this section exists to obtain: a quantity computed from a candidate answer that *proves* the answer optimal. The gap $\max_x d(x,\xi) - p$ is a certificate, not a convergence report.

Two further facts come free. Equality puts the support exactly where $d(x,\xi^\star)$ is largest, so the optimal design runs at the settings of greatest standardized prediction variance. D-optimality and G-optimality coincide. And by Carathéodory's theorem applied to the convex hull of the $M(x)$, an optimal measure supported on at most $p(p+1)/2+1$ points always exists: at most 11 here.

Computing it. The implementation uses the multiplicative algorithm of Silvey, Titterington and Torsney [14],

$$\xi_i \;\longleftarrow\; \frac{\xi_i\,F_i}{\sum_j \xi_j F_j}, \qquad F_i = \operatorname{tr}\big[M(\xi)^{-1}M(x_i)\big] \;\text{ for }\; \Phi = \log\det, \tag{27}$$

which stays on the simplex by construction: weights remain non-negative and normalized without projection, and the criterion rises at each step. Note that $\sum_i \xi_i F_i = p$ identically, so (27) moves mass toward designs whose $d(x,\xi)$ exceeds p, which is precisely the set that (26) says must be empty at the optimum. The update and the certificate are the same quantity.

Vertex-direction steps with a line search were tried first and cannot converge here. The step required near the optimum falls below any fixed search grid, and the iteration stalls with a positive gap.

Convergence of (27) degrades as candidates crowd together. A design just off the support is nearly as good as one on it, and its weight decays in proportion. Certifying a parabola on $[-1,1]$ took 6400 iterations over 51 candidates, 25 000 over 101, and 329 000 over 401, with the criterion value correct to six figures long before any of those.

An exhausted budget therefore yields a converged value without a proof. That is a different thing from a wrong answer, and the reported gap distinguishes them.

The certified design for these datasets. Over 480 candidates spanning $R_{\max} \in [50, 1200]\,\mu$m and stretch $[3, 20]$, with $M(x) = J^\mathsf{T}J$ for the four qSLS parameters at the posterior median, the algorithm certifies at a gap of 9×10^{-10} in 1406 iterations: $\max_x \operatorname{tr}[M^{-1}M(x)] = 4.00000000$ against $p=4$. The optimal measure is

$R_{\max}$	stretch	weight
$1015\,\mu$m	6.70	0.470
$161\,\mu$m	19.26	0.303
$50\,\mu$m	17.78	0.227

against a best single design of $1015\,\mu$m at stretch 6.70. The measure exceeds it by 6.99 in $\log\det$, a D-efficiency ratio of $\exp(6.99/4) = 5.74$; equivalently the confidence ellipsoid is $\exp(6.99/2) \approx 33$ times smaller in volume, and a typical parameter's interval is $\exp(6.99/8) = 2.4$ times tighter. No single design can match it, and this is not a claim about the search: (26) certifies that none exists.

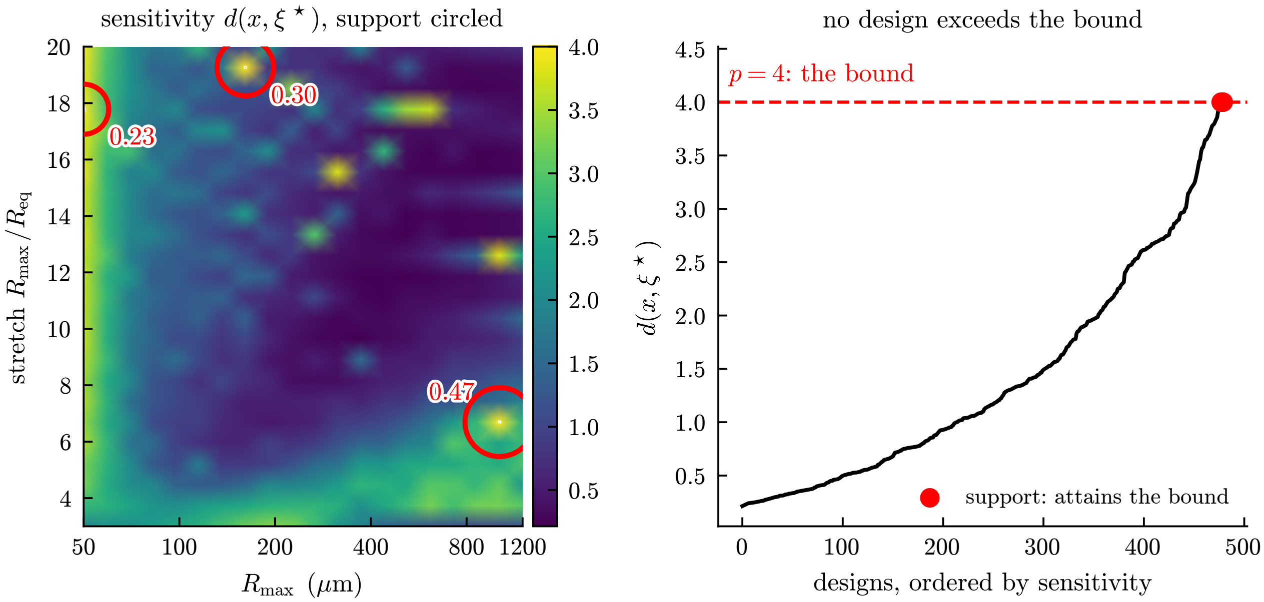

Figure 12: The equivalence-theorem certificate. Left: the sensitivity $d(x,\xi^\star) = \mathrm{tr}[M(\xi^\star)^{-1}M(x)]$ over the design plane, with the three support points circled and labeled by weight; the dashed contour is the level $p=4$. Right: the same values sorted. No design exceeds the bound and the support attains it exactly, which is the proof, $\max_x d = 4.00000000$ against $p=4$, gap 9×10^{-10}. Nothing analogous exists for the single-design searches elsewhere in this document.

The mixture is the resolution of the conflict. The support is worth reading physically. Nearly half the runs go to a large, gently driven bubble and the rest to small, hard-driven ones, the two corners that pull in opposite directions. Separating g from α wants a gentle collapse, because the distinguishing term of Z_e decays as λ^{-4}; exciting the relaxation mode wants a fast one.

A single design must choose between them. A measure need not. So the trade-off that motivated the Pareto front dissolves once the design space is the one the experiment actually admits, the front was answering a question made hard by an artificial restriction.

Discrimination in the same frame. Log evidence is additive over independent experiments, so an expected log Bayes factor per run is linear in ξ and therefore concave. The compound criterion

$$\Phi_\beta(\xi) = (1-\beta)\log\det M(\xi) + \beta\, u^\mathsf{T}\xi, \qquad \beta\in[0,1], \tag{28}$$

is then concave for every β and inherits the certificate unchanged. Its directional derivative is the matching combination of (25) and $u_x - u^\mathsf{T}\xi$. This is the compound-criterion construction of Atkinson and of Cook and Wong [15, 16], the principled form of the trade this study otherwise handled by search.

Sweeping β traces the Pareto front, and each criterion being concave in ξ makes that front convex. This is why the classical literature scalarizes with weighted sums where this study needed a Chebyshev form. The non-convexity that forced the elaborate scalarization was an artifact of optimizing over a single design point.

Temperature is the only axis on the table that rotates rather than amplifies. Section 3.7 proves that a design change which merely amplifies signal sends $F\to sF$, leaving κ untouched, so degeneracy can only be removed by a design that rotates the eigenvectors. Every candidate in this chapter is $(R_{\max}, \text{stretch})$, and both are drive settings. Temperature is not: it moves the material's own $g\alpha$, it costs nothing but a bath, and it has never been a design variable here.

The obstruction is that each temperature has its own parameters, so three datasets at three temperatures are three separate fits with nothing to design *for*. Temperature becomes an axis only under a model that shares parameters across it, and the weakest such model that is still physics is Arrhenius, $\log\theta_a(T) = c_a + b_a(1/T - 1/T_{\text{ref}})$: two coefficients per axis, six in all, with the chain rule turning a trace's sensitivity to $\log\theta_a$ into sensitivities to (c_a, b_a), the second scaled by $(1/T - 1/T_{\text{ref}})$.

Under that model the comparison is not close, and not subtle. *With temperature held, the six-parameter information is singular and the design is refused.* That is near-tautological once stated (a slope cannot be measured at one point, so the b_a columns are exact multiples of the c_a columns and the rank is three, not six) and it is exactly the abstract claim of Section 3.7 made concrete: no optimization over geometry, at any budget, identifies the temperature dependence. Only varying temperature does. Freeing it certifies at a gap of 8×10^{-10} on six settings spanning 12 °C to 36 °C, with $\log\det = 5.15$ and $\kappa = 2.8 \times 10^9$ where the held design has no finite κ at all.

This is the most conditional result in the chapter and is reported as such. Three temperatures fitting six coefficients leaves one residual degree of freedom per axis, so the Arrhenius form is assumed rather than tested by its own fit; its residuals, ± 0.005 in $\log \mu$, ± 0.05 in $\log g\alpha$ and ± 0.13 in $\log \lambda_1$, are a consistency note and nothing more.

Tested another way, the assumption fails. An Arrhenius axis is the weakest form of time–temperature superposition: warming slides the relaxation spectrum along the time axis without changing its shape. That has a sharp consequence the fit does not impose. Every characteristic time must shift by one factor a_T, and every modulus by $b_T = \rho T/\rho_{\mathrm{ref}} T_{\mathrm{ref}}$, which is within 6 % of unity here. The identified coordinates carry *two* independent times, λ_1 and the viscosity over the modulus $\mu/g\alpha$, built from coordinates λ_1 never touches, so their agreement is a free test.

They disagree. Carrying the per-trial covariance through, a_T from λ_1 is 0.657 and 0.875 at 23 and 33 °C while a_T from $\mu/g\alpha$ is 2.04 and 3.55: a gap of up to 6.1 standard errors, and the two move in *opposite* directions with temperature. The vertical shift fails on its own, b_T measured at 0.583 and 0.427 where entropy elasticity requires 1.00 to 1.06 — the modulus does the opposite of what a fixed network must do, by a factor of fifteen.

That is what a thermoreversible gel looks like. Warming gelatin does not slide its spectrum, it melts helices and changes the network (Section 22.1), so no shift factor exists to be fitted. The temperature axis of this section therefore rests on a superposition its own data refuse, which is a stronger objection than the one the paragraph above records, and it points at what should replace it: a design over a *structural* coordinate, the fraction of network still bonded, rather than over a rate shift.

The setpoint is not the geometry

Every design in this chapter names a target and treats it as a knob: 416 µm at stretch 10.9, 1015 µm at 6.7, a batch of eight settings. The acquisition data says it is not a knob. Per event, $R_{\max}$ has a coefficient of variation of 0.25, 0.28 and 0.23, and the stretch a further 0.05 to 0.07. A quarter is not a tolerance: it is the difference between 400 and 500 µm, and it means a certified optimum sitting on a ridge is worth only what can actually be hit.

The repair is a change of variable and costs nothing. What the experimenter chooses is a setpoint; what arrives is a draw. One run at setpoint s therefore supplies

$$M(s) \;=\; \mathbb{E}_{x \sim p(\cdot|s)}\big[J(x)^{\mathsf{T}} J(x)\big], \tag{29}$$

and every property this chapter relies on survives it: an expectation of information matrices is a matrix, $M(\xi) = \sum_i \xi_i M(s_i)$ is still linear in ξ, so the criterion is still concave and (26) still certifies. The certificate is identical. Only the answer moves.

scatter	nominal peak, delivered	scatter-aware optimum, delivered	regret
$\times\frac{1}{2}$	514 µm at 6.92, 8.982	636 µm at 6.92, 9.646	0.664
measured	514 µm at 6.92, 9.034	786 µm **at** 6.92, 10.254	**1.220**
$\times 2$	514 µm at 6.92, 7.473	786 µm at 5.62, 9.031	1.558

Both columns are scored against what actually happens, the expectation, so the difference is what ignoring the scatter costs rather than an accounting artifact. At the measured scatter the recommended bubble is *half again larger*, 786 against 514 µm, and the regret is 1.220 nats. For a whole certified batch the same comparison costs 1.046: a measure built on the nominal information delivers 9.877 where one built on (29) delivers 10.923.

Those widths were assumed, and they have since been measured. The coefficients above come from a three-row table, and (29) was evaluated over independent log-normal draws, a width, a shape and an independence assumption, none of them checked. All three are checkable against 437 per-event $(R_{\max}, R_{\mathrm{eq}})$ pairs recorded in micrometres, before the normalisation that removes absolute size from every trace fitted here. The measured coefficients are 0.223 and 0.067 against the assumed 0.25 and 0.06, and the correlation between $\log R_{\max}$ and $\log s$ is -0.14: the widths hold and the independence is very nearly right. Resampling the measured pairs in place of the assumed law, on a common grid:

law over the deviations	scatter-aware optimum	regret	support shared
assumed log-normal	678 µm at 6.70	0.434	3 of 5
log-normal at the measured widths	678 µm at 6.70	0.436	3 of 5
measured pairs, shuffled	866 µm at 6.70	0.334	4 of 5
measured pairs, joint	866 µm **at** 6.70	**0.267**	5 of 5

The conclusion of this section survives and its arithmetic does not. Under every law the regret is real and positive and the recommendation moves, so the *claim* never depended on the assumed distribution; but the assumed law overstates the regret by 63% and places two of five certified settings where the measured law does not. Delivered information is nearly identical either way (10.240 against 10.208 nats), so what the wrong law costs is misplacement rather than loss, and since the settings are what a collaborator runs, that is the part worth getting right.

One caveat belongs with this and not in a footnote: these 437 events are polyacrylamide, at a mean $R_{\max}$ of 359 µm, while the design they are applied to is gelatin at 277 to 312 µm. That a scatter *law* transfers across materials is the standing assumption of Section 9 and is not something these events establish. What they do establish is that the widths and the independence assumed above were sound.

Read that against the chapter's other effects. Choosing the geometry under the wrong noise model costs at most 1.34 nats (Section 14); restoring testability costs 0.01 to 0.24; the spread across robust batches is 0.21. *Ignoring that the setpoint is not the geometry costs about as much as the entire noise-model correction*, and unlike that correction it requires no new measurement and no new theory, only averaging the information over a distribution that has already been measured.

The direction is worth reading physically. Scatter pushes the recommendation *up* in $R_{\max}$, because the information ridge is asymmetric: undershooting a small bubble loses more than overshooting a large one, so a setpoint with room beneath it is worth more than a sharper peak without. The batch spreads accordingly, putting 0.467 of its runs at 636 and 0.150 at 786 µm, with the small hard-driven corner kept at reduced weight.

The scatter itself is measured rather than assumed, and swept at half and twice, under the standing caveat of Section 9: whether a laser produces the same spread at a setpoint nobody has run is exactly the kind of transfer this chapter cannot verify from three datasets at one size.

So should the scatter be paid to remove? No. The obvious follow-on is an investment question: a steadier laser or a better-controlled nucleation site tightens the scatter, so how many nats does that buy? Priced on the same grid, sweeping the coefficient of variation with the other coordinate held at its measured value:

$R_{\max}$ cv	best single setpoint	delivered	under an optimal measure
0 (perfect)	531 µm	9.100	**10.434**
0.10	576 µm	9.124	10.365
0.25 (measured)	736 µm	9.623	10.242
0.50	678 µm	9.760	10.049

The two columns disagree in *sign*, and which one applies is the whole point. Run as a single setting, scatter *helps*, perfect control is worth -0.523 nats, because the laser's imprecision accidentally spreads a design that would otherwise be a point, and Section 9 is an argument that spread designs beat points. Run as a measure, scatter hurts: perfect control is worth $+0.191$.

The sign of the second column is not a measurement but a theorem. The reachable $\mathbb{E}[M]$ under any scatter is a convex combination of $\{M(x)\}$, hence inside the hull that (26) already optimises over, so a measure can emulate any scatter and cannot do worse. Only the *magnitude* is informative, and the magnitude is what settles the investment.

0.191 nats is small, second from last in the price list of Section 14, inside the range restoring testability spans, and a twentieth of the bubbles-against-frames split. Perfect control of the stretch is worth less still, +0.100, and is last.

Account for the scatter; do not pay to remove it. Designing as though the setpoint were exact costs about a nat, and fixing that is free. Making the bubbles actually alike costs an apparatus and returns a fifth of a nat, because a certified measure wants a spread anyway and is largely indifferent to who chooses it.

Robust to the bias bound, which is the prior this chapter lacked

The limitation below, information evaluated at a point estimate of θ, offers its own repair, averaging over a prior, and this chapter never took it because there was no prior to average over. Equation (21) supplies something better than a prior: a computed set containing the truth unless the unseen half of the discrepancy exceeds the seen half. On the 15 °C dataset it is $\times 1.58$ in μ, $\times 1.30$ in $g\alpha$ and $\times 3.11$ in λ_1.

The bound reaches the design. Across the nine corners of that box the best single geometry is one of *five* distinct answers, running from 50 to 307 µm, so the recommendation is not stable under the parameter uncertainty this document itself computes, and reporting it at the fit alone overstates what is known.

Both robustifications keep the certificate, for the reason Section 9 already used: averaging concave criteria is concave, and a pointwise minimum of concave criteria is concave.

design built on	settings	at the fit	average over the box	worst corner
nominal, at the fit	4	**10.951**	10.708	8.795
Bayesian, $\max \mathbb{E}[\log\det]$	5	10.536	**11.101**	9.044
maximin, $\max \min \log\det$	6	9.779	10.312	**9.464**

Each design is best in its own column, which is the check that the three criteria are being optimised rather than merely evaluated. Designing at the point estimate costs 0.393 nats on average and 0.669 in the worst case, the same order as restoring testability (0.24) or controlling $R_{\max}$ perfectly (0.19), and well below the noise model (1.34) or the allocation (3.9).

One honest qualification on the last row. The Bayesian design certifies at a gap of 10^{-9}; the maximin does *not* converge, stopping at a gap of 4.4. A pointwise minimum is not smooth, and the subgradient iteration that Section 9 relies on is not the right tool for it. Its value is a lower bound reached rather than an optimum proved, and it is reported as such, the same distinction that section draws between an exhausted budget and a wrong answer.

What this does not fix. Three limitations, stated plainly. The information matrices are evaluated at a point estimate of θ, so the design is locally optimal in the parameters though globally optimal in ξ, which Section 9 now prices and partly repairs, at 0.39 to 0.67 nats.

The candidate set is still a grid, so the certificate proves optimality *over that grid*. That scan has now been run, and it fails.

The certificate does not reach past its own grid. On the 224-point grid the measure certifies exactly: $\max_x d(x, \xi^\star) = 4.000000$ against $p = 4$, at a gap of 8×10^{-10}. Evaluated on a grid eight times finer, the same measure gives

$$\max_x d(x, \xi^\star) = 69.56 \quad \text{against} \quad p = 4,$$

exceeding the bound by 65.6 at 552 µm and stretch 17.82. The measure is optimal among the settings it was offered and is *not* optimal over the continuum.

The exceedance is not numerical. It survives a hundredfold tightening of the solver tolerance, $\|M\| = 1.0086 \times 10^6$ at $\mathrm{rtol} = 10^{-7}$ and at 10^{-9}, identical to five digits, and the trajectory itself varies smoothly through it, the collapse deepening monotonically from 0.0379 to 0.0287 across the feature. What spikes is the sensitivity, not the solve.

It is, though, extremely narrow. Scanned in stretch at fixed radius, d sits at 0.4 to 0.9 almost everywhere and rises to 9.69 at 17.80 and 64.06 at 17.82, falling back to 0.52 by 18.00. These are ridges a coarse grid steps over, where a small parameter change flips the post-collapse trajectory between qualitatively different histories.

Averaging over the setpoint scatter helps and does not fix it. Since Section 9 establishes that a geometry is not a knob, the stretch arrives with a coefficient of variation of 0.06, wider than these ridges, the candidate a design can actually ask for is $\mathbb{E}[M]$, and much of the spike averages out. The exceedance falls from 65.6 to 8.4. It does not reach zero, so the ridges are not merely information nobody can request; some part of them survives being asked for imprecisely.

The consequence is a real limitation on this chapter's strongest claim, and it is the honest version of the caveat this list previously carried. *As computed, every certified batch here is optimal over its candidate set and none is proved optimal over the continuum.* The scan says where to refine, and refining there does move the answer.

Refined, the certificate very nearly reaches the continuum. The vertex-direction step is the classical one, add $\arg\max_x d(x, \xi)$ and re-certify, and the only care it needs here is that each round must be scanned against an *independent* set. Refining until $d \leq p$ on the set being refined proves nothing, since the optimiser guarantees that by construction. Scanning the midpoint lattice of the base grid instead, which no original candidate occupies, plus a zoom around the current support:

round	candidates	$\max_x d$, independent scan	excess over p	points above p
0	1840	6.897	2.90	12
1	2114	4.045	0.045	5
2	2181	4.030	0.030	1
3	2181	**4.017**	**0.017**	1

From 65.6 on the grid the measure was proved on, through 2.90 on the eightfold grid, to 0.017, a factor of nearly four thousand, monotone, with the violating set shrinking from twelve points to one. That last point is not a new ridge: it sits at 1023 µm and stretch 6.70 and moves by less than a micrometre per round toward a support point already at 1023.6 µm, which is what the residual of a discretised vertex-direction method looks like. The support grows from four settings to six, and the 937 µm ridge flagged above is now *in* it, carrying weight 0.033.

So the qualifier stands as a statement about what was computed and not as a claim about the model: the certified batches of this chapter are optimal over their candidate sets, and the distance from there to the continuum is a refinement anyone can run, not a barrier.

The compound front, which was previously not run. It was blocked on an estimator rather than on theory: Φ_β is concave for every β, but the per-design log Bayes factor suffered the boundary pinning of Section A. The linearised separation of Section 15, the model difference orthogonal to the material span, has no inner optimisation and no boundary to pin against, so it can play the utility.

β	settings	gap	$\log\det M$	separation	of each maximum
0	4	7×10^{-10}	**31.92**	179.4	100%, 19.9%
0.10	3	6×10^{-10}	24.45	887.9	76.6%, 98.3%
0.50	3	1×10^{-12}	20.74	902.1	65.0%, 99.9%
1	1	0	18.82	**903.0**	58.9%, 100%

Certified along the whole front. The trade is real rather than dissolved, no β attains both maxima, but it is steeply asymmetric, and that is the practical finding: *a tenth of a weight on discrimination buys* 98.3% *of it for a quarter of the estimation.* A collaborator who wants both should take a small β, not a balanced one, and the support moves from four settings to three rather than collapsing to the single design $\beta = 1$ would give.

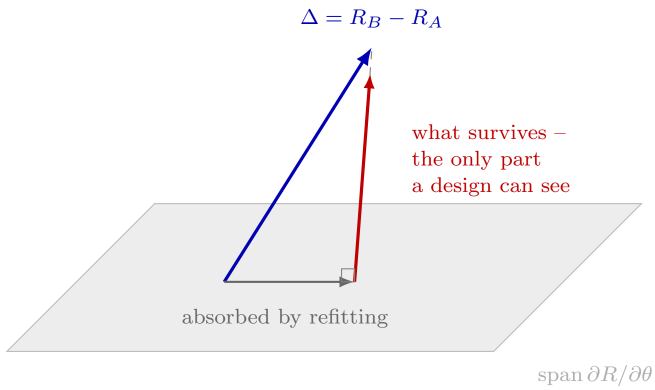

Figure 13: Why a model difference can be invisible at every design. Changing a model perturbs the trace by $\Delta = R_B - R_A$. The component lying in the span of the material sensitivities is reproduced by adjusting $g, \mu, \lambda_1, \alpha$, refitting absorbs it, and no experiment recovers it. Only the orthogonal remainder is evidence that the model changed. Measured on the 15 °C dataset and reported in Section 6: the bubble-dynamics operator is 77.7% absorbed, leaving under two noise units, which is why its ranking proved so fragile; the constitutive and thermal axes leave 8.1 and 8.5.

10 One criterion for every question

Section 15 gives a recipe and Section 3.7 shows the criteria in it disagree. Both leave a question unanswered: the material is scored by $\log\det M$, the model label by a Schur complement, the initial radius by a separate study, and none of the three share a scale, so *which* question a batch should serve cannot be asked. They are one quantity.

The value of an experiment

What an experiment is worth is how much it moves your beliefs, and about *what* is an argument rather than a fixed choice. Write A for the quantity of interest (parameters, a model label, the initial radius, the noise scale, any subset) and the value of a design is the mutual information between the data and A,

$$U_A(\xi) \;=\; \mathbb{E}_y\big[\mathrm{KL}\big(p(A \mid y, \xi) \,\|\, p(A)\big)\big] \;=\; I(A; Y \mid \xi). \tag{30}$$

For a Gaussian likelihood in prior-standardized coordinates (the unit-cube scaling Section 3.4 already uses, in which the prior covariance is the identity) the posterior precision is $N = I + M(\xi)$ and (30) is available in closed form. Partitioning into the coordinates A and the nuisance coordinates B,

$$U_A(\xi) \;=\; \tfrac{1}{2} \log\det \Big[N_{AA} - N_{AB} N_{BB}^{-1} N_{BA} \Big], \tag{31}$$

the Schur complement of everything you are not asking about. Every criterion in this document is (31) at a different A: all coordinates gives Bayesian D-optimality, the material block gives Section 3.7's $\log\det$, the mixing coordinates of Section 15 give D_s-optimality, and `req_scale` alone asks whether the initial radius is driving the conclusion. Answers in nats are comparable across those, and with weights π_A they add, $\Phi(\xi) = \sum_A \pi_A U_A(\xi)$.

The choice that was hiding in $\log\det M$

That the criteria are one family also exposes a decision nobody made. For any partition,

$$\log\det M \;=\; \log\det M_{\theta\theta} \;+\; \log\det S, \qquad S = M_{\varepsilon\varepsilon} - M_{\varepsilon\theta} M_{\theta\theta}^{-1} M_{\theta\varepsilon}, \tag{32}$$

so maximizing $\log\det M$ over the augmented set, as Section 15 does, is maximizing material log-volume *plus* discrimination log-volume with equal weight. That is a defensible compromise and it is not neutral, which is what it had been taken to be.

Three properties, and one that removes a failure

(31) is concave in ξ for every A. The Schur complement has the variational form $u^\mathsf{T} S u = \min_v (v,u)^\mathsf{T} N(v,u)$, a pointwise minimum of functions linear in N, hence matrix-concave; log det is concave and matrix-monotone, so the composition is concave, and $M(\xi)$ is linear in ξ. The equivalence theorem of Section 9 therefore still applies. With $P = \left(\begin{smallmatrix} I_A \\ -N_{BB}^{-1} N_{BA} \end{smallmatrix}\right)$ the envelope theorem gives the directional derivative, and $\xi^\star$ is optimal exactly when

$$d(x,\xi^\star) = \tfrac{1}{2}\operatorname{tr}\big[S^{-1}P^\mathsf{T} M(x) P\big] \;\le\; \tfrac{1}{2}\Big(|A| - \operatorname{tr}\big[S^{-1}P^\mathsf{T} P\big]\Big) \quad \text{for all } x, \tag{33}$$

with equality on the support. The bound is computed rather than looked up: it is p for log det and depends on ξ here, but it is still a proof and not a convergence report.

The prior term I in N is what makes (31) finite when M is singular. Under $\log\det M$ a design that cannot determine every parameter scores $-\infty$ and cannot be ranked at all, which is how 22 of 48 candidate geometries became unscorable in Section A. Here such a design scores low, which is the truth about it.

One caution follows from (32) being nonlinear: S is not linear in M, so $\log\det S\big(\sum_i \xi_i M_i\big) \neq \sum_i \xi_i \log\det S(M_i)$. Scoring candidates one at a time and taking the best N is therefore wrong, a design that discriminates well alone but leaves θ loose has its advantage eaten when the material is refitted from the same batch, and only the Schur complement of the *summed* information sees that.

11 From nats to bubbles

Nats say how much a batch teaches. They do not say whether it decides anything, which is what a collaborator with a fixed budget is asking. That has a closed form, and it is exact in the Gaussian-linear limit rather than asymptotic.

Let s be the per-run information about the coordinate separating two models, after everything else is fitted out. Information adds over independent runs, so N runs give $S = Ns$. Writing d for the component of the whitened model difference orthogonal to the material span, so that $S = \|d\|^2$ as in Figure 13, data generated under one model give a profiled log Bayes factor

$$\Lambda \;=\; \tfrac{1}{2}\big(\|d + n_\perp\|^2 - \|n_\perp\|^2\big) \;=\; \tfrac{S}{2} + d\cdot n_\perp, \qquad \text{so} \qquad \Lambda \sim \mathcal{N}\big(\tfrac{S}{2},\, S\big), \tag{34}$$

since $d \cdot n_\perp$ has variance $\|d\|^2 = S$. The experiment has not been run and its Bayes factor is unknown, but its *distribution* is known before anyone books the laser. The chance of clearing δ nats in favour of the truth is $\Phi\big((S/2-\delta)/\sqrt{S}\big)$, and inverting for confidence $1-\alpha$ gives the budget

$$N \;\ge\; \frac{\Big(z_{1-\alpha} + \sqrt{z_{1-\alpha}^2 + 2\delta}\Big)^2}{s}. \tag{35}$$

At $\delta = 5$ nats and 95% confidence the numerator is 27.14. An s of zero (the difference wholly absorbed, the infinite variance of Section 6) sends (35) to infinity, which is the statement that no experiment of this shape ever settles the question.

The same unit serves the parameter question. After N runs the posterior standard deviation of coordinate k is $\big[(I+NM)^{-1}\big]_{kk}^{1/2}$ in prior-standardized coordinates, so "pin g to a tenth of its prior width" is also a number of bubbles. That shared denominator is what makes the two comparable; nats alone are not, because they measure different things about different objects.

12 Model error, and what no criterion reaches

Every A above is a *coordinate*. "Is every model in the catalogue wrong?" is not one, and Section 5.1 answers it *yes* on all three datasets. No column of the Jacobian reaches model inadequacy, and worse, (30) maximized freely walks away from the batch that could detect it: a criterion linear in ξ is optimized by a single setting repeated N times, which has no pure error and no lack-of-fit numerator.

Testability is therefore a *constraint* on the batch, not a term in the objective:

$$\max_\xi \sum_A \pi_A U_A(\xi) \quad \text{subject to} \quad |\operatorname{supp}\xi| \ge k > p, \qquad n_i \ge J \ge 2. \tag{36}$$

The replication half is cheap and does nothing here: the number of settings is a property of the *measure*, and apportionment only allocates runs within the support it is handed, so forcing three replicates on the operator design leaves the batch at two settings and df $= -2$ still.

The support half is a cardinality constraint and is not convex, so it cannot join the objective without forfeiting (33). `pyimr.measure.constrained_measure` buys most of the certificate back in two steps. Which settings to guarantee (the set $\mathcal{K}$, of size k) is decided by the certificate itself: those the free solution already funds, then the candidates of largest sensitivity, the ones (26) shows the free problem came closest to wanting. Naming them is not enough, since an optimizer merely restricted to a larger candidate set reproduces the free optimum by leaving the new settings at zero and reports no cost at all; they are guaranteed instead by the substitution

$$\xi = c\,\mathbf{1}_{\mathcal{K}} + (1-kc)\,\eta, \qquad c>0, \quad kc<1, \tag{37}$$

which maps $\{\xi : \xi_i \geq c \ \forall i \in \mathcal{K}\}$ onto the whole simplex in η. The search is then exactly the one (33) already certifies, the floor holds by construction rather than by projection, and every candidate, guaranteed or not, still competes for the remaining mass $1-kc$. What is certified is optimality *among measures meeting the floor*, which is weaker than (33) and is reported as such.

On the operator design the constraint is worth paying. The free batch takes two settings for 4.39 nats; held to five it spreads over six geometries and df $= +2$ for 0.10 nats, and held to seven it reaches df $= +3$ for 0.04, both certified below 10^{-9}. That seven costs less than five is not a paradox: the default floor is half an equal share, so asking for more settings also asks less of each, and the two demands are not nested. At a fixed c they are, and the price rises with k as it must.

Unless it is made one

"No criterion in Section 9 takes model error as an argument" is the sentence above, and it need not stand. Section 10 treats model inadequacy as unreachable because it is not a *coordinate*, no column of the Jacobian points at it, which is why testability enters as the cardinality constraint (36) rather than as anything optimized for. But Section 7 does not merely bound $\hat{\delta}$; it extracts it as a vector.

Write $m(\theta) + \eta\,\hat{\delta}$ and treat η exactly as Section 15 treats the model label: one more column of J. Its variance after the material is fitted out,

$$\mathrm{Var}(\eta) \;=\; \Big(\|d\|^2 - d^{\mathsf{T}} P d\Big)^{-1}, \qquad P = J(J^{\mathsf{T}}J)^{-1}J^{\mathsf{T}}, \tag{38}$$

is the part of the discrepancy a design can *see* rather than absorb. The two uses point the same way, which is what makes it worth having: testing the model wants $\hat{\delta}$ visible, and keeping the parameters clean wants it unabsorbed, and both are the same angle.

Carrying it to an unrun geometry needs the transfer of Section 14, under the standing caveat of Section 9, and the premise is checkable rather than assumed. Resampled onto a common t/t_c grid, the three datasets' $\hat{\delta}$ overlap at 0.44, 0.42 and 0.81 against 0.05 for unrelated curves of that length, the same test, and the same verdict, that justified transferring σ by phase.

Scoring (38) at the four geometries Section 14 put through two synthetic mismatches:

geometry	$R_{\max}$	stretch	visible share	$g\alpha$ err (thermal)	(two-mode)
performed	277 µm	7.09	**0.998**	3.8%	5.0%
E-optimal	199 µm	3.55	0.972	5.9%	4.9%
discrimination-optimal	100 µm	20.0	0.876	**37.1**%	**60.7**%
measure support	60 µm	18.0	**0.763**	5.2%	43.5%

A single measured $\hat{\delta}$ cannot know which mismatch is the real one, so the most it can predict is what the two have in common. It does: the rank correlation of visible share against the parameter error is -0.40 under thermal truth and -0.60 under two-mode truth, both of the predicted sign, from four points. That is weak evidence and it is the right amount to claim, what it establishes is that the criterion is not refuted, not that it is calibrated.

The criterion also disagrees with the one the chapter has been using, which is the reason to carry it: material log det is largest at 89 µm and stretch 20, where 83.7% of $\hat{\delta}$ is visible, while visibility is largest at 1200 µm and stretch 20 at 99.0%. The worst geometry on the grid hides two fifths of it.

13 What a certified batch delivers

Over the 227 candidate geometries of Section 15 at a budget of twelve, with the material held at the 15 °C fit:

batch optimized for	material (nats)	operator (nats)	settings	lack-of-fit df
material	**19.51**	3.78	7	+3
operator	17.03	**4.39**	2	−2
portfolio, 1:4	18.92	4.24	6	+2

all three certified at gaps below 10^{-9}. The geometry does move with the question, the operator question wants 785 µm at stretch 6.9, which the material-optimal batch never visits, but the portfolio buys 97% of each for 3%, and is the only aggressive option that stays testable *at this noise model*.

That qualification is not decoration. Every number in the table takes $\sigma = 0.018\, R_{\max}$ at every sample, and Section 14 shows the datasets contradict it by two to three orders of magnitude within a single trace. Recomputed under the measured profiles the portfolio's six settings become four and its df becomes zero on two of the three, so the last clause of the paragraph above holds only under the flat scale that motivated it. The repair costs two tenths of a percent and is given there, along with the batch this document actually recommends.

A note on the search. The multiplicative algorithm of Section 9 relies on $\mathrm{tr}[GM(\xi)]$ being the constant p, which holds for $\log\det$ and for nothing else; on the operator design it stalls at a gap of 0.155. It reports itself uncertified, which is the certificate working, but mirror descent on the simplex is what converges here, to 1.8×10^{-10} and a higher value.

What all of this is worth against correlated noise

Every number above is information, and Section 5.2 shows this likelihood counts more of it than it has. The correction belongs here rather than as a caveat, because it enters (35) directly: s falls, and the required budget rises by the same factor.

Replacing independent samples with the exponential covariance $\Sigma_{ij} = \sigma_i \sigma_j e^{-|t_i - t_j|/\tau}$ makes the whitened information $J^{\mathsf{T}}\Sigma^{-1}J$, and everything above inherits it unchanged, the criterion, the certificate and (35) are all statements about M, whatever M was computed from. The effect is not a single deflation. Measured at $\rho = 0.908$: a constant offset loses a factor of 0.053, a smooth half-cycle 0.049, a five-cycle oscillation 0.176, while point-to-point jitter is *amplified* 20.7-fold. Model differences are smooth, so the deflation is what applies to every margin in this document, but it is a fact about the shape of those differences rather than a property of the likelihood.

Applied rather than quoted, it changes neither the design nor, here, its sign. The paragraph above is a statement about idealized shapes. Recomputing every geometry with $M = J^{\mathsf{T}}\Sigma^{-1}J$ at $\tau/t_c = 0.292$ (the correlation length transferred by phase, for the reason Section 14 gives for σ and under Section 9's caveat) settles what it does to the answer, and the answer is two negatives and one positive.

The recommended geometry does not move, under any of the three measured noise profiles: the diagonal and correlated likelihoods pick the same design in all three cases. So the batches of this chapter are not misdirected by the likelihood they were computed under, which is the more important of the two questions and the one the deflation figures leave open.

Nor is the operator question deflated. The discrimination coordinate is *amplified*, by a median of ×3.23, ×2.85 and ×2.51 across the three profiles, over a range running from ×0.52 to ×10.4. There is no contradiction with the figures above: those are the deflations of a constant offset and a smooth half-cycle, and the operator differences on this grid are not smooth enough to sit there.

The mechanism, however, is exactly the one the figures imply, and that is the positive result. Ranked against the fraction of each difference's energy above eight cycles across the dataset, the amplification correlates at +0.66, +0.64 and +0.65. A correlated likelihood pays for a difference that is *fast* and charges for one that is smooth, so which feature of a design is worth buying does change even where the ranking does not.

14 What the design is worth

The certified measure of Section 9 is a theorem about a criterion. What it is worth depends on which variable that criterion is allowed to move, and the geometry everything above optimises turns out not to be the important one.

Every effect this chapter prices is in nats of the same posterior, so they can be ranked on one axis. Doing that once, before the two largest are derived, is the point of this section:

what it buys	nats	where
splitting a frame budget between bubbles and samples	**3.40–3.96**	Section 14
choosing the geometry under the measured noise model	1.34	Section 14
accounting for the scatter in the setpoint	1.220	Section 9
designing against the bias bound rather than a fit	0.39–0.67	Section 9
restoring a testable batch	0.01–0.24	(36)
perfect control of $R_{\max}$	0.191	Section 9
perfect control of the stretch	0.100	Section 9

The ordering is the recommendation. The top entry is the one variable no batch in this chapter was allowed to move, and it is worth more than everything below it combined; the bottom two are the ones that cost an apparatus. The rest of this section derives the top two, in that order.

Frames are a design variable, and there are far too many of them

Section 22.1 reports 201 samples carrying the information of about ten. Redundancy is a design variable: a camera trades frame rate against window length, against exposure, and against how many bubbles fit in a session. Scoring frames by $\log\det J^{\mathsf{T}}J$ at the fitted qSLS, with the same number of observations spent each way:

observations	uniform	optimally placed	gain
201	18.54	25.72	6.0×
50	11.73	20.15	8.2×
12	5.98	14.42	8.3×

The D-optimal measure over the 201 candidate times is certified at a gap of 10^{-10} on *six* support points. Fifty observations placed on those six carry more information than all 201 spread uniformly.

That is a measurement of slack, not a recommendation. A design on six distinct times leaves no degrees of freedom to notice the model is wrong, and the lack-of-fit test above is exactly what those degrees of freedom buy. The honest reading is that frames are cheap relative to their information, so the camera should be traded for something scarce, bubbles, or window, or the replication that makes (16) possible.

Bubbles against frames, priced

That trade can now be made quantitative, because Section 5.2 measures Σ. Every batch in this document is chosen over *geometries*, with the number of trials J and the samples per trial N fixed at whatever the datasets happened to use. Under independent noise that omission is harmless, information is proportional to JN and only the product matters. Under trial-to-trial variation it is not, and the asymmetry is exact. Averaging J trials of an N-sample trace and applying the Woodbury identity to (17),

$$F(J,N) \;=\; J\left(\Sigma + F_N^{-1}\right)^{-1}, \qquad F_N \;=\; G^{\mathsf{T}}\Sigma_\varepsilon(N)^{-1}G, \tag{39}$$

with F_N the information a single N-sample trace supplies. Trials enter linearly and never saturate. Samples enter only through F_N, and as $N\to\infty$ the bracket tends to Σ, so $F \to J\Sigma^{-1}$ and further frames buy *nothing*. Past a point, more samples measure one bubble ever more precisely rather than the population.

At a fixed frame budget $S = JN$ this bites: $F = (S/N)(\Sigma + F_N^{-1})^{-1}$, and once F_N^{-1} is small against Σ the prefactor S/N is all that survives, and it *falls* with N. Measured on the datasets, at each one's own total:

dataset	as run	best split at the same frames	gain
15 °C	$J = 18,\ N = 201$	$J = 72,\ N = 50$	+3.93 nats
23 °C	$J = 14,\ N = 192$	$J = 56,\ N = 48$	+3.96 nats
33 °C	$J = 7,\ N = 198$	$J = 28,\ N = 49$	+3.40 nats

Doubling the samples buys 0.035 to 0.188 nats. Four times the bubbles at a quarter the frames each, for the same total, buys three and a half to four.

That number should be read against every other design effect in this document. Choosing the geometry under the wrong noise model costs at most 1.34 nats (Section 14); restoring testability costs 0.01 to 0.24; the spread across robust and per-dataset batches is 0.21. **The allocation nobody has been optimizing is worth three to ten times any of them**, and it is the one variable the batch optimization of Section 15 has never been allowed to move.

Is the gain the clock rather than the material? It rests entirely on Σ_θ being non-zero, as $\Sigma_\theta \to 0$, (39) becomes JF_N, only the product JN matters, and there is nothing to allocate. And Section 5.2 has since shown a quarter of the trial variance to be a clock error rather than a material difference. Σ_θ was estimated by fitting each trial separately, so whatever of that the parameters could absorb, they did.

Rescaling every trial onto the dataset's median collapse time, a processing change costing nothing, and refitting gives a Σ_θ with the timing taken out:

dataset	J	μ raw	aligned	$g\alpha$ raw	aligned	λ_1 raw	aligned
15 °C	18	14.1%	14.9%	31.5%	**13.2**%	51.0%	93.8%
23 °C	14	29.2%	30.3%	18.4%	35.9%	46.3%	65.5%
33 °C	7	20.8%	20.4%	53.4%	52.9%	86.0%	79.3%

and the allocation recomputed on it moves by +0.589, +0.043 and −0.416 nats against gains of 5.78, 5.67 and 5.35. *The recommendation survives*: the contamination is at most a tenth of the effect on one dataset, nothing on another, and of the wrong sign on the third.

The test is weaker than it looks, and the Σ table says why. Alignment mostly *raises* the spread rather than lowering it, λ_1 at 15 °C goes from 51% to 94%, because rescaling a trace's clock and resampling it distorts the trace in ways the material parameters then have to absorb. Only $g\alpha$ at 15 °C behaves as the hypothesis predicts. So this does not support the contamination reading, and it is not clean enough to refute it either; what it does establish is that the allocation recommendation is not obviously an artifact of the clock, which is what it was asked.

Three caveats, and they do not cancel. The trials column of (39) gains exactly $p \log 2$ per doubling because F is linear in J, that is arithmetic, and the measurement is the sample column beside it. F_N is scaled linearly in N here, as though samples within a trace were independent; Section 22.1 measures N_{eff} near ten of 201, so the true F_N is flatter still and cutting N costs *less* than this says, conservative in direction, but not to be pushed past $N \approx N_{\text{eff}}$, where the collapse stops being resolved. And N is not a free dial: frame rate and window length set it, so a smaller N means a slower camera or a shorter trace.

It folds into the certified problem, once the candidates are priced. This section previously stopped here, on the grounds that folding the split into the geometry optimization needs Σ at a geometry nobody has run. It folds in exactly. Let a candidate be a *pair* (x, N), a geometry and a number of samples per trace, and let ξ allocate *trials* rather than runs. Then (39) gives

$$F \;=\; \sum_i J_i \big(\Sigma + F_{N_i}(x_i)^{-1}\big)^{-1} \;=\; J \sum_i \xi_i M_i, \tag{40}$$

still linear in ξ, so $\log\det$ is still concave and Equation (26) still applies. What breaks is the tacit assumption that candidates cost the same: an N-sample trace costs N frames. Normalizing by cost rather than by count, $\sum_i \xi_i c_i = 1$, ranges ξ over the set with vertices e_i/c_i, which is convex; the substitution $\eta_i = \xi_i c_i$ maps it onto the standard simplex with $M(\xi) = \sum_i \eta_i (M_i/c_i)$. So the search is the one `optimal_measure` already certifies, run on the matrices divided by their prices, and the certificate returns in its cost-normalized form,

$$\xi^\star \text{ is optimal} \iff \operatorname{tr}\big[M(\xi^\star)^{-1} M(x)\big] / c_x \;\le\; p \quad \text{for all } x, \tag{41}$$

with equality on the support: a candidate earns runs when its information *per unit price* clears the bar. `pyimr.measure.budgeted_measure` is this, and it composes with the cardinality floor of (37) because both are substitutions in the same η.

Measured under (40) and certified by (41), on the 15 °C dataset at its own 3618 frames, against the geometry-only formulation with N pinned at 201:

one bubble costs	joint (nats)	geometry only	gained	N funded
0 frames	27.70	17.72	**+9.99**	6
25 frames	22.78	17.37	+5.41	6
100 frames	19.18	16.51	+2.67	12
400 frames	15.33	14.43	+0.90	25

all certified below 10^{-9}. The price of one bubble against one frame is not knowable from here, so it is swept; the gain exceeds the 1.34 nats Section 14 attributes to choosing the geometry at every price below about 400 frames a bubble. Holding the batch to five settings, so (16) keeps a numerator, costs 0.008 to 0.034.

Σ is the assumption, and it is both weaker and less load-bearing than the paragraph above assumed, though it inherits the standing caveat of Section 9 like everything else here. Weaker, because Σ is a population covariance of *material* parameters (a property of the gel and of how repeatably the laser nucleates, not of how large the bubble is) and the three datasets measuring it differ in *temperature*, not geometry, sitting at 277, 298 and 312 µm. There is no evidence of geometry dependence because geometry was never varied. Less load-bearing, because swept over $\times\frac{1}{2}$ to $\times 2$ the recommended N moves once, from 12 to 8.

What does bind is validity. At the two cheapest bubble prices N pins at the bottom of the grid, and that is a limit of the model rather than an optimum: F_N is scaled linearly in N here while N_{eff} is near ten of 201, so below that the linear form claims resolution the dataset does not have.

Nor do they survive the noise model

Every Fisher matrix, expected gain and batch in this chapter is built on a single number: $\sigma = 0.018\,R_{\max}$ at every sample. Fitting has never used it, the likelihood of Section 3.2 takes the per-sample spread, so this is a gap between the two halves of this document rather than an error inside either. The datasets disagree with the constant by two to three orders of magnitude *within one trace*: 124×, 882× and 1316× between the smallest and largest σ_i, around medians of 0.018 to 0.023.

At the same Jacobian, the constant overstates $\log\det F$ by 1.3 to 2.2 nats and understates each parameter's standard deviation by up to a factor of two. What matters more is that it changes the answer. Transferring each dataset's measured profile to a candidate geometry by phase (reading σ as a function of t/t_c, which Section 5.2 justifies by showing the profile is dynamical rather than instrumental) the single best geometry moves from 514 µm at stretch 6.9 to 416 µm at 10.9, and the constant's choice gives up as much as 1.34 nats of 8.89, fifteen percent.

The transfer is the assumption here, so it was tested rather than asserted. Under a raw per-sample profile the datasets split, 15 °C keeping the constant's geometry. Median- filtering the spread before transferring (a standard deviation from seven to eighteen trials carries 17 to 29% relative error and should not be interpolated raw) makes all three datasets agree on 416 µm at 10.9, and none agrees with the constant. The disagreement that survives is with the flat scale, not among the datasets, which is the opposite of what a transfer artifact looks like. Transferring by absolute time instead picks a different geometry again and is wrong for a statable reason: t_c spans a factor of twenty-four across the candidates, so the same clock time lands at a different point of the collapse.

Testability is what it really costs. The portfolio of Section 10 spends twelve runs over six geometries and carries df = +2. Under two of the three measured profiles it spreads over *four* and df = 0: the lack-of-fit test has no power at all. So (36) is not a formality the portfolio satisfies unaided, and the floor has to be imposed. Holding the measure to five settings with `constrained_measure` restores df = +1 for 0.010 and 0.084 nats of 36.5, two tenths of one percent, and every constrained measure is certified under its floor.

One batch, chosen against the hardest case. Three profiles give three batches, and a recommendation has to be one. Averaging the gain across profiles is the move Section 10 already makes across questions, and it fails here: the weighted batch has the *worst* worst case of the five compared, 0.211 nats behind, below the flat-scale batch it was meant to improve on. Maximizing an average does not maximize a minimum. What works needs no machinery, one profile binds for every batch, so the worst case is always that column and the maximin answer is the argmax under it. When one member of a family is uniformly hardest, robust design is design against the hardest member.

The batches differ by 0.211 nats of 33.4, six tenths of a percent, so information barely separates them; settings run from five to eight and df from +1 to +4, so testability does. The batch recommended here wins on both:

$R_{\max}$ (µm)	stretch	runs
50	20.0	1
272	17.4	1
416	10.9	2
514	5.6	2
514	6.9	1
636	5.6	1
786	6.9	1
1200	6.9	3

eight settings, df = +4. Read it with Section 14: how the twelve runs are split between bubbles and frames is worth more than which geometries they visit.

15 Choosing the next N experiments

Everything above scores designs. None of it answers the question a collaborator actually asks: *we can run N more bubbles, what should they be?* This section is that recipe. It is five steps, four of which are already derived above and one of which is new.

Step 0: ask which questions are still open

A batch that separates models the datasets have already separated buys nothing. Given the log evidence $\log Z_m$ of each rival on the data in hand, write the margin against the best, $\Delta_m = \log Z_m - \max_k \log Z_k$, and call a rival *decided* when $\Delta_m < -\delta$ for a threshold δ we take as five nats. Survivors carry the posterior they have earned,

$$w_m = \frac{e^{\Delta_m}}{\sum_{k \text{ live}} e^{\Delta_k}}, \qquad w_m = 0 \text{ for decided } m. \tag{42}$$

This step does more than save runs. The criterion below is *local*, it linearizes the model difference, which is the right approximation for rivals that are close and the wrong one for rivals far apart. But rivals far apart are easy to separate and are exactly the ones (42) decides. What survives screening is the close pairs, which is the regime the linearization is valid in, so the two failure modes cancel rather than compound.

Step 1: make the model choice a parameter

For a live rival m against the incumbent, write $R(\theta, \varepsilon_m) = (1-\varepsilon_m)R_{\text{inc}}(\theta) + \varepsilon_m R_m(\theta)$, so that $\partial R/\partial \varepsilon_m = R_m - R_{\text{inc}}$ at $\varepsilon_m = 0$. The model question becomes one more column of the Jacobian. With the material sensitivities whitened by the measurement deviation, the augmented Jacobian at a candidate setting x is

$$\tilde{J}(x) = \Big[\ \underbrace{\sigma^{-1}\partial R/\partial\theta}_{p \text{ columns}}\ \Big|\ \underbrace{\sqrt{w_1}\,\sigma^{-1}\Delta_1\ \cdots\ \sqrt{w_q}\,\sigma^{-1}\Delta_q}_{q \text{ rivals}}\ \Big], \qquad M(x) = \tilde{J}(x)^{\mathsf{T}}\tilde{J}(x). \tag{43}$$

The $\sqrt{w_m}$ is what makes (42) bite: a rival at a tenth of the posterior contributes a tenth of the information about its own coordinate, and a decided rival contributes none.

Nothing is sampled here. This is the point at which an expected log Bayes factor would normally be estimated, and Section A explains why that estimator fails on datasets this long, the effective sample size collapses to one. The derivative is exact, costs one Jacobian, and needs no prior on the rival, which also removes the prior-placement trap of Section 9: only the difference direction enters, never the rival's own parameters.

Step 2: optimize over measures, not over settings

A batch is a measure. Let ξ put weight ξ_i on candidate x_i; the information is $M(\xi) = \sum_i \xi_i M(x_i)$, *linear* in ξ, so $\log\det M(\xi)$ is concave and

$$\xi^\star = \arg\max_{\xi} \; \log\det M(\xi) \tag{44}$$

has no local optima. Because M carries both blocks, (44) balances estimating the material against determining which model produced the trace without any weighting parameter to choose.

This is deliberately not the compound criterion of Section 9. That form blends log det with a utility, and the blend is concave only because the utility is *linear* in ξ. The natural discrimination utility, the posterior variance of ε, is not linear, so blending it would forfeit the certificate. Putting ε inside the matrix keeps one concave criterion and the proof that comes with it.

Step 3: certify

By Kiefer–Wolfowitz, $\xi^\star$ solves (44) exactly when no candidate setting offers an uphill direction:

$$d(x, \xi^\star) = \operatorname{tr}\big[M(\xi^\star)^{-1} M(x)\big] \le p + q \quad \text{for all } x, \tag{45}$$

with equality on the support. The gap $\max_x d(x, \xi^\star) - (p+q)$ is a proof, not a convergence report, and it is what distinguishes this from every single-design search in this document.

Step 4: spend the budget

$\xi^\star$ is a measure and the experiment runs integers. For a budget of N, apportion by efficient rounding [17]: give every support point one run and allocate the rest by the rule that maximizes $\min_i n_i/(N\xi_i^\star)$, so no support point is rounded away. The rule carries no guarantee at small N, which is the regime here, so the loss is *measured* rather than assumed:

$$\text{D-efficiency} = \left(\frac{\det M(n/N)}{\det M(\xi^\star)}\right)^{1/(p+q)}. \tag{46}$$

What the batch buys is then read off the inverse. The variance of the coordinate that says which model produced the trace, after the material has absorbed everything it can, is the Schur complement

$$\operatorname{Var}(\varepsilon_m) = \Big[\big(M_{\varepsilon\varepsilon} - M_{\varepsilon\theta} M_{\theta\theta}^{-1} M_{\theta\varepsilon}\big)^{-1}\Big]_{mm}, \tag{47}$$

and an infinite entry is the statement that no design separates that pair at all, the difference lies entirely in the span of $\partial R/\partial\theta$ and refitting reproduces it.

The answer for these datasets

Screening (42) on the operator axis leaves five live rivals against Gilmore/Mie–Grüneisen. Over 48 candidate geometries spanning 50 µm to 1200 µm and stretch 3 to 20, (44) certifies at a gap of 10^{-9} on five support points, and a budget of $N = 12$ apportions to

$R_{\max}$	stretch	runs
1200 µm	9.8	6
1200 µm	6.4	3
307 µm	16.6	1
50 µm	20.0	1
484 µm	6.4	1

at 99.6% of the measure's own efficiency. Against spending all twelve at the geometry already performed, (47) falls by 16.0, 17.7 and 9.3 times for Gilmore/Tait, Keller enthalpy/Tait and Keller enthalpy/Mie–Grüneisen respectively.

Two caveats travel with that table. Every design that integrates is scorable here, where the sampled criterion could rank only 26 of the same 48, but the criterion is still evaluated at one material and one incumbent, so it is locally optimal in exactly the sense Section 14 shows can mislead under model error. And the screening that produced the rival list rests on margins computed under an independent likelihood; Section 16 shows that deflating them by N/N_{eff} changes which rivals are live at all.

16 Which questions are open

Step 0 of Section 15 asks which rivals the datasets have already decided. Run on this package's three axes, the answer is stark and then immediately contingent. The margins it runs on are the ones Part III reports, and this section is where they are used rather than where they are derived.

At face value almost nothing is still in question. Of the constitutive candidates, one is live on every dataset; the rest trail by up to 4573 nats. Of the six operators scored, two survive on every dataset. Each dataset admits a single thermal treatment.

That reading would mean the design of Section 15 is aimed at a question already settled, and by a wide margin. It does not survive the likelihood.

Every log evidence in this document assumes the residuals are independent, and Section 22.1 measures N_{eff} near 10 of 201. Evidence differences scale with the count of independent observations, so each margin is inflated by roughly N/N_{eff}. Deflating by that factor and re-screening:

axis	live at face value	live deflated by N/N_{eff}
constitutive (17 candidates)	1	2
bubble dynamics (6 operators)	2–5	5–6
thermal treatment (3)	1	3

SLS returns, five or six operators return, and every thermal treatment returns.

Screening is where this matters most, and the reason is structural rather than numerical. Elsewhere the correlation inflates a quantity we report and compare, an evidence, a χ^2/N, and the ranking survives because every model is inflated alike. Section 22.1 says exactly that: the ranking survives, the absolute claim does not. But a screen converts a margin into a *yes or no*, and a twentyfold factor moves margins across a threshold wholesale. Here the inflation does not cancel, because the comparison is no longer between models but between a margin and a constant.

So which model questions are open is at present undetermined, and undetermined by the likelihood rather than by anything about the models.

But the design need not wait for it. The reading above makes the correlated likelihood a precondition for the design work, and it is not one. What is fragile is the *threshold*, not the deflation: elsewhere every model is inflated alike and the ranking survives, while (42) compares a margin to a constant, and a factor of twenty moves margins across a constant wholesale.

Section 14 met this shape of problem once already and its finding is the recipe, averaging over a family gave the worst worst case of the five batches compared, and maximizing against the hardest member won on every count. Here the family is the one-parameter deflation $N/N_{\text{eff}} \in [1, 25]$ and the hardest member is the largest, since it revives the most rivals. Scoring each batch by the share of the genuinely live questions it funds, under both readings of the truth:

axis	batch designed at face value	batch designed at full deflation
constitutive	0.978	**1.000**
bubble dynamics	0.595	**1.000**
thermal treatment	0.410	**1.000**

worst case over the two truths. The maximin batch has none: it funds every live question whichever reading is right. The face-value batch funds two fifths of the thermal questions if the deflation is real.

The asymmetry is why, and it is not an artifact of the scoring. A batch designed at full deflation and met by a face-value truth spends 0.595 of its effort on rivals that were already decided, wasted, and recoverable by simply not believing the answer about them. A batch designed at face value and met by a deflated truth never asks about the rivals it wrote off, and no amount of later analysis recovers a question that was never funded. Wasted effort and an unasked question are not symmetric costs, and the maximin choice is the one that trades the first for the second.

Part III

The comparison, worked

The ranking is what supplied Part II with its identified coordinates, its noise profile, its Σ and its $\hat{\delta}$, and it is reported for that purpose. It is also the case the design is built against: the margins below are large under the likelihood that produced them and small under the one Section 16 argues for, which is the circumstance a certified design exists to repair.

17 Which model the data prefer

Table 4 and Figure 14 give the outcome. The quadratic Zener model `qSLS`, strain-stiffening elasticity with stress relaxation, wins at every temperature. The ordering beneath it separates into three tiers: models with both ingredients, models with relaxation only, and memoryless models.

Table 4: Best χ^2/N *on the selection grid* by model class, the best node, not the best fit; Section 5.1 refits and reaches 0.962 for qSLS at 15 °C. Values near 1 indicate a fit within experimental repeatability.

	15 °C	23 °C	33 °C
`qSLS` (stiffening + relaxation)	**1.35**	**1.33**	**1.23**
`SLS` (relaxation, linear elasticity)	2.94	1.85	1.41
viscoelastic fluids	5.3–5.7	4.5–4.7	1.9–2.1
memoryless	12.6–16.7	,	,

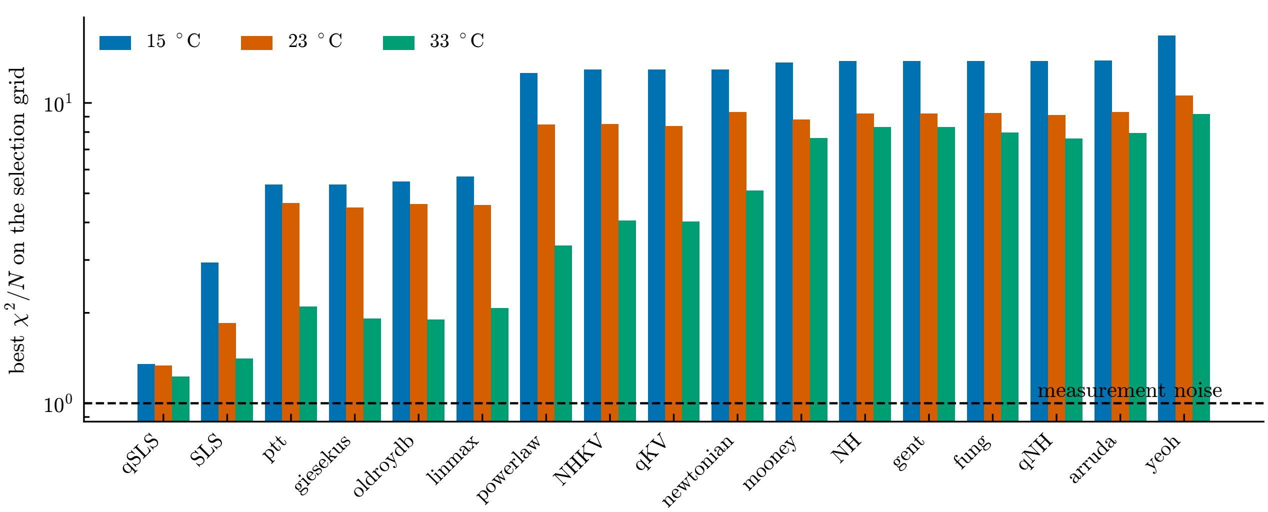

Figure 14: Best χ^2/N on the selection grid for every candidate, all three temperatures, logarithmic scale. The dashed line is the measurement-noise floor.

Figure 15 overlays the winning model on the data it was selected against.

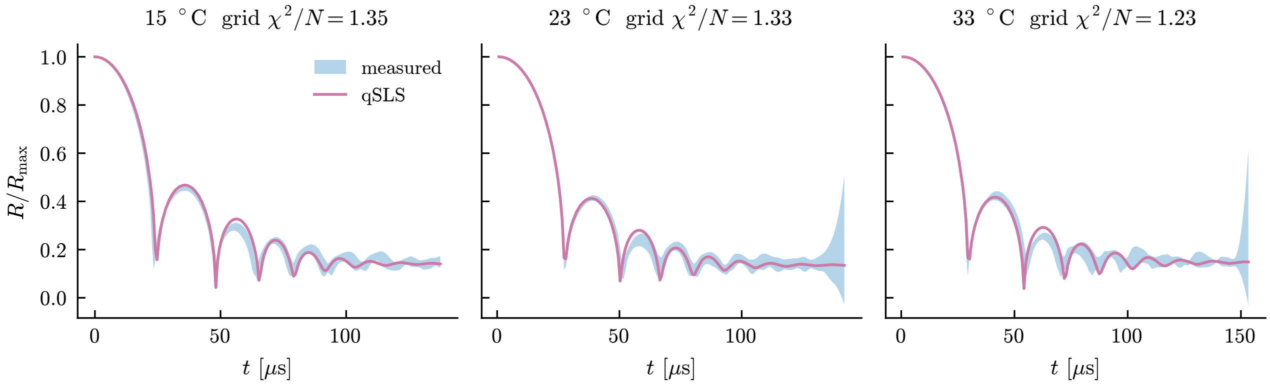

Figure 15: The selected model against the measured band it was fitted to.

17.1 Reading the posteriors

The model posteriors saturate. The winner takes essentially all the mass and the runners-up are reported as 10^{-29} or smaller. Do not read these magnitudes literally.

Evidence differences scale with sample count. With roughly 3600 samples, a modest per-sample difference in fit compounds into an astronomical Bayes factor. The defensible reading is ordinal: the winner is decisive and the ordering means something, the magnitudes do not. Quote the residual χ^2/N. Section C.6 sets out which normalisation each reported log evidence uses, and why numbers from different sections must not be differenced.

18 The bubble-dynamics operator

Every comparison above varies the material and holds the bubble-dynamics equation at Keller–Miksis. That is an assumption. With the material at its fit, changing only the operator moves the trace by four to fourteen times the median noise, which exceeds any constitutive difference measured here.

The operators are not candidate materials, so they are compared differently. We hold the candidate fixed and vary the forward solve. The parameter space is then identical across the set, the Occam terms cancel, and the difference in $\log Z$ is a Bayes factor between operators.

18.1 What the operators are

The set is a product of two choices rather than a list of equations, and reading it as a list hid a defect. The enthalpy forms share one equation: they differ only in which equation of state supplies the wall enthalpy and whether the sound speed is the constant c_∞ or the local wall value. The name *Keller–Miksis* was doing double duty, standing for both the pressure form and the constant-sound-speed enthalpy form, which are different equations.

Separating the axes gives four dynamics that need an equation of state and two that do not:

dynamics	order	closure
Rayleigh–Plesset	incompressible	none
Keller–Miksis	$O(M)$, pressure form	none
Keller enthalpy	$O(M)$, enthalpy form, $\lambda = 0$	Tait, Mie–Grüneisen or NASG
Herring	$O(M)$, enthalpy form, $\lambda = 1$	Tait, Mie–Grüneisen or NASG
Gilmore	Kirkwood–Bethe, local c	Tait, Mie–Grüneisen or NASG
Lezzi–Prosperetti	$O(M^2)$, implicit in $\ddot{R}$	Tait, Mie–Grüneisen or NASG

Fourteen operators, of which six were available when the results below were computed. Keller enthalpy and Herring are the $\lambda = 0$ and $\lambda = 1$ members of the one-parameter family of Prosperetti and Lezzi [18], not two theories; NASG [19] adds the covolume Tait lacks and reduces to Tait exactly at zero covolume.

The second-order equation, and two typos in reading it. Equation 8.7 of Lezzi and Prosperetti [20] is the only member here that is not first order in the Mach number, and the only one implicit in the acceleration: an $R^2\ddot{R}^2/c^2$ term makes it quadratic, so it is solved rather than divided, on the branch that stays finite as $c \to \infty$. It is used at the authors' own recommended $(\lambda, \theta) = (0.5, 0)$.

A secondary source we worked from first prints two errors in the pair, and both were caught before implementation, the first because $\lambda = 0$ must reduce to the Keller form, the second because the term multiplying $\dot{h}_B$ is dimensionally impossible as printed. Neither is a criticism of that paper so much as an argument for reading the source: coefficients are exactly the thing a transcription loses.

The same equation carries a group of $O(M^2)$ far-field terms that vanish only for a steady drive. This implementation drops them and *refuses* a time-varying one, rather than quietly losing the order the equation exists to supply.

Validating a coefficient no asymptotic argument reaches. Members of a family agree to one order beyond the family's accuracy, so varying λ or θ tests only the terms they multiply, and the constants 14/5 and 16/15 multiply neither. With no compressible reference solver available, the gate is direct: take the acceleration the solver returns, substitute it into 8.7 transcribed independently, and require the residual to vanish. It does, to 10^{-11}, and six mutations of the coefficients each break it.

That gate exists because a weaker one passed. The first check on the first-order family, that its members agree to $O(M^2)$, gave the right ratios and survived a deliberate error, since members agree at that order for *any* constant in the $\dot{R}^2$ coefficient. The test that works compares against the independently written pressure form, which has no free constant in it, and it needs the sound speed raised together with the equation of state or the two forms describe different liquids.

The prior box sets the answer. Scored with the bounds of Section 3.4, Keller enthalpy/Mie–Grüneisen ranks first at +4.02 nats and Gilmore/Tait second to last at −27.66. Three of the six fits sit on $g = 10^2$ or $\lambda_1 = 10^{-7}$, their own floors. Widening the box to $g \in (10^0, 10^6)$ and $\lambda_1 \in (10^{-9}, 10^{-2})$ reverses the ranking: Gilmore/Tait leads at +21.93 and Keller enthalpy/Mie–G falls to +7.32.

Widening moves the pinning rather than removing it. Five of six fits then sit on $\alpha = 10^3$ with g collapsed to 1.1 Pa to 3.5 Pa. Figure 26 shows the mechanism: the optimizer slides along the $g\alpha$ valley until it meets an edge.

Identified coordinates. Fixing g/α and fitting the product $g\alpha$ removes the pinning entirely, with none of 54 fits on a bound. We repeat at three ratios spanning two decades and on all three datasets. An ordering counts as settled when one dataset decides it by more than a nat and none contradicts it.

ordering	15 °C	23 °C	33 °C
compressible > Rayleigh–Plesset	+162 to +182	+74 to +79	+74 to +75
Gilmore/Tait > Keller–Miksis	**+17.1**	**+4.2**	**+1.2**
Keller enthalpy/Tait > Keller–Miksis	+5.8	+2.0	
Keller enthalpy/Mie–G > Keller–Miksis	+4.5	+1.5	

Twelve of the fifteen pairwise orderings are settled. Compressibility is required on every dataset. The pressure form assumed by every candidate in this package is beaten by three separate operators, and by Gilmore/Tait on all three datasets.

Under the likelihood the residuals have, that ranking dissolves. Every margin above assumes independent samples, and Section 16 deflates such margins by $N/N_{\text{eff}} \approx 20$. Applied here that puts Rayleigh–Plesset at −4.0 and −3.8 on two datasets, inside the five nats (42) calls live, while the conclusions call compressibility the most robust statement in this work. Both cannot stand, and the scalar is the wrong instrument for deciding which: it is derived for a sum of correlated terms and applied as though every model difference had the same shape, which Section 10 measures they do not.

So the evidences are recomputed with Σ in the likelihood, one exponential covariance per dataset, shared by every operator, since evidences taken under different covariances differ by $\log|\Sigma|$ and are not a Bayes factor at all. The correlation length is swept, because whitening amplifies fine structure by roughly $1/(1-\rho^2)$ and a single ρ is a single arbitrary amplification.

15 °C, nats vs. the best	independent	$\rho = 0.8$	$\rho = 0.9$	$\rho = 0.95$	fitted
Rayleigh–Plesset	−185.7	**−4.3**	−5.6	−8.5	−5.1
Keller–Miksis	−20.8	−0.15	−0.19	−0.29	−0.17
Keller enthalpy/Tait	−14.6	−0.14	−0.18	−0.29	−0.17
Gilmore/Tait	−2.2	0	0	0	0
Gilmore/Mie–G	0	−0.15	−0.20	−0.31	−0.18

Two things change and one does not. *The compressible operators become indistinguishable*: they span 0.31 nats at 15 °C and 0.33 at 23 °C, against the 17.1 and 4.2 the independent likelihood reports. Whatever separated them was information the samples did not carry. And *compressibility itself sits at the threshold rather than far above it*: Rayleigh–Plesset lands between −3.3 and −9.2 across the two datasets and the swept ρ, crossing the five-nat line depending on which ρ is taken.

What does not change is the sign. No correlated fit prefers the incompressible model, and Gilmore remains at or within a fifth of a nat of the top everywhere. The defensible statement is therefore narrower than the one above: *compressibility is preferred on every dataset and decisive on none of them at* 15 *or* 23 °C*, and the six operators are not ordered by these data at all.*

Section 22.1 sets 33 °C aside rather than reporting it: there Rayleigh–Plesset lands at -10^4 while two compressible operators separate by 200 to 550 nats, out of family with both other datasets by three orders of magnitude, which is the signature of an ill-conditioned covariance and not of physics.

A search budget produced an earlier answer. At six multistart restarts the same comparison placed Gilmore/Tait fourth on two datasets and first on the third, and reported the operator as undetermined. It also settled no ordering at 33 °C, which we read as a dataset without discriminating power. At ten starts the 33 °C fits reach $\chi^2/N = 0.42$ where six starts found 1.151.

Every operator had been held in the same poor basin. A dataset that cannot decide and a search that has not converged produce the same output. Only a better search separates them.

The initial radius is not exact. Every fit above holds R_{eq} at its inferred value, which asserts that it is known exactly. It is not: R_{max} is read off the images, but R_{eq} is inferred, and the margins in the table are small enough that this matters. Perturbing R_{eq} alone and projecting off the material span, as in Section 6, an error of 1.68 % leaves the same residual as replacing Keller–Miksis with its Mie–Grüneisen enthalpy form.

That is not a license to ignore the ranking. The two remainders lie at 58.4°, so an initial-radius error and an operator change are separable in principle, an experiment can tell them apart, and Section 6 shows the initial state absorbs under 5 % of the operator difference. What it means is narrower and worse: a quantity we set to zero is the size of the effect being reported. The remedy is to fit R_{eq} rather than assert it, which Section 19 does.

The perturbation is superlinear (the residual per unit fractional error runs 79, 93, 138 across half, one and two percent) so 1.68 % is solved for by re-solving rather than scaled from any one of those rows. Read linearly off the 1 % row it would come out a third larger.

19 Fitting the initial radius

Section 18 reports operator margins of a few nats while holding R_{eq} fixed, and a 1.68 % error in R_{eq} is worth as much as the operator change those margins measure. A quantity that large cannot be asserted. So we fit it, as a nuisance parameter with a prior. The question is not whether the evidence moves, adding a parameter always moves it, but whether the *ordering* does. An operator preferred by 17 nats with R_{eq} pinned, and not preferred once it floats, was never an operator result.

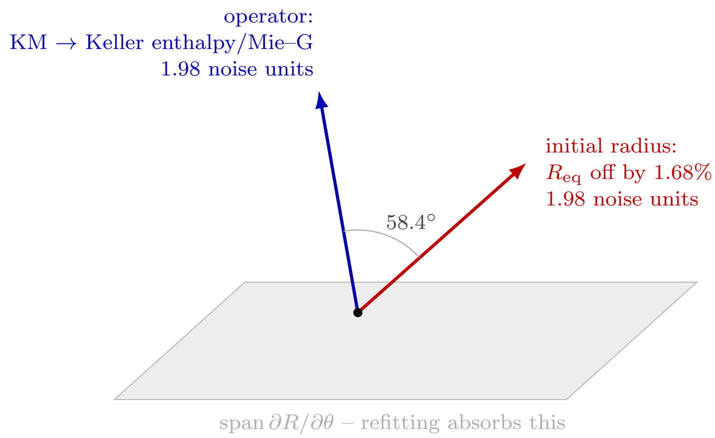

Figure 16: The size of the operator result, against the size of an error bar this work set to zero. Both arrows are what survives refitting the material, the part of a perturbation that lies outside the shaded span, and so the only part any experiment can see. They are drawn at their measured length and their measured separation. A 1.68 % error in the inferred equilibrium radius leaves exactly as much residual as replacing the bubble-dynamics operator does. The 58.4° between them is the reason this is a call to fit R_{eq} rather than to abandon the ranking: at that angle the two are separable in principle, and Section 6 confirms the initial state absorbs under 5 % of the operator difference.

The prior width is the whole argument, and nothing measures it. Too wide and R_{eq} absorbs the operator difference, which reads as "the operators are indistinguishable" while being a statement about the box. Too narrow and the pinned answer is reproduced by construction. We therefore sweep the half-width over 3 % to 25 % rather than choose one. A conclusion that holds across that range is a conclusion; one that flips inside it is a prior.

A convergence gate with no tuning constant. Every wider box contains every narrower one, and all of them contain R_{eq} at its inferred value, which is the pinned fit. The best attainable χ^2/N can therefore only fall as the width grows. A cell that rises has not found its own optimum. This matters because it caught a real failure: at ten multistart restarts, five of 72 cells failed it, one reporting −117 nats between a +13 and a +7, which reads as a prior effect and is a search that stopped early. At 24 restarts two cells fail, and both are excluded below rather than read as physics.

	15 °C		23 °C		33 °C	
operator	pinned	±25%	pinned	±25%	pinned	±25%
Rayleigh–Plesset	−165.0	**−178.6**	−76.9	**−81.8**	−74.4	**−74.2**
Keller–Miksis	0	0	0	0	0	0
Keller enthalpy/Tait	+6.1	+0.8	+2.1	+0.1	+0.8	+0.8
Gilmore/Tait	+18.5	**+5.0**	+4.3	+0.5	+1.9	−0.0
Keller enthalpy/Mie–G	+4.7	+0.3	+1.6	+0.1	+0.6	+0.9
Gilmore/Mie–G	+20.7	**+11.2**	+4.3	,	+1.1	−0.3

nats against Keller–Miksis; ", " marks the one cell the gate excluded. Of 17 orderings the pinned fit called decisive, 15 stay decisive and keep their sign at every width. None reverses. Two fall below five nats at some width, both at 15 °C.

The conclusions of Section 18 therefore survive, but not equally. *Compressibility is required* at every width and on every dataset, by 74 to 179 nats, the most robust statement in this work. *Gilmore beats the pressure form* at 15 °C, still by 5 to 11 nats with R_{eq} free. But at 23 °C and 33 °C every margin above Keller–Miksis collapses to under a nat. Those two datasets never had operator discrimination; pinning R_{eq} is what made it look as though they did.

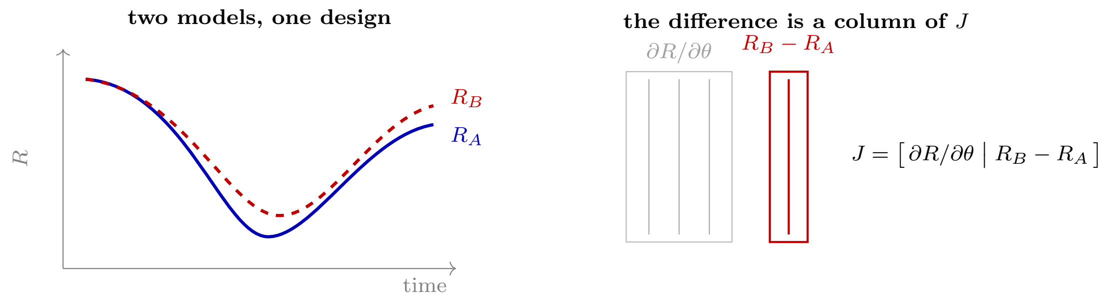

Figure 17: Turning a model choice into a parameter. Writing $R(\theta, \varepsilon) = (1 - \varepsilon)R_A(\theta) + \varepsilon R_B(\theta)$ gives $\partial R/\partial\varepsilon = R_B - R_A$ at $\varepsilon = 0$, so the difference between two models becomes one more column of the Jacobian. Discrimination and estimation then live in a single information matrix, and D-optimality over the augmented set balances learning the material against learning which model produced the trace. Nothing about the criterion changes, so the certificate still applies. The variance of ε after the material is fitted out is what says whether a design can tell the models apart at all.

The data want a larger equilibrium radius. The fitted scale is the same story told forwards, and it is the most consistent number in this study. At ±3% every one of the 18 cells pins against the upper edge. Given room, they agree closely: 1.104 to 1.115 at 15 °C, 1.074 to 1.086 at 23 °C, and 1.032 to 1.040 at 33 °C, across six operators and two prior widths.

So R_{eq} is inferred about 3 % to 11 % too small, and the fit says so whichever operator it is asked through. The residual improves accordingly, χ^2/N falls from 0.97 to 0.62 at 15 °C, 1.03 to 0.90 at 23 °C, and 0.44 to 0.38 at 33 °C. That the offset shrinks as the gel warms is a pattern we can measure but not yet explain, and it is larger than the 1.68 % that makes R_{eq} worth as much as the operator choice.

Where it actually wants to sit

The sweep above asks whether a conclusion survives letting R_{eq} float. It never asks where R_{eq} goes, and the answer is not the stored value. Fitting it on the mean trace of each dataset over a box wide enough that nothing is cut off:

dataset	stored stretch	implied	offset	χ^2/N	lack of fit
15 °C	7.09	6.40	+**10.8**%	0.971 → 0.632	16.79 → 8.80
23 °C	7.37	6.80	+8.3%	1.029 → 0.901	2.95 → 2.15
33 °C	6.83	6.59	+3.6%	0.439 → 0.379	1.80 → 1.49

Every dataset prefers a larger equilibrium radius, by three to eleven percent against the 1.68 % this section opens by calling worth an operator change.

Read against the source, most of that is not an offset. The stretch is not a single measurement. Chu et al. [6] tabulate it as a mean over the experiments in each bin, with a spread: 7.09 ± 0.36 over 18 events, 7.37 ± 1.28 over 14, and 6.83 ± 0.49 over 7. (Nucleation and imaging follow Abeid et al. [21]; the tabulated quantities are that study's own.) Against those the preferred values sit at 1.9, 0.45 and 0.49 standard deviations. Only 15 °C is outside the tabulated uncertainty at all, and it is marginal.

The definition matches too, which was the other thing worth checking. That source defines R_∞ as the equilibrium bubble-wall radius, the quantity the solver wants, so the ratio is not a different measurement pressed into service, and there is no unit or convention error to correct.

So the honest reading is not that the stretch is wrong. It is that a *population mean is being applied to every trial*, and the population is wide: the same table reports $R_{\max}$ at 277 ± 48, 298 ± 59 and 312 ± 32 µm, so individual bubbles differ by ten to twenty percent in the absolute scale that sets the Reynolds, Weber and Deborah groups. Every fit in this document uses one value for all of them.

Two consequences follow. The per-trial spreads of Section 5.2, 1.9, 6.7 and 3.1 percent in R_{eq}, are *smaller* than the scatter the source reports for the same quantity, so they are consistent with it rather than evidence of

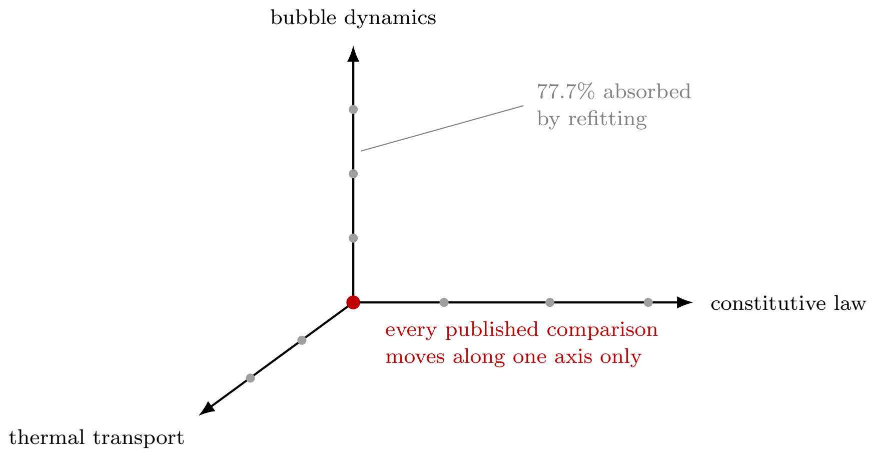

Figure 18: Three choices go into every prediction, and they are made independently: the constitutive law, the bubble-dynamics equation, and whether heat transfer is modeled. Published comparisons vary the first and leave the other two at the origin. Some of each change can be canceled by refitting stiffness and viscosity, which makes it invisible no matter how the experiment is run; what remains is 22% of an operator change, 57% of a constitutive change and 53% of a thermal one, and no two of those remainders point in similar directions. All three are therefore distinguishable in principle. The 77.7% is the CLOSEST pair of operators rather than the axis: across the six dynamics of Section 18.1 absorption runs from there down to 19.5%, and the second-order equation leaves twenty-two times as much as the pair drawn here.

anything new. And the absolute $R_{\max}$ of each event, which varies by more than the ratio does, is not fitted anywhere here and is not recoverable from the traces, since each is normalized by its own maximum before it reaches us.

What must be said plainly is what the reported parameters rest on. Freeing R_{eq} moves μ by 11 to 23 percent and $g\alpha$ by 9 to 41, and it moves λ_1 by a factor of *five and a half* at 15 °C. Those are the numbers this document reports, and they are conditional on a stretch known only to ±5 to ±17 percent.

20 Scoring the thermal treatment

Section 6 finds the thermal treatment the largest of the three model axes: it moves the trace by 16 noise units and keeps 8.5 after the material absorbs what it can. Scoring it means running the comparison of Section 18 with the thermal treatment varied and the material axes held identical, so the Occam terms cancel and the difference is a Bayes factor between physics rather than between parameter counts.

The first attempt failed, and how it failed was diagnostic. At ten multistart restarts and again at twenty-four, a 2.4× change of budget, the 23 °C rows moved by at most 0.01 nats while the others moved by 15, 153, 19 and 68. Decisively, *more restarts fitted worse*: at 33 °C bubble + medium went from $\chi^2/N = 0.428$ to 1.073. Best-of-24 cannot be worse than best-of-10 at a converged optimum, so the search was landing in different basins rather than refining one, and a third increase of the budget was not the remedy.

The remedy was a better start. Each thermal fit is now seeded from its own dataset's converged cold optimum instead of sampling a four-dimensional box afresh (`docs/writeup/thermal.py`), at twenty-four restarts. That run takes 2.7 h on six workers and recovers what the cold-started search had lost: 33 °C bubble + medium returns to 0.428, and 15 °C bubble + medium improves by 15.8 nats.

It also loses a basin the earlier run had found. At 15 °C the *bubble* treatment lands 15.0 nats worse warm-started than cold-started, because seeding from the cold optimum is a good start for a treatment close to cold and a bad one for a treatment that is not. So this is not a converged search either; it is a search whose failures have moved. Both runs are minimisations of the same objective, so the table below reads best-of-both, which is valid and is what a third run would be compared against.

dataset	cold	bubble	bubble + medium
15 °C	0	+8.6	**+21.6**
23 °C	0	+2.7	**+8.0**
33 °C	0	−5.0	**+1.7**

nats against the cold model. The bubble + medium column is positive on every dataset and is the same number in both runs at 23 °C and 33 °C, so it is a property of the data rather than of the optimiser. The bubble column changes sign across datasets and moved by 15 nats between runs at 15 °C; it is search noise and we decline to read it.

What the score is worth

Read raw, +31.2 nats combined is overwhelming. It is not, and Section 16 says why: these margins come from a likelihood that treats 201 samples as 201 independent observations, and the residuals of these very fits have lag-one autocorrelation between 0.907 and 0.962. Deflating each dataset by its own AR(1) effective sample size, $N_{\text{eff}} = N(1-\rho)/(1+\rho)$:

dataset	raw	lag-one	N_{eff} of 201	deflated
15 °C	+21.57	0.915	8.9	+0.96
23 °C	+8.01	0.962	3.9	+0.16
33 °C	+1.66	0.907	9.8	+0.08
combined	+31.24			**+1.19**

The datasets are independent, so the deflated margins add. A Bayes factor of 3.3 is substantial on Jeffreys' scale and an order of magnitude short of the five nats Section 16 requires before calling a question closed. The 23 °C dataset, whose fit was the most stable of the three, is inflated 52-fold and contributes least.

The conclusion is therefore the same as the first attempt's and rests on nothing the first attempt could see. The thermal treatment is favored (consistently, on every dataset, by the one column that is stable under both a change of optimiser and a change of budget) and it is not established. What separates those two statements is not more restarts. It is 1.19 nats against a threshold of five, which is a statement about how much data of this kind is worth, and it is the quantity Section 15 designs against.

21 Temperature dependence

The most informative thing in the comparison is not any single number but a trend (Figure 19). The advantage that the stiffening term buys over pure relaxation shrinks monotonically as the gel warms: the gap between `SLS` and `qSLS` falls from 2.94 versus 1.35 at 15 °C to 1.41 versus 1.23 at 33 °C. A warmer, softer gel needs less strain-stiffening to explain its dataset. Because this is a trend across three independent datasets rather than a single fitted value, it is harder to attribute to a fitting artifact than any individual χ^2. It is not, however, free of one: Section 22.1 finds that switching on the vapour pressure this package leaves off halves the Arrhenius slope of $g\alpha$, so about half of what follows is a graded pressure error rather than the gel.

22 Conclusions

A way to choose the next experiment that comes with a proof is what this work adds. The case for needing one is that the datasets in hand cannot settle what they are being asked.

What a dataset of this kind supports. The shear modulus is not identified separately from the stiffening exponent, so a reported modulus is a statement about the prior box, and Section 7 shows this is not a bound that can be widened away, since releasing one wall merely slides the fit onto another. A residual at the measurement scatter is not adequacy: the samples carry a twentieth of the information the likelihood counts, and a replicated lack-of-fit test rejects every candidate on every dataset. And because every candidate is

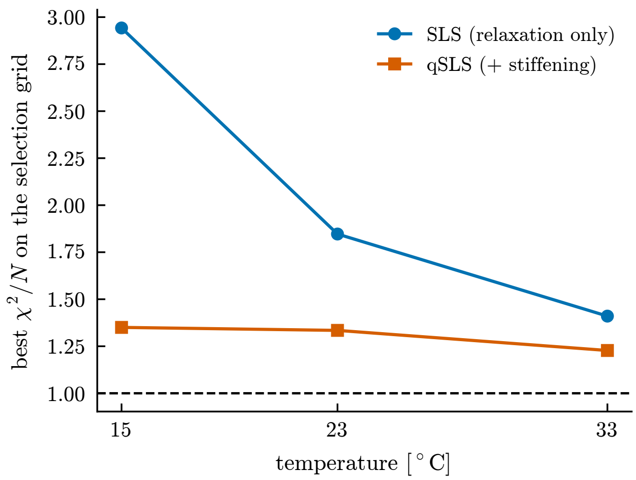

Figure 19: What the stiffening term buys, against temperature. The advantage of `qSLS` over `SLS` narrows as the gel warms.

rejected, the fitted parameters carry a model-form budget one to two orders of magnitude wider than their posterior, which (21) computes rather than laments.

What the design adds. Over measures the problem is convex, so its answer is certified and not merely converged. Every extension here was built to keep that certificate: averaging over the scatter a setpoint delivers, normalising candidates by what they cost, averaging over the bias bound rather than a point estimate, and making the measured discrepancy a coordinate. One criterion serves the material, the model label and the nuisance parameters alike, so their answers are comparable in nats, which is what makes the ranking below possible at all.

And what it says to do differently. The useful conclusion is about what to optimize rather than what the optimum is. The certified formulation was applied to the geometry, which its own numbers rank last: pricing the candidates so the bubbles-against-frames split enters the same convex problem buys 0.9 to 10 nats against at most 1.3 for the geometry, and the certificate survives the change intact because normalizing by cost is the same simplex in different coordinates. Three further questions the chapter raised are now answered, and two of the answers are negative. The screening cliff Section 16 calls a precondition is removed by a maximin over the deflation, at no worst-case cost. Staging a budget is worth nothing, because one bubble at a geometry chosen for the operator question settles every live rival, which says the operator margins reported here are small for want of a geometry, not for want of runs. And the temperature dependence cannot be identified at all without varying temperature, which no design in this literature does.

The comparison, and how much of it survives. A quadratic Zener solid is the best-fitting candidate at every temperature and under every perturbation applied to it, and neither strain stiffening nor stress relaxation suffices without the other. Two claims around it needed qualifying once the equilibrium radius was fitted rather than asserted. Compressibility is required on every dataset at every prior width, by 74 to 179 nats, the most robust statement in this work. Gilmore still beats the pressure form at 15 °C, but at 23 and 33 °C the operator margins collapse to under a nat, so those datasets never discriminated operators at all and pinning R_{eq} is what made it appear they did. The fit wants R_{eq} 3 % to 11 % larger than inferred, consistently across six operators and three datasets, and we cannot yet explain the trend with temperature.

Read with Section 16, that is the case for the design rather than a result standing beside it. The margins are decisive under a likelihood that counts 201 samples as 201 observations and are not under the one these residuals imply, and no amount of further analysis of these three datasets resolves which reading is right. Only a different experiment does.

What remains genuinely open is the one thing a criterion still cannot take as an argument: which physics is missing. Section 12 gets as far as the measured $\hat{\delta}$ allows, and the transfer it rests on is supported rather than established.

22.1 Limitations of this work

Eleven, and they are of different kinds: one set of numbers reported so it is not mistaken for evidence, one region of parameter space the winning model cannot be integrated in, one estimate the quadrature cannot make, one material property that comes out implausible for gelatin, one candidate that was never genuinely exercised, one property of the material this document did not account for, and five defects in how the fits are set up that were measured after the rest of it was written and are not corrected in it.

The 33 °C correlated fits are not usable. Recomputing the operator evidences with Σ in the likelihood behaves on two datasets and not on the third. At 15 and 23 °C the compressible operators collapse to within a third of a nat of each other and Rayleigh–Plesset lands a few nats below, which is a sensible picture of data that cannot order six similar equations. At 33 °C the same computation puts Rayleigh–Plesset at -1.1×10^4 to -2.8×10^4 and separates two compressible operators by 200 to 550 nats.

Three orders of magnitude out of family with the other two datasets is not a physical result. The mechanism is visible in the construction: whitening by an exponential covariance amplifies whatever fine structure a residual has by about $1/(1-\rho^2)$, and a residual with a little high-frequency content left in it therefore produces an enormous χ^2 that says more about the conditioning of Σ than about the model. The dataset with the fewest trials, seven against eighteen, has the noisiest spread and so the least reliable whitening. These numbers are reported here so they are not mistaken for evidence, and excluded from every conclusion.

The dropped points are outside the model, not outside the solver. Roughly one point in six of the winner's parameter box cannot be integrated, and that is not a numerical shortcoming. In that region the model expands to $R/R_0 = 2132$ after its collapse, identically at relative tolerances of 10^{-3}, 10^{-4} and 10^{-5}. So it is a converged property of the equations, not an artifact. The bubble begins at its maximum radius and its energy is finite, so growth of that size is the constitutive model failing. Refusing those points is the correct outcome.

The count of admissible points is pinned near 1958 of 2401 sampled whatever the settings. Looser tolerances and a 20× step budget convert failures into runaways, never into usable solves. This corrects an earlier reading of the same failures as a stiffness problem. The dropped points cannot be recovered by numerical effort, and no claim about the winner's fit should be made in that region.

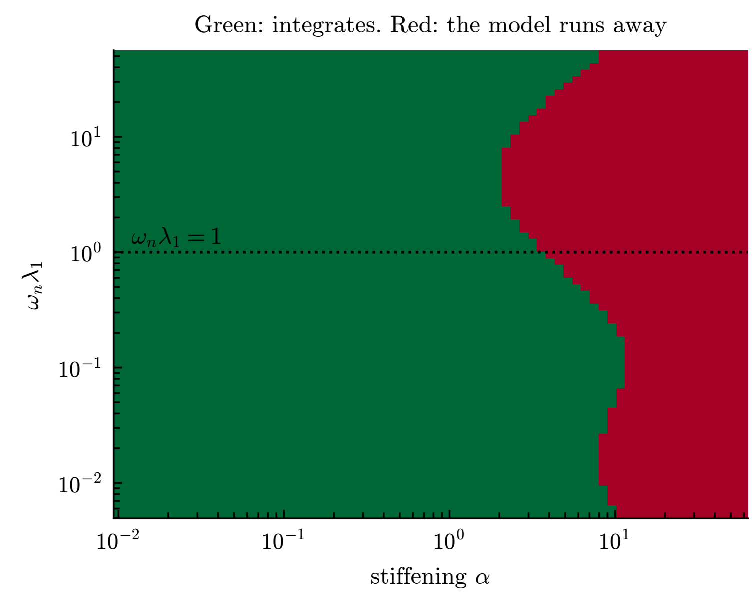

Figure 20: Where the winner can be integrated, at the fitted modulus and viscosity. Failure is governed principally by the stiffening, and the threshold falls to its lowest where the relaxation time is comparable to the bubble's own period, but it is a boundary that moves, not an isolated band.

The ranking survives; the parameter estimates do not. Independent sequential Monte Carlo runs give the winner 2173.55 ± 0.19 in log evidence against 2071.90 ± 0.38 for the relaxation-only model, a gap of 101.6 against run-to-run noise near 0.4, so the comparison is resolved by a wide margin. The tensor grid cannot say this on its own: it returns a single number with no uncertainty.

The same grid cannot estimate parameters. Refining it from 12 to 20 points per axis moves the log evidence by 17.3, and moves the best-fit point from $g = 658$, $\alpha = 1.52$ to $g = 144$, $\alpha = 8.86$. The best cell is simply whichever node lands nearest the ridge.

Refinement cannot fix this. The posterior standard deviations are 0.0018, 0.0025, 0.0055 and 0.078 in unit coordinates, against a grid spacing of 0.053. The posterior therefore occupies about 10^{-7} of the prior volume, and a uniform grid would need of order 10^{10} points. Discretization error of 17.3 sits well inside the 101.6 gap, so the ranking is safe; the reported best-fit parameters are not. Figure 21 shows which models lose points.

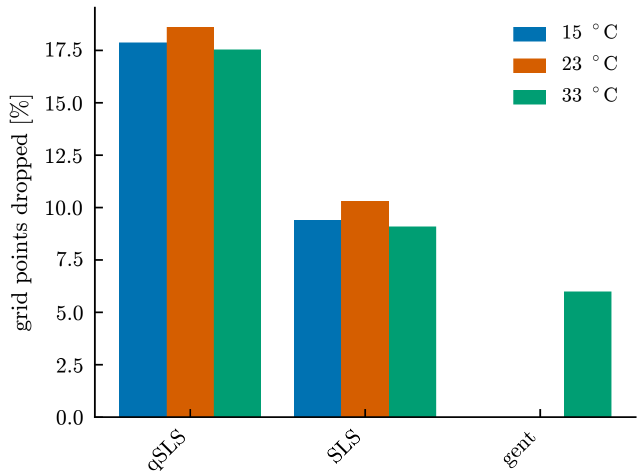

Figure 21: Fraction of each model's parameter grid that could not be integrated. Models not shown lost no points.

Consequences for the numbers reported here. Three methods that resolve the posterior (sampling, a refined grid, and continuous optimization) agree that g lies near 10^2 Pa. The coarse grid used for the ranking reports 2154 Pa. The ranking does not depend on this; any material property quoted from the grid does.

A modulus that size is also implausibly soft for gelatin. That is the degeneracy above making itself felt: the fit buys a lower residual by trading modulus against stiffening, along a direction the data barely constrain.

The datasets want a collapse five to nine percent slower than gelatin's density gives. Two densities enter every fit and they are not the same quantity. The clock carries $998\,\mathrm{kg\,m^{-3}}$, which is correct: the acquisition file normalises time by $R_{\max}\sqrt{\rho/p_\infty}$ with $\sqrt{p_\infty/\rho} = 10.07\,\mathrm{m\,s^{-1}}$ (Section 5.2), so the clock's job is to undo the experimenters' convention and it does. The equation of motion carries $\rho = 1064\,\mathrm{kg\,m^{-3}}$, the polyacrylamide value, applied to 5 % gelatin at roughly $1015\,\mathrm{kg\,m^{-3}}$.

Holding the clock and sweeping only the medium density, every dataset has a clean interior optimum, at 1200, 1120 and $1120\,\mathrm{kg\,m^{-3}}$ against gelatin's 1015. That is ten to eighteen percent too dense, and since $t_c \propto \sqrt{\rho}$ it is a collapse time five to nine percent too slow. An interior optimum says the density is *identified*; it does not say it is a density, and a well-posed optimum at a value no gel can take is a sink for model error rather than a measurement. So `RHO`= 1064 has been buying fit quality it is not entitled to, and setting the density to gelatin's true value is correct physics that makes every fit worse. What should be carried from this is the deficit, which nothing in the model explains, and not any density.

The fit target is the mean of jittered traces, which is not a trajectory. Every fit here targets the sample mean over trials at fixed lab time. The trials' first-collapse times scatter by about two percent, so averaging across that jitter fills and widens the trough: measured against the mean of the per-trial minima, the collapse minimum of the target is 19.6, 9.9 and 14.0 % too shallow. No spherical solution of any candidate passes through that curve at any parameter value, because the curve is not a solution of the equations.

Warping each trial onto a common first collapse removes the bias, completely at 23 and 33 °C and most of the way at 15. It does not touch the correlated residual of Section 5.1: lag-one stays at 0.918, 0.967 and 0.910. So this defect is real and is *not* the explanation for the structure Section 7 bounds, and the ranking is untouched, with the quadratic Zener solid best on every dataset under all four constructions of the target.

One warning belongs with that, because it inverts the obvious way to judge the repair. A sham warp, each trace given a neighbour's knots so the resampling is identical and only the alignment is broken, *fits better* than the correct one on every dataset (χ^2/N of 0.440 against 0.962 at 15 °C). Mismatched knots add jitter, jitter widens the trough, and a wider trough is easier to fit. Fit quality is therefore anti-diagnostic here, and the argument for repairing the target is that a mean of jittered traces is not a trajectory, not that the repaired fit looks better.

The vapour and mass-transfer model is not resolved at the grid this package uses. Section 20 runs at $N_t = 11$ spectral nodes and never varies it. Measured against the finest grid affordable, in the noise units Section 6 states the thermal axis in, the discretisation error at that grid is 0.7 *noise units* with `bubtherm` alone and 6.0 once vapour and mass transfer are switched on, against 8.5 for the entire thermal axis after the material absorbs what it can. The first is negligible and the second is the same order as the physics it would measure.

It is a spatial error and not a time-stepping one, which is what makes it awkward: tightening the ODE tolerance from 10^{-6} to 10^{-10} moves the collapse depth by about one percent at fixed N_t, while going from $N_t = 7$ to 11 moves it by 41. The solver reports on the converged axis and is silent about the one that is not. Refinement is slow, 71.5, 17.7, 14.9 and 6.0 noise units at $N_t = 5, 7, 9, 11$, and the $N_t = 15$ reference those are measured against has not itself settled.

That is the reason a fit through the full model cannot be trusted here, and it is a stronger reason than expense. An objective carrying six noise units of numerical error is not repaired by more restarts, and an attempt to score it produced exactly what an unresolved objective produces: fits that failed their own nesting gate, a viscosity five hundred times water's, and a reversed $g\alpha$ trend. *The thermal comparison of Section 20 is sound as run, because `bubtherm` alone is resolved at this grid; it is the completion of it that is not yet computable.*

The thermal comparison scores incomplete physics, and completing it is not affordable. Section 20 varies `bubtherm` and `medtherm` and sets neither `vapor` nor `masstrans`. The validator requires all three together, so what it scores is gas conduction with no latent heat and no condensation, which is the dominant thermal effect at the collapse of a laser bubble in water. Its +31.2 nats compares two incomplete thermal models.

Completing it costs more than this study can spend, and the reason is worth recording because it is not obvious from the physics. One forward solve takes 15 ms cold, 372 ms with `bubtherm`, and 3356 ms with vapour and mass transfer: a factor of 223, and 9 over `bubtherm` alone. At the restart budget this surface needs a single such fit is an eleven-hour job. Attempted at a quarter of that budget the search fails its own nesting gate — full physics fits *worse* than the `bubtherm` model it contains, returns a viscosity five hundred times water's, and reverses the sign of the $g\alpha$ trend — so the question is open rather than answered, and the price is the finding.

What can be said without fitting is the direction. Holding the material at its fit and solving on a grid twenty times finer than the record's, the full physics moves the first collapse *later* by 1.14, 1.97 and 3.31 % at the three temperatures. That is the sign the inflated density above would be compensating for, graded with temperature as the viscosity anomaly would require, and between a quarter and a half of the 5–9% wanted. `bubtherm` alone contributes about 0.2 %, so nearly all of it is the vapour and mass transfer Section 20 leaves out.

What gelatin is, and why that changes how μ should be read. This document has treated gelatin as a generic soft solid, and it is not one. Gelatin is built from triple helices: three polypeptide chains wound together and held by hydrogen bonds, on chains of markedly unequal length, entangling in a way that has no well-defined crosslink sites — closer to interlaced strands that come and go than to the fixed covalent network of a polyacrylamide. Warming does not break that network at one temperature. Helices unzip stochastically into single strands, and an unzipped run is fluid-like if it is short and can still carry elasticity if it is long enough to entangle (J. Estrada, personal communication; see also te Nijenhuis [22]).

Two consequences bear directly on numbers reported above. The population of floppy single strands *grows* as the gel warms, so a fitted viscosity that rises with temperature is what this structure predicts rather than a contradiction: the activation energy of $-19.3\,\mathrm{kJ\,mol^{-1}}$ is unphysical only for a μ that is a liquid viscosity obeying an Arrhenius law, and here μ is closer to a reading of how much of the network has come apart. Arrhenius is then the wrong model for it, which is a stronger objection to the fit of Section 9's temperature axis than the sign of its slope.

Our own numbers are consistent with that picture and do not establish it. The identified μ and $g\alpha$ move oppositely as it requires, $0.0437 \to 0.0662$ against $1106 \to 472\,\mathrm{Pa}$, but the simplest form of it — one helix fraction with $g\alpha$ tracking the helices and μ the coils — makes μ linear in $g\alpha$, and the two intervals give slopes differing by a factor of 4.5. Three temperatures and an unknown fraction cannot settle that either way, which is an argument for more temperatures rather than for either reading.

Vapour is off, and half the temperature trend goes with it. `SimulationConfig.vapor` defaults off and nothing in this package sets it, so the bubble carries only its non-condensable charge, about 25 Pa at $R_{\max}$. The saturated vapour pressure at the three temperatures is 1702, 2770 and 4916 Pa. The driving pressure is therefore wrong by 1.7, 2.7 and 4.9 %, which is not a constant a refit absorbs into the material but a ramp along the axis Section 21 reads as physics.

Switching vapour on, which needs neither `bubtherm` nor `masstrans` for its leading term, improves every fit: lack of fit falls from 16.79, 2.95 and 1.80 to 15.79, 2.72 and 1.60, and χ^2/N with it. That makes this the one defect in this list whose correction the data actually want. And it halves the trend: the Arrhenius slope of $g\alpha$ falls from +3564 to 1797 K, with the 33 °C value moving from 513 to 794 Pa. *Roughly half of the temperature dependence this document reports is the graded pressure error and not the gel.*

Two things do not move. The far-field temperature is fixed at 298.15 K on all three datasets, which is wrong and is also inert here: T_8 enters only through p_v and through conductivities gated on `bubtherm`, so with both off the fits are identical to the last printed digit. And the identified μ still *rises* as the gel warms, an Arrhenius slope of $-2317\,\mathrm{K}$ and an activation energy of $-19.3\,\mathrm{kJ\,mol^{-1}}$ against -18.1 before. No liquid or gel has a negative activation energy for viscosity, water being +16; whatever μ is absorbing, it is not the missing vapour.

One candidate was never genuinely exercised. The Gent solid appears in the comparison but cannot be tested on these datasets. Its energy diverges as $I_1 - 3 \to J_m$, and a Gent solid soft enough to be distinguishable from Neo-Hookean predicts a collapse it cannot sustain, so it locks up. Where it does integrate, it differs from Neo-Hookean by 1.7×10^{-4} of $R_{\max}$ against measurement noise near 2×10^{-2}, two orders of magnitude below what the data can resolve. Any statement about strain-stiffening laws here excludes it.

A How the design criterion was arrived at

The criterion of Section 10 replaced two earlier formulations. Both are recorded here: they are correct, and the reasons they were set aside are the reasons the present form looks the way it does.

Discrimination as an integral rather than a minimum

At this stage the design results of Section 9 scored a design by T, defined through a *global minimum* over the rival's parameters. That definition is the trouble, and the trouble is structural rather than computational. Here we replace the minimum with an integral, say why that is more robust, and report what it gives.

The model label is a parameter. Write $M \in \{M_1, M_2\}$ for the model index with prior $p(M)$, and let each model carry its own parameter θ with prior $\pi(\theta \mid M)$. Nothing distinguishes M from any other unknown, so the information a design d yields about it is the mutual information

$$I(M; Y \mid d) = \mathbb{E}_M\, \mathbb{E}_{Y|M,d} \left[\log \frac{p(Y \mid M, d)}{p(Y \mid d)} \right], \qquad p(Y \mid d) = \sum_M p(M)\, p(Y \mid M, d), \tag{48}$$

the expected reduction in entropy of M, exactly as the expected information gain above is the expected reduction in entropy of θ. Discrimination and estimation are one criterion applied to different components of the augmented unknown (θ, M). They are not competing philosophies: the earlier tension between T- and E-optimality is about *which component we care about*, not about two schools.

The quantity entering (48) is the marginal likelihood, or evidence,

$$p(Y \mid M, d) = \int p(Y \mid \theta, M, d)\, \pi(\theta \mid M)\, \mathrm{d}\theta, \tag{49}$$

the same object as (4) but now at a hypothetical design and for hypothetical data. It is an integral over θ, and that single fact is what the rest of this subsection turns on.

The criterion actually computed. Rather than (48) we score a design by the expected log Bayes factor in favor of the model that generated the data,

$$U(d) = \mathbb{E}_{Y\sim p(\cdot|M_1,d)} \left[\log p(Y \mid M_1, d) - \log p(Y \mid M_2, d)\right] = D_{\mathrm{KL}}(p(\cdot \mid M_1, d) \,\|\, p(\cdot \mid M_2, d))\,. \tag{50}$$

Recognizing it as a Kullback–Leibler divergence between the two prior-predictive distributions settles its properties. By Gibbs' inequality $U(d) \geq 0$, with equality exactly when the two predictives agree almost everywhere: *a model cannot discriminate against itself, at any design.*

That is sharp and falsifiable, and it is what the implementation is tested against. Two independently drawn banks from one family must score zero; any systematic excess is bias.

Equation (48) relates by $I(M; Y \mid d) = \frac{1}{2}[D_{\mathrm{KL}}(p_1 \| \bar{p}) + D_{\mathrm{KL}}(p_2 \| \bar{p})]$ with $\bar{p}$ the equal mixture. So (50) is the same comparison read against the rival rather than the mixture, the more direct statement of "how strongly will this experiment favor the truth".

Why an integral is robust where a minimum is not. Both criteria require an inner computation that can go wrong. The asymmetry is in how the error propagates.

For T, any optimizer returns $\hat{T} = \|m_1 - m_2(\hat{\theta}_2)\|/\sigma$ at whatever point it stopped, and since T is a minimum, $\hat{T} \geq T$ always. If $\hat{\theta}_2$ lies in the correct basin but is imprecise, stationarity gives $\hat{T} - T = O(\|\hat{\theta}_2 - \theta_2^\star\|^2)$ and the error is negligible. If it lies in the *wrong basin*, the error is $O(1)$: a fixed, unbounded overstatement with nothing in the returned value to indicate it. This is what was measured, Nelder–Mead and Gauss–Newton returning 2.39 and 1.52 at one design and reversing at another.

For the evidence, suppose the posterior has modes $k = 1, \ldots, K$ contributing $Z = \sum_k Z_k$, and a procedure finds only a subset $\mathcal{S}$. Then

$$\log \hat{Z} - \log Z = \log\left(1 - \sum_{k \notin \mathcal{S}} \frac{Z_k}{Z}\right), \tag{51}$$

so the error is controlled by the *posterior weight* of what was missed. Missing a subdominant mode costs $O(Z_k/Z)$ and missing a negligible one costs nothing. Because the contributions add, a multistart that finds several modes can sum them rather than choose between them, and each additional mode found monotonically improves the estimate. There is no analogue for a minimum, where finding more candidates only ever changes which single one is reported.

This is the whole argument for the change, and it is worth stating plainly: multimodality is a *failure mode* for a minimum and merely a *property of the integrand* for an integral.

Estimating the evidence I: prior sampling, and why it fails here. The direct estimator draws $\theta_s \sim \pi(\theta \mid M)$ and averages the likelihood,

$$\hat{Z} = \frac{1}{S}\sum_{s=1}^{S} p(Y \mid \theta_s, M, d), \qquad w_s = p(Y \mid \theta_s, M, d), \tag{52}$$

which is unbiased for Z, though $\log \hat{Z}$ is biased low by Jensen's inequality. Its reliability rides on the spread of the weights, summarized by the effective sample size

$$S_{\mathrm{eff}} = \frac{\left(\sum_s w_s\right)^2}{\sum_s w_s^2} \;\in\; [1, S], \tag{53}$$

which equals S when all draws contribute equally and 1 when a single draw dominates. It costs nothing to compute. It is also the only thing standing between this estimator and a confidently reported number with no content.

That it must fail here can be seen in advance. With N samples the log-likelihood scales as $\log p(Y \mid \theta) \simeq -\frac{1}{2}N\bar{d}(\theta)$ for a mean squared discrepancy $\bar{d}$, so weights are $w_s \propto \exp[-\frac{1}{2}N\bar{d}(\theta_s)]$ and the ratio of the two largest is exponential in N. Useful draws are those landing inside the posterior. Its width is $O(N^{-1/2})$ in each of p directions, so the expected number of useful draws is

$$\mathbb{E}[\#\text{useful}] \;\sim\; S \cdot \frac{\text{vol(posterior)}}{\text{vol(prior)}} \;\sim\; S\, N^{-p/2}. \tag{54}$$

For the present study $N = 201$, $p = 4$ and $S = 220$, giving $220 \times 201^{-2} \approx 5 \times 10^{-3}$, fewer than one useful draw.

The measurement agrees exactly. $S_{\text{eff}} = 1.0$ at *every* design tried, with reported values between 793 and 10 979 nats that are pure noise. The standard errors ran 56 to 130 throughout and looked entirely respectable. Without (53) the failure is invisible.

Estimating the evidence II: Laplace. Expand $\log[p(Y \mid \theta)\pi(\theta)]$ to second order about its maximum $\hat{\theta}$ and do the resulting Gaussian integral. As in (5),

$$\log Z \simeq \log p(Y \mid \hat{\theta}) + \log \pi(\hat{\theta}) + \tfrac{p}{2}\log 2\pi - \tfrac{1}{2}\log\det H, \qquad H = -\nabla^2 \log p(Y \mid \theta)\big|_{\hat{\theta}}. \tag{55}$$

For Gaussian data $H = J^{\mathsf{T}}J + \sum_i r_i \nabla^2 r_i$. Dropping the second term (the Gauss–Newton approximation, valid when residuals are small or the model nearly linear) leaves $H \simeq J^{\mathsf{T}}J$, which the fit already computed. Parameterizing so the prior is uniform on the unit cube makes $\pi(\hat{\theta}) = 1$, and with whitened residuals $r = (m - y)/\sigma$,

$$\log Z \simeq -\tfrac{1}{2}\|r\|^2 - \tfrac{N}{2}\log\left(2\pi\sigma^2\right) + \tfrac{p}{2}\log 2\pi - \tfrac{1}{2}\log\det\left(J^{\mathsf{T}}J\right). \tag{56}$$

The first two terms are the best fit; the last two are the Occam factor, and their sign is what prevents a more flexible model from winning on fit alone. Crucially for the present purpose, (56) does not care how sharp the likelihood is, which is precisely the regime that destroyed (52). It is affordable only because the fit uses analytic Jacobians: derivative-free minimization of the same objective costs 55 s against 7.6 s, and the Jacobian needed for $\det(J^{\mathsf{T}}J)$ is a by-product of the solve rather than an extra cost.

The two criteria are consistent where both can be computed. The criteria are not independent, and the relation between them is a check on both. If the true model fits to within noise, $\chi_1^2 \approx N$, while a rival that cannot fit leaves a residual of root-mean-square $T\sigma$, so $\chi_2^2 \approx NT^2$. Substituting into the leading term of (56),

$$\log \mathrm{BF} \;\approx\; \tfrac{1}{2}\left(\chi_2^2 - \chi_1^2\right) \;\approx\; \tfrac{N}{2}\left(T^2 - 1\right) \;\xrightarrow{T \gg 1}\; \tfrac{N}{2}T^2, \tag{57}$$

up to the Occam terms, which T has no way to express. Testing (57) against the measured values:

design	T (vs SLS)	$\frac{N}{2}T^2$ predicted	measured U (nats)
199 μm, stretch 3.55	0.56	31.5	35.4 ± 12.2
895 μm, stretch 4.93	1.52	232	150 ± 73

Two computations sharing no machinery (one a constrained least-squares minimum, the other a marginal likelihood by Laplace) agreeing to within their stated uncertainty. The residual gap at 895 μm is of the right sign and size for the Occam penalty that qSLS pays for its fourth parameter.

What it costs, and where it is not yet trustworthy. Two prices are paid for removing the minimum, and both are real.

Prior sensitivity. Z depends on $\pi(\theta \mid M)$ through the normalization of (49). Widening a prior by a factor c in each of p directions divides the evidence by roughly c^p without changing the fit at all. This is the Lindley–Jeffreys effect [23, 24], and it is not a defect to engineer away: a model whose prior is diffuse really is less supported by data that pin it down.

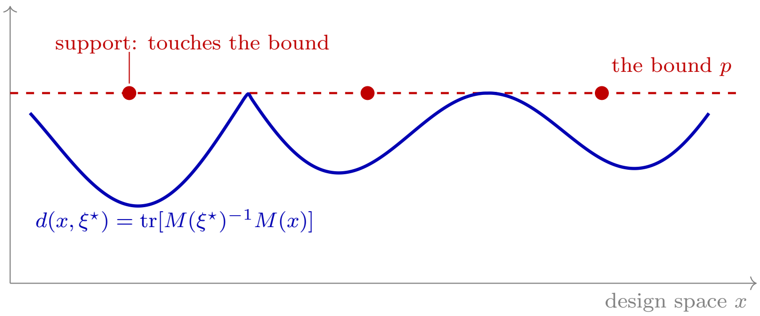

Figure 22: The certificate, which is what makes an optimal design a proof rather than a stopping rule. Kiefer and Wolfowitz: a measure $\xi^\star$ is D-optimal *if and only if* its sensitivity $d(x, \xi^\star)$ never exceeds the parameter count p anywhere in the design space, and attains p exactly on the points carrying weight. Both halves are visible at once (a curve capped by a level it touches and never crosses) so a search reports not merely that it stopped improving but that there is nowhere left for a better design to hide. The gap $\max_x d(x, \xi^\star) - p$ is the proof's residue; 10^{-9} in the designs here.

But it makes the prior part of the criterion, where T needed none, so the prior must be stated. Centering each rival's prior on its fit to the *existing* dataset produced a criterion measuring prior placement instead of discriminability, with every Laplace expansion invalid. The results here use the wide, physically motivated box of the model-selection grid, identical across models on every shared parameter.

Boundary pinning. Equation (55) presumes an interior maximum. In 50–90% of the fits at each design the rival's best parameters lie on a face of the prior box, where the expansion does not apply and the value is not to be trusted.

That it happens is itself informative. A rival that can imitate qSLS only by driving a parameter to the edge of physical plausibility has, in a meaningful sense, been discriminated against. But saying so needs a criterion built for a constrained maximum, not a quadratic expansion about an interior one. Which parameter pins, and whether the pinning is physical or an artifact of the box, is the outstanding step.

The honest summary: the integral criterion is better founded than T and not yet more reliable on this problem. The multimodality that made T untrustworthy is genuinely gone. What replaces it is a dependence on the prior, now explicit rather than hidden, and a boundary condition the current estimator does not handle.

Which criterion the conflict is between

The scaling argument of Section 3.7 says a design change that merely amplifies signal leaves the condition number κ untouched. Degeneracy is a property of the eigenvectors, and only a design that rotates them can remove it.

For these datasets, E-optimality preferred a larger bubble (895 μm, $\lambda_{\min} = 12.8$). Conditioning and eigenvector composition preferred a gentler collapse at the present size (277 μm at stretch 5, $\kappa = 4.2 \times 10^3$, the modulus–stiffening share of the worst direction falling from 87% to 66%). Both are correct answers to different questions, and (15) says which one matters here.

So there are two distinct disagreements in play, and they have different resolutions. The classical criteria disagree among themselves because they summarize one spectrum differently; that is a choice about what one wants to learn. Estimation and discrimination disagree because they are about different quantities altogether. The next subsection resolves the second, and in doing so dissolves the first.

B Supporting checks

Three checks the design chapter depends on and does not have room for: that rounding a measure to whole runs costs what it is reported to cost, that a design optimised under one model can be worse than the one already performed when that model is wrong, and that staging a budget across two rounds buys nothing here.

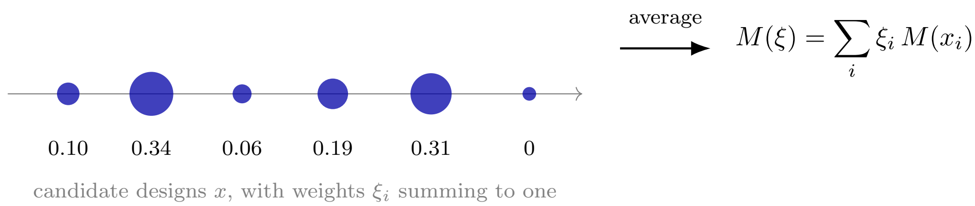

Figure 23: A design is a measure, not a point. Each candidate x_i carries an information matrix $M(x_i)$, and a design assigns weights ξ_i summing to one, read as the fraction of runs spent at that candidate. The averaged information $M(\xi)$ is linear in ξ, so any concave function of it is concave in the design, and the maximization becomes convex. That convexity is what makes Figure 22 possible: a local optimum is global, and a certificate can therefore exist. Weights of zero are the common case, so the support of the optimal measure is usually a handful of designs out of hundreds.

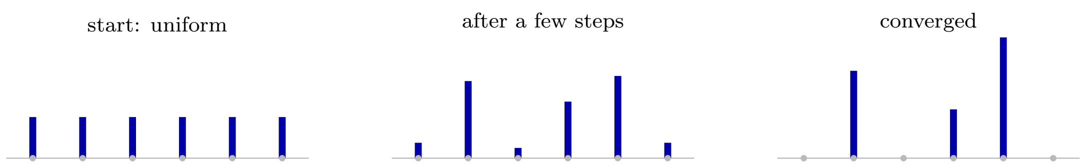

Figure 24: How the optimal design is found. Every candidate starts with equal weight, and each step multiplies a candidate's weight by how informative it is relative to the current average, then renormalizes. Weights therefore stay positive and keep summing to one by construction, and the criterion increases at every step. Candidates that are not worth running decay to zero, so the answer is a handful of settings out of hundreds. The final weights are the fraction of bubbles to run at each setting, and Figure 22 is what certifies that no other allocation does better.

A measure assigns weights; an experiment runs integers

The optimum splits runs three ways. With twelve bubbles that is 5.4/3.7/2.9, and the equivalence theorem certifies optimality over *measures*, an exact design is not one, so the certificate does not reach it. `pyimr.measure.apportion` closes the gap by the efficient rounding of Pukelsheim and Rieder, and because that rule carries no guarantee at small budgets, it reports the D-efficiency it achieved rather than asserting one.

On the four-point design of cubic regression, where every quantity is closed form: budgets divisible by four lose nothing, and six runs give 2/2/1/1 at an efficiency of 0.9428, which is $\left[(2/6)^2(1/6)^2/(1/4)^4\right]^{1/4}$ exactly. The rule earns its complexity on awkward budgets, at weights 0.7/0.2/0.1 and four runs, proportional rounding gives 3/1/0, dropping a support point and leaving the information matrix singular.

The recommended geometries do not survive model error

Optimal designs are computed assuming the model is right. Ours is not: (16) says so at every temperature. The literature on sloppy systems warns that a model can fit worse on data from its own optimal experiment, and that is testable before a bubble is spent.

The test must not generate data from the model being fitted: that is the assumption under examination, and a study that makes it finds nothing. So the truth carries physics the fitted model lacks, and cold one-mode qSLS is fitted to it at four geometries. Two mismatches are used, because one would not distinguish a property of the design from a property of the missing physics: thermal transport, which Section 6 sizes at 16 noise units, and a second relaxation mode, at 14.4.

	thermal truth			two-mode truth		
geometry	χ^2/N	$g\alpha$ err	model err	χ^2/N	$g\alpha$ err	model err
performed, 277 µm at 7.09	1.231	**3.8**%	0.53	0.905	5.0%	**0.11**
E-optimal, 199 µm at 3.55	2.533	5.9%	1.24	0.877	4.9%	0.19
discrimination-optimal, 100 µm at 20.0	2.197	**37.1**%	1.17	1.915	**60.7**%	1.06
measure support, 60 µm at 18.0	0.892	5.2%	0.15	0.909	**43.5**%	0.25

One result holds under both mismatches, and it is the strongest thing here. The **discrimination-optimal geometry is worse on both counts, both times**: it fits 1.8 and 2.1 times worse than the geometry already performed, and recovers $g\alpha$ 9.8 and 12.2 times worse. A design chosen to separate constitutive models degrades the parameter it was supposed to help measure, whichever physics is missing.

The rest does not survive the change of mismatch, and saying so is the point of running two. Under thermal error the E-optimal geometry loses on both counts; under constitutive error it is indistinguishable from the performed one (0.97 and 0.98). The 60 µm support point reverses more sharply: it carries the least model error under thermal truth and recovers $g\alpha$ 8.7 times worse under two-mode truth.

So the sloppy-systems warning is not simply confirmed. What these datasets support is narrower and, for a collaborator, worse: *which design is safe depends on which physics is missing, and we do not know which is missing*. Equation (16) says the model is inadequate on every dataset; it does not say in what direction. A design cannot be certified robust without naming the model error it is robust to, and no criterion in Section 9 takes model error as an argument. Its *magnitude* is available even though its direction is not: Equation (21) bounds what the unseen half can do to the parameters a design is optimising, which is the difference between an unquantified worry and a term one could eventually carry. Section 12 carries it, by making the measured $\hat{\delta}$ a coordinate rather than a bound, which is as far as one direction gets you, and is why the qualification in that subsection is about calibration rather than about existence.

The mechanism is visible in the model-error columns. A criterion computed under the correct-model assumption sends the experiment where the model is most *sensitive*, which is often where it is most *wrong*: the thermal mismatch grows from 0.53 noise units at the performed geometry to 1.24 at the E-optimal one, and the fitted parameters absorb it. That mechanism is general. Which geometry it punishes is not.

Staging the budget buys nothing, and why that is worth reporting

The recipe of Section 15 commits all twelve runs before any is taken, which is the wrong default when Section 16 says the rival list itself is uncertain. Equation (34) prices the alternative without any new machinery: the log Bayes factor of a stage of N_1 runs is $\mathcal{N}(N_1 s_m/2,\ N_1 s_m)$ before anyone books the laser, so the batch can be simulated, re-screened at the same five nats, re-certified for the survivors, and finished.

Swept over budgets of 2 to 12 and every split of each, the gain is +0.000 rivals at every split. The reason is visible one column left: the one-shot arm already decides all four live operator rivals at a budget of *two*, at 971 µm and stretch 5.62. There is nothing for a second stage to reallocate.

That is a statement about the geometry rather than about staging, and it is the useful part. The operator margins this document reports from the datasets are small, under a nat at 23 and 33 °C once R_{eq} floats (Section 19), because the datasets sit at 277 µm and stretch 7.09, not because the question is hard. At a geometry chosen for it, one bubble settles it.

Two things had to be fixed before this number meant anything, and both are worth naming because both looked like results. Run without Step 0, the rival list includes Rayleigh–Plesset, whose separation from the compressible operators spans eleven decades and saturates every budget, compressibility is not in doubt, by 74 to 182 nats, and screening is what removes it. And the constitutive axis has no live rival at 15 °C at all: SLS trails by 1496 nats, or 74 deflated, so a test that appeared to run on it was matching `SLS` as a substring of `qSLS` and scoring the incumbent against itself.

C Statistical background

This appendix develops, from first principles, the results the main text uses. It is self-contained and assumes only calculus and elementary probability.

C.1 Inference as conditioning

Write θ for parameters, y for data, M for a model. The product rule $p(\theta, y) = p(y \mid \theta)\, p(\theta) = p(\theta \mid y)\, p(y)$ rearranges to Bayes' theorem,

$$p(\theta \mid y, M) = \frac{p(y \mid \theta, M)\, \pi(\theta \mid M)}{p(y \mid M)}, \qquad p(y \mid M) = \int p(y \mid \theta, M)\, \pi(\theta \mid M)\, \mathrm{d}\theta. \tag{58}$$

The denominator is a normalizing constant in θ, which is why it is ignored when estimating parameters and why it is the entire content of model comparison: applying (58) at the level of models gives

$$\frac{p(M_1 \mid y)}{p(M_2 \mid y)} = \underbrace{\frac{p(y \mid M_1)}{p(y \mid M_2)}}_{\text{Bayes factor } B_{12}} \cdot \frac{p(M_1)}{p(M_2)}. \tag{59}$$

Everything the data say about which model to prefer is in B_{12}. Jeffreys' conventional reading takes $\log B > 1$ as substantial and $\log B > 5$ as decisive; the gap reported here is over 100, which is why its numerical resolution had to be checked rather than assumed.

C.2 The score, and two definitions of information

Let $\ell(\theta) = \log p(y \mid \theta)$. The *score* is $u = \nabla_\theta \ell$. Since $\int p(y \mid \theta)\, \mathrm{d}y = 1$ for every θ, differentiating under the integral gives

$$\mathbb{E}_\theta[u] = \int \frac{\nabla p}{p}\, p\, \mathrm{d}y = \nabla \int p\, \mathrm{d}y = 0, \tag{60}$$

so the score has zero mean at the true parameter. Its covariance defines the Fisher information, $F = \mathbb{E}[uu^\mathsf{T}]$. Differentiating the same identity twice gives the equivalent and more useful form

$$F = \mathbb{E}\big[-\nabla^2 \ell\big]\,, \tag{61}$$

the expected curvature. The two agree because $\nabla^2 \log p = \nabla^2 p/p - (\nabla p/p)(\nabla p/p)^\mathsf{T}$ and the first term integrates to zero. Curvature and variance-of-score are the same object, which is the reason a sharply peaked likelihood and a precisely determined parameter are the same statement.

C.3 The Cramér–Rao bound

For an unbiased estimator $\hat\theta(y)$, differentiate $\mathbb{E}[\hat\theta] = \theta$ under the integral to get $\mathbb{E}[\hat\theta\, u^\mathsf{T}] = I$. Applying the Cauchy–Schwarz inequality to the pair $(\hat\theta - \theta,\ u)$ in the scalar case gives $\mathrm{Var}(\hat\theta)\, \mathbb{E}[u^2] \geq 1$, i.e. $\mathrm{Var}(\hat\theta) \geq 1/F$; the multivariate statement is

$$\mathrm{Cov}(\hat\theta) \succeq F^{-1}. \tag{62}$$

So F^{-1} is a floor on achievable precision, attained asymptotically by maximum likelihood: $\sqrt{N}(\hat\theta_N - \theta) \to \mathcal{N}(0, F_1^{-1})$ under regularity. This is the license for reading F^{-1} as a covariance and its eigenvectors as the axes of a confidence ellipsoid, and the regularity conditions are exactly what near-singular F threatens, which is why sloppiness is worth detecting rather than tolerating.

C.4 Gaussian data, and why $F = J^\mathsf{T} J$

With independent Gaussian observations of known deviation, $\ell = -\frac{1}{2}\sum_i (y_i - m_i(\theta))^2/\sigma_i^2 + \text{const}$. Then

$$\frac{\partial \ell}{\partial \theta_j} = \sum_i \frac{y_i - m_i}{\sigma_i^2}\frac{\partial m_i}{\partial \theta_j}, \qquad -\frac{\partial^2 \ell}{\partial \theta_j \partial \theta_k} = \sum_i \frac{1}{\sigma_i^2}\left[\frac{\partial m_i}{\partial \theta_j}\frac{\partial m_i}{\partial \theta_k} - (y_i - m_i)\frac{\partial^2 m_i}{\partial \theta_j \partial \theta_k}\right]. \tag{63}$$

Taking the expectation kills the second bracket because $\mathbb{E}[y_i - m_i] = 0$, leaving $F = J^\mathsf{T} J$ with $J_{ij} = \sigma_i^{-1}\partial m_i/\partial \theta_j$. Note what was discarded: the curvature of the model. At the optimum the residuals are small and the Gauss–Newton form is accurate; far from it, or with a badly misspecified model, F and the observed Hessian differ, and the discrepancy is itself a diagnostic.

C.5 The Laplace approximation and the Occam factor

To evaluate $Z = \int e^{\ell(\theta)}\pi(\theta)\, \mathrm{d}\theta$, expand about the maximum $\hat\theta$ where $\nabla \ell = 0$:

$$\ell(\hat\theta + \delta) = \ell(\hat\theta) - \tfrac{1}{2}\delta^\mathsf{T} H \delta + O(\|\delta\|^3), \qquad H = -\nabla^2 \ell(\hat\theta). \tag{64}$$

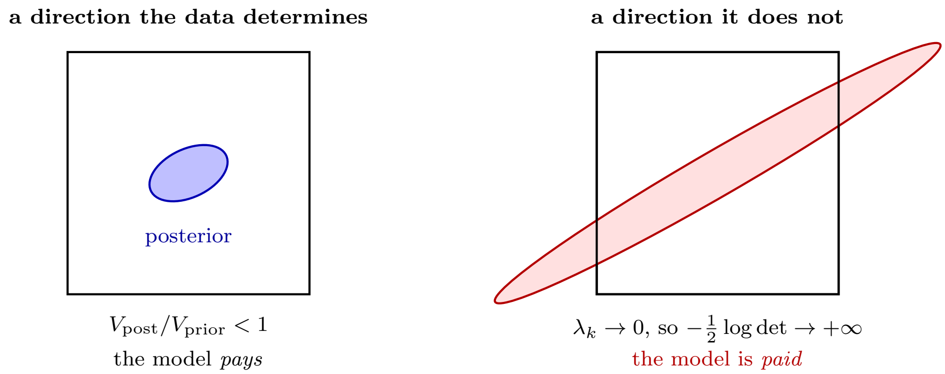

Figure 25: The Occam factor is the share of the prior the data leaves standing, $\prod_k \sqrt{2\pi/\lambda_k}$ over the eigenvalues of $J^\mathsf{T} J$. Left: every direction is narrower than its prior, so a flexible model pays for volume it does not use. Right: a direction the data does not determine has $\lambda_k \to 0$, the fitted Gaussian extends far outside the box the prior occupies, and the expansion returns a factor *greater* than one, the evidence rises when a useless parameter is added. Capping each direction at unity, (68), restores what the geometry already required; on the models of Section 5.3 the uncapped form favored the two-mode law by 29.7 nats exactly where its second arm did nothing.

Holding π constant over the peak and using $\int e^{-\delta^\mathsf{T} H\delta/2}\mathrm{d}\delta = (2\pi)^{p/2}|H|^{-1/2}$,

$$Z \simeq e^{\ell(\hat{\theta})}\,\pi(\hat{\theta})\,(2\pi)^{p/2}\,|H|^{-1/2}. \tag{65}$$

The correction is $O(N^{-1})$ relative for regular problems. Writing the posterior volume as $V_{\text{post}} = (2\pi)^{p/2}|H|^{-1/2}$ and the prior density as $1/V_{\text{prior}}$ where it is flat, the second factor is $V_{\text{post}}/V_{\text{prior}}$: the fraction of the space the data leave standing. A model with more parameters has a larger V_{prior} and pays for it unless the fit improves enough to compensate. Nothing was added by hand; the penalty is the Jacobian of the integral.

Taking logarithms of (65), using $H \approx NF_1$ for N independent observations, and dropping terms that do not grow with N,

$$-2\log Z \simeq -2\ell(\hat{\theta}) + p\log N + O(1), \tag{66}$$

which is the Bayesian information criterion. BIC is therefore not an alternative to the evidence but a coarse asymptotic of it, valid when the posterior is Gaussian, the parameters identified, and N large. AIC answers a different question, expected out-of-sample Kullback–Leibler divergence, and is not an approximation to Z at all. Where a model is sloppy, H is near-singular, $|H|^{-1/2}$ is enormous, and the asymptotic form fails while the integral remains well defined.

C.6 What a nat means here

Three sections report a log evidence for the same model on the same dataset and they disagree by a factor of four: -2289.86 from the grid quadrature, $+529.26$ from the capped Laplace expansion of Section 5.3, and $+2173.55$ from the sequential Monte Carlo of Section 22.1. Every threshold in this document is stated in nats, so this has to be said rather than left to inference.

An evidence is defined only up to the normalisation of its likelihood, and the term that moves is $-\frac{1}{2}\sum_i \log(2\pi\sigma_i^2)$. Recomputed at one fit on the 15 °C dataset:

σ convention	χ^2/N	$-\chi^2/2$	$-\frac{1}{2}\sum\log 2\pi\sigma^2$	Occam	$\log Z$
trial spread	0.971	-97.6	$+640.6$	-15.9	**527.1**
standard error, $/\sqrt{18}$	17.478	-1756.6	$+931.1$	-20.3	-845.7

The two conventions differ by 1373 nats on one residual, 290 of it the $N\log\sqrt{J}$ shift and the rest the χ^2, which scales by J rather than shifting. This is the same choice Section 5.2 shows moving χ^2/N from 0.974 to 17.54, and it moves $\log Z$ far more.

Against that decomposition the three reported numbers resolve as follows. The capped Laplace is reproduced to 2.15 nats, so Section 5.3 uses the trial spread and the expansion of (68). The grid quadrature is not this integral at all: Section 2 adds a marginalised noise scale, a redundancy weight and a BIC-style penalty, and quadratures a box rather than expanding about a mode, which accounts for its being 2817 nats away. The sequential Monte Carlo sits 1646 above the Laplace on the same convention and is *not* reconciled here.

What saves the document's conclusions is that every comparison it draws is made *within* one pipeline, SMC against SMC for the 101.6-nat constitutive gap, Laplace against Laplace for the enrichment margins. Those are valid. What is not valid, and what a reader should not attempt, is differencing a number from one section against a number from another. The five-nat threshold of (42) is meaningful only against margins computed the same way, and it is applied to Laplace-scale margins throughout.

C.7 Capping the Occam factor at the prior

The failure just noted is not merely a loss of accuracy; it has a sign, and the sign is wrong. Work in coordinates where the prior is uniform on the unit cube, so $V_{\text{prior}} = 1$ and $H = J^\mathsf{T} J$. The Occam factor is a product over the eigendirections of $J^\mathsf{T} J$,

$$\frac{V_{\text{post}}}{V_{\text{prior}}} = \prod_{i=1}^{p} \sqrt{\frac{2\pi}{\lambda_i}}, \tag{67}$$

each factor being the posterior width in that direction against a prior of width one. A direction the data does not determine has $\lambda_i \to 0$ and contributes a factor diverging to $+\infty$. The evidence then *increases* with the addition of a parameter that does nothing, which is the opposite of what the Occam factor exists to do.

The pathology is entirely an artifact of the Gaussian approximation. The fitted Gaussian has width $\sqrt{2\pi/\lambda_i} \gg 1$ in that direction and so extends far outside the cube the prior actually occupies, while the true integral, taken over the cube, simply returns the prior mass. Imposing what the geometry already requires (a posterior cannot be wider than the prior it came from) gives

$$\log \frac{V_{\text{post}}}{V_{\text{prior}}} = \sum_{i=1}^{p} \min\left(0,\ \tfrac{1}{2}\log\frac{2\pi}{\lambda_i}\right), \tag{68}$$

so a direction the data cannot see costs nothing and buys nothing. Where every $\lambda_i > 2\pi$ each minimum selects its second argument and (68) reduces identically to the logarithm of (67): the cap is a strict generalization, altering an answer only where the uncapped form was not entitled to one. Section 5.3 gives the size of the error it removes, which on the models considered there is 29.7 nats.

This matters wherever models of differing dimension are compared and some parameters are weakly identified, which, for the sloppy spectra of Section C.8, is the normal case rather than the exceptional one.

C.8 Sloppy models

Write the eigen-decomposition $F = V\Lambda V^\mathsf{T}$ with $\lambda_1 \geq \cdots \geq \lambda_p$. The misfit near the optimum is $\Delta\chi^2 = \sum_k \lambda_k c_k^2$ for $\delta = \sum_k c_k v_k$, so moving a distance c along v_k costs $\lambda_k c^2$ and the 95% contour lies at $c_k = \sqrt{3.84/\lambda_k}$. Multiparameter models of this kind generically show spectra spanning many decades, with a few stiff directions and many sloppy ones, the observation of Gutenkunst [25], Sethna and co-workers, and the reason parameter values in such models are often far less determined than their predictions.

The distinction to keep is between a rank-deficient F, where a direction is *structurally* unidentifiable and no experiment helps, and a merely ill-conditioned F, where the direction is determined but weakly. In logarithmic parameters, an eigenvector is a power-law combination, and the invariant it leaves fixed is read off its components.

C.9 Correlated errors

Let r be the residual vector with covariance C. The Gaussian log-likelihood is

$$-2\ell = r^\mathsf{T} C^{-1} r + \log|C| + N\log 2\pi, \tag{69}$$

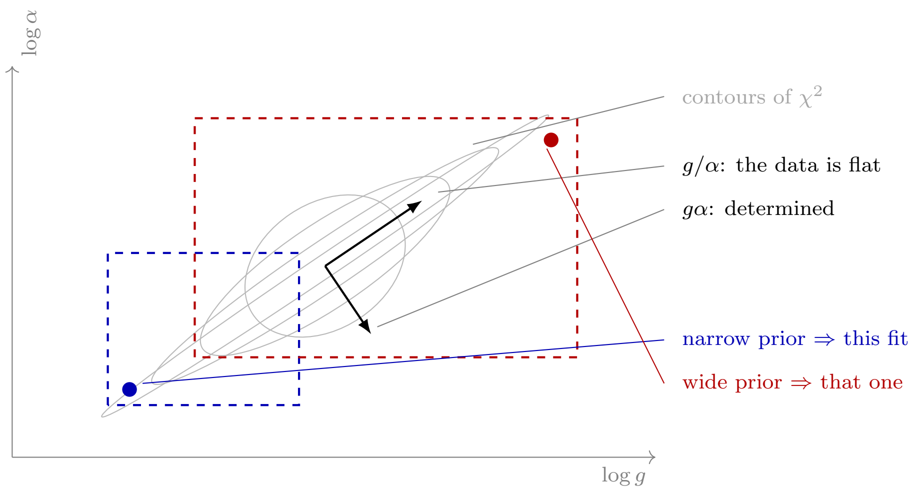

Figure 26: Why a "best fit" for g is a statement about the prior. The likelihood depends on the product $g\alpha$, so the misfit surface is a valley running along constant $g\alpha$ and the data is flat along g/α. An optimizer slides until it meets the edge of the box, and *where* it stops is set by the box: the two dashed rectangles give the two filled points, which differ in g by more than thirtyfold. This is the mechanism behind the fifty-nat swing in the operator ranking of Section 18 when the bounds were widened, and behind the order-of-magnitude disagreement in g between methods in Figure 9.

and with the Cholesky factorization $C = LL^\mathsf{T}$ this becomes $\|L^{-1}r\|^2 + 2\sum_k \log L_{kk} + \text{const}$: whitening turns generalized least squares back into ordinary least squares on transformed residuals. Ignoring the off-diagonal terms, i.e. using $\operatorname{diag}(C)$, does not merely lose efficiency, it misstates the amount of information. For a stationary sequence with autocorrelation ρ_k,

$$\operatorname{Var}\left(\sum_{i=1}^{N} r_i\right) = N\sigma^2 \left(1 + 2\sum_{k=1}^{N-1}\left(1 - \tfrac{k}{N}\right)\rho_k\right) \xrightarrow{N\to\infty} N\sigma^2 \left(1 + 2\sum_{k\geq 1}\rho_k\right), \tag{70}$$

so the sequence carries the information of $N_{\text{eff}} = N/(1 + 2\sum \rho_k)$ independent draws. The bracket is the spectral density of the residual sequence at zero frequency, which is the cleaner way to see why a long-range correlation is so costly.

Correlated residuals in a *fitted* model usually indicate model discrepancy rather than correlated measurement noise. Kennedy and O'Hagan [7] formalize this by writing $y = m(\theta) + \delta(\cdot) + \varepsilon$ with δ a Gaussian process, and Brynjarsdóttir and O'Hagan show that omitting δ leaves parameter estimates biased *and* overconfident, while an unconstrained δ absorbs the physics. Neither failure is visible in χ^2.

C.10 Optimal experimental design

A design ξ determines what is measured and hence $F(\xi, \theta)$. Since F is a matrix and designs must be ranked, a scalar functional is required, and the classical choices are

$$\Phi_D = \log\det F, \quad \Phi_A = -\operatorname{tr} F^{-1}, \quad \Phi_E = \lambda_{\min}(F), \quad \Phi_c = -c^\mathsf{T} F^{-1} c, \tag{71}$$

each maximized. In Bayesian terms, Φ_D is the expected Kullback–Leibler gain from prior to posterior under a Gaussian approximation, which is why it is the default: it maximizes information about *all* parameters jointly. That default is wrong whenever one direction is the question, because $\log\det F = \sum_k \log \lambda_k$ is dominated by the large eigenvalues.

Because F depends on θ, a local design presupposes an answer. The standard repairs are to average over a prior, $\Phi(\xi) = \mathbb{E}_\theta[\Phi(F(\xi, \theta))]$, or to take the worst case over a plausible set. Sequential design uses the posterior from completed experiments as the prior for the next, which is what is done here.

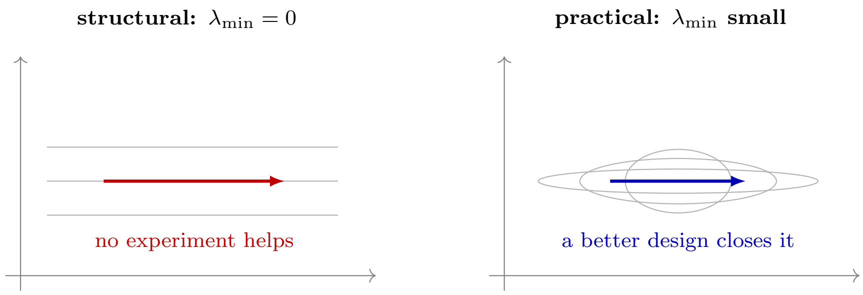

Figure 27: Two failures that look alike in a fit and differ completely in what can be done about them. Left: the misfit is exactly flat along a direction, F is singular, and the parameter combination is structurally unidentifiable. No experiment recovers it, and only a reparameterization or a fixed value helps. Right: the misfit is closed but very elongated, F is invertible and ill-conditioned, and the combination is determined weakly. Here a design that rotates the ellipse can close it, which is what Section 9 sets out to do. On these datasets $\lambda_{\min} = 5.2 \times 10^{-2} \neq 0$, so the problem is the second kind.

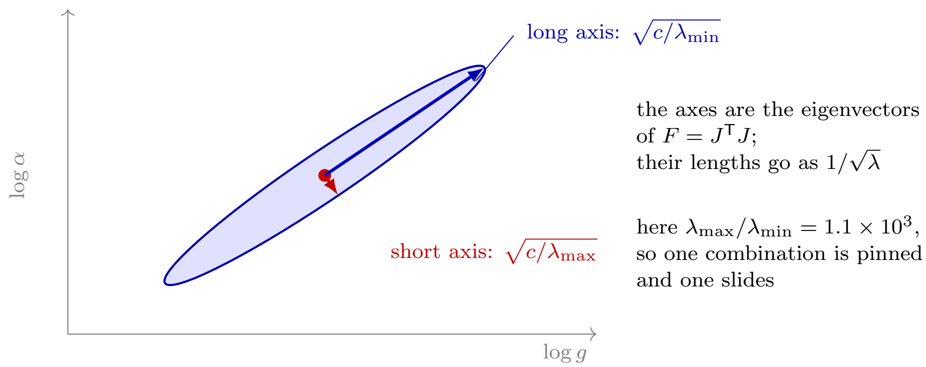

Figure 28: Why some parameter combinations are determined and others are not. Near the best fit the misfit rises as a quadratic form, so the contour of constant misfit is an ellipse whose axes point along the eigenvectors of $F = J^\mathsf{T} J$ and whose half-lengths go as $1/\sqrt{\lambda}$. A large eigenvalue is a short axis: move that way and the fit degrades at once. A small one is a long axis: move that way and the data barely object. On these datasets the ratio is 1.1×10^3, and the long direction is g up with α down at fixed product, which is the ridge of Figure 26 seen from above. The ellipse is drawn at an aspect of about $8 : 1$; the measured aspect is $\sqrt{1.1 \times 10^3} \approx 33 : 1$, which would render as a line.

When the model itself is uncertain, all of the above presuppose the wrong thing. T-optimality (Atkinson and Fedorov) maximizes instead the lack of fit of the inferior model,

$$\Phi_T(\xi) = \min_{\theta_2} \left\| m_1(\hat{\theta}_1; \xi) - m_2(\theta_2; \xi) \right\|^2, \tag{72}$$

which asks which experiment the candidates most disagree about. Its virtue here is that it shares no assumptions with the Fisher criteria, so agreement between them is evidence rather than restatement.

D Reproducing

```
.venv/bin/python examples/measured_selection.py <dataset> [thermal Nt] [workers]
.venv/bin/python docs/writeup/collect.py 16     # datasets results.json
.venv/bin/python docs/writeup/figures.py        # draws every figure from it
```

Every number and figure in this document comes from `results.json`; none is transcribed by hand.

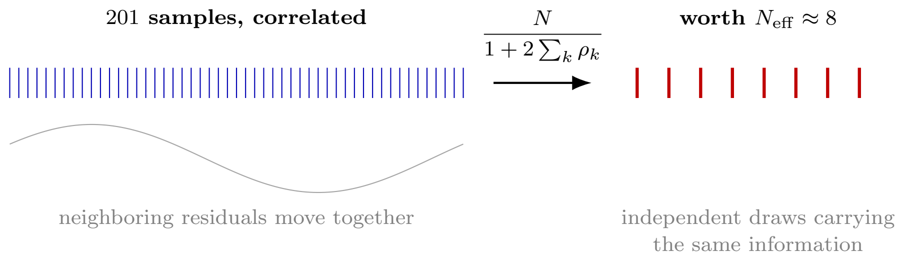

Figure 29: Why 201 samples do not carry 201 samples' worth of evidence. The residuals here are strongly correlated from one sample to the next, so neighboring points repeat information rather than adding it. A correlated sequence carries what $N_{\text{eff}} = N/(1 + 2\sum_k \rho_k)$ independent draws would, and on these fits N_{eff} is near 8. Both χ^2/N and the evidence assume the samples are independent, so each overstates the information by roughly $N/N_{\text{eff}} \approx 25$. A margin of a few nats between two models is inside that factor.

Code and data availability

Every number and figure in this document is reproduced by scripts in the repository that contains it: `figures.py`, `figures_geometry.py`, `figures_identifiability.py`, `figures_measure.py` and `figures_trial.py` draw the figures, `fit_quality.py` the fit table and `trial_variation.py` the trial-spread analysis, each from the tracked `results.json`. The design measures use `pyimr.measure`, the discrimination criteria `pyimr.discriminate`, and the trade-off front `pyimr.pareto`.

The design results added in Sections 12 and 14 to 16 come from `allocation_measure.py` (the priced joint measure), `discrepancy_design.py` ($\hat{\delta}$ as a coordinate), `correlated_design.py` (the correlated likelihood), `screen_maximin.py` (the deflation maximin), `two_stage.py` (staging) and `temperature_design.py` (the Arrhenius axis). The measured datasets themselves are not redistributed here.

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